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## trTeControl

**Address:** 0x42001000

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 32 bits

**Reset Value:** 0x01000068

**Description:** None

| Field Name              | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|-------------------------|------------|-----------|----------|--------|-------------|-------------|
| trteactive              | 0          | 1         | false    | WARL   | 0x0         |             |
| trteenable              | 1          | 1         | false    | WARL   | 0x0         |             |
| trteinstracing          | 2          | 1         | true     | WARL   | 0x0         |             |
| trteempty               | 3          | 1         | true     | WARL   | 0x1         |             |
| trteinstmode            | 4          | 3         | false    | WARL   | 0x6         |             |
| rsvd87                  | 7          | 2         | -        | WPRI   | -           | Reserved    |
| trtecontext             | 9          | 1         | false    | WARL   | 0x0         |             |
| rsvd1010                | 10         | 1         | -        | WPRI   | -           | Reserved    |
| trteinstriggerenable    | 11         | 1         | false    | WARL   | 0x0         |             |
| trteinststalloroverflow | 12         | 1         | true     | WARL   | 0x0         |             |
| trteinststallena        | 13         | 1         | false    | WARL   | 0x0         |             |
| rsvd1414                | 14         | 1         | -        | WPRI   | -           | Reserved    |
| trteinhibitsrc          | 15         | 1         | false    | WARL   | 0x0         |             |
| trtesyncmode            | 16         | 2         | false    | WARL   | 0x0         |             |
| rsvd1918                | 18         | 2         | -        | WPRI   | -           | Reserved    |
| trtesyncmax             | 20         | 4         | false    | WARL   | 0x0         |             |
| trteformat              | 24         | 3         | false    | WARL   | 0x1         |             |
| rsvd3227                | 27         | 5         | -        | WPRI   | -           | Reserved    |

## trTeInstFeatures

**Address:** 0x42001008

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 32 bits

**Reset Value:** 0x40000000

**Description:** None

| Field Name                | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|---------------------------|------------|-----------|----------|--------|-------------|-------------|
| trteinstnoaddrdiff        | 0          | 1         | false    | WARL   | 0x0         |             |
| trteinstnotrapaddr        | 1          | 1         | false    | WARL   | 0x0         |             |
| rsvd72                    | 2          | 6         | -        | WPRI   | -           | Reserved    |
| trteinstenrepeatedhistory | 8          | 1         | false    | WARL   | 0x0         |             |

|             |    |    |       |      |     |          |
|-------------|----|----|-------|------|-----|----------|
| rsvd159     | 9  | 7  | -     | WPRI | -   | Reserved |
| trtesrcid   | 16 | 12 | false | WARL | 0x0 |          |
| trtesrcbits | 28 | 4  | false | WARL | 0x4 |          |

### trTeInstFilters

**Address:** 0x4200100C

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** None

| Field Name      | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|-----------------|------------|-----------|----------|--------|-------------|-------------|
| trteinstfilters | 0          | 16        | false    | WARL   | 0x0         |             |
| rsvd3216        | 16         | 16        | -        | WPRI   | -           | Reserved    |

### trTeFilter0Control

**Address:** 0x42001400

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** None

| Field Name               | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|--------------------------|------------|-----------|----------|--------|-------------|-------------|
| trtefilterenable         | 0          | 1         | false    | WARL   | 0x0         |             |
| trtefiltermatchprivilege | 1          | 1         | false    | WARL   | 0x0         |             |
| rsvd322                  | 2          | 30        | -        | WPRI   | -           | Reserved    |

### trTeFilter0MatchInst

**Address:** 0x42001404

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** None

| Field Name                     | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|--------------------------------|------------|-----------|----------|--------|-------------|-------------|
| trtefiltermatchchoiceprivilege | 0          | 8         | false    | WARL   | 0x0         |             |
| rsvd328                        | 8          | 24        | -        | WPRI   | -           | Reserved    |

### c\_ResetVector

**Address:** 0x42005300

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Reset Vector

| Field Name  | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|-------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| rsvd_1_0    | 0          | 2         | true     | WARL   | 0x0         | Reserved.                                                  |
| resetvector | 2          | 55        | true     | WARL   | FIXME       | Reset Vector                                               |
| rsvd_62_57  | 57         | 6         | true     | WARL   | 0x0         | Reserved.                                                  |
| lock        | 63         | 1         | true     | WARL   | 0x0         | Lock bit, when set, write to ResetVector field is ignored. |

### c\_fe\_error\_header\_vendor

**Address:** 0x42008000

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000821

**Description:** Vendor and implementation ID

| Field Name          | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                 |
|---------------------|------------|-----------|----------|-----------|-------------|-------------------------------------------------------------------------------------------------------------|
| vendor_id           | 0          | 32        | false    | Read-Only | 0x821       | Provides the JEDEC manufacturer ID of the provider of the component hosting the error bank                  |
| imp_id              | 32         | 16        | false    | Read-Only | 0x0         | Provides a unique identity, defined by the vendor, to identify the component and revisions of the component |
| wpri_reserved_63_48 | 48         | 16        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                |

### c\_fe\_error\_header\_bank\_info

**Address:** 0x42008008

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0100000000420000

**Description:** Error bank information

| Field Name          | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                 |
|---------------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| inst_id             | 0          | 16        | false    | Read-Only | 0x0         | Identifies a unique instance of an error bank within a package or die, ideally in the whole system; Not being used, tied to 0                                                               |
| n_err_recs          | 16         | 6         | false    | Read-Only | 0x2         | Indicates the number of error records implemented by the error bank                                                                                                                         |
| layout              | 22         | 2         | false    | Read-Only | 0x1         | Indicates the layout of the registers in the error bank and the error records. The layout encoding 0 indicates the registers are arranged and have meaning as defined by the specification. |
| wpri_reserved_55_24 | 24         | 32        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                |
| version             | 56         | 8         | false    | Read-Only | 0x1         | Version of the arch register layout specification implemented by the error bank                                                                                                             |

### c\_fe\_error\_header\_valid\_summary

**Address:** 0x42008010

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000001

**Description:** Summary of valid error records

| Field Name   | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                    |
|--------------|------------|-----------|----------|-----------|-------------|------------------------------------------------------------------------------------------------|
| sv           | 0          | 1         | false    | Read-Only | 0x1         | Indicates that the valid_bitmap provides a summary of the valid bits from the status registers |
| valid_bitmap | 1          | 63        | true     | Read-Only | 0x0         | Summary of the valid bits from status registers in the error bank                              |

### c\_fe\_error\_record\_0\_control

**Address:** 0x42008040

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000001

**Description:** Control register of error record 0

| Field Name           | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                        |
|----------------------|------------|-----------|----------|-----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| else                 | 0          | 1         | false    | WARL      | 0x1         | If 1, the hardware unit logs and signals errors in the error record. If 0, the hardware unit will not log RAS errors                                                                                               |
| cece                 | 1          | 1         | false    | WARL      | 0x0         | If 1, hardware will count corrected errors in the cec field of the status register. If 0, cec does not count and retains its value.                                                                                |
| ces                  | 2          | 2         | false    | WARL      | 0x0         | Used to enable signaling of corrected errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal.            |
| ueds                 | 4          | 2         | false    | WARL      | 0x0         | Used to enable signaling of uncorrected deferred errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| uecs                 | 6          | 2         | false    | WARL      | 0x0         | Used to enable signaling of uncorrected critical errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| wpri_reserve_d_31_8  | 8          | 24        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| eid                  | 32         | 16        | true     | WARL      | 0x0         | Error injection delay                                                                                                                                                                                              |
| sinv                 | 48         | 1         | true     | WARL      | 0x0         | Status register invalidate. When written to 1, causes the valid (v) field of the status register to be cleared.                                                                                                    |
| srdep                | 49         | 1         | true     | WARL      | 0x0         | Set read in progress. When 1, causes read in progress (rdip) to be set                                                                                                                                             |
| wpri_reserve_d_55_48 | 50         | 10        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |

|               |    |   |       |           |     |          |
|---------------|----|---|-------|-----------|-----|----------|
| custom_unused | 60 | 4 | false | Read-Only | 0x0 | Reserved |
|---------------|----|---|-------|-----------|-----|----------|

## c\_fe\_error\_record\_0\_status

**Address:** 0x42008048

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Status register of error record 0

| Field Name          | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                            |
|---------------------|------------|-----------|----------|-----------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v                   | 0          | 1         | true     | WARL      | 0x0         | If 1, error record holds a valid error                                                                                                                                                 |
| ce                  | 1          | 1         | true     | WARL      | 0x0         | Set to 1 if the detected error was corrected                                                                                                                                           |
| ued                 | 2          | 1         | true     | WARL      | 0x0         | Set to 1 if the error could not be corrected but was deferred                                                                                                                          |
| uec                 | 3          | 1         | true     | WARL      | 0x0         | Set to 1 if the error could not be corrected or deferred and needs critical handling by RAS handler                                                                                    |
| pri                 | 4          | 2         | true     | WARL      | 0x0         | Priority of the error. Indicates relative priority among errors of the same class.                                                                                                     |
| mo                  | 6          | 1         | true     | WARL      | 0x0         | Set to 1 if more errors of the same class and severity occur                                                                                                                           |
| c                   | 7          | 1         | true     | WARL      | 0x0         | Containable bit. If 1, indicates that the error has not propagated beyond the boundaries of the hardware unit                                                                          |
| tt                  | 8          | 3         | true     | WARL      | 0x0         | Transaction type. 0=Unspecified or NA, 1=UNSPECIFIED, 2-3=Reserved for future standard use, 4=Explicit read, 5=Explicit write, 6=Implicit read, 7=Implicit write                       |
| iv                  | 11         | 1         | true     | WARL      | 0x0         | Information valid, set to 1 if error reports additional information in the info register                                                                                               |
| at                  | 12         | 4         | true     | WARL      | 0x0         | Address type, indicates the type of address reported in the addr register. 0=None, 1=Supervisor Physical Address, 2=Guest Physical Address, 3=Virtual Address, 4-15=Component Specific |
| siv                 | 16         | 1         | true     | WARL      | 0x0         | Supplemental information valid, set to 1 if error reports supplemental information in the suppl_info register                                                                          |
| tsv                 | 17         | 1         | true     | WARL      | 0x0         | Timestamp valid, set to 1 if the error record holds a valid timestamp in the timestamp register                                                                                        |
| lv                  | 18         | 1         | true     | WARL      | 0x0         | Locator valid, set to 1 if the error record holds a valid timestamp in the locator register                                                                                            |
| wpri_reserved_19_19 | 19         | 1         | true     | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                           |
| scrub               | 20         | 1         | true     | WARL      | 0x0         | Set to 1 to indicate that the storage location has been updated with the corrected value. Only valid for corrected errors.                                                             |
| ceco                | 21         | 1         | true     | WARL      | 0x0         | Corrected error counter overflow                                                                                                                                                       |
| wpri_reserved_22_22 | 22         | 1         | true     | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                           |

|      |    |    |      |      |     |                                                                                                                                                                                                               |
|------|----|----|------|------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rdip | 23 | 1  | true | WARL | 0x0 | Read in progress. Set to 1 when a new error is recorded in an invalid status register. Cleared to 0 when a valid status is overwritten. When control.sinv is set, status.v is cleared to 0 only if rdip is 1. |
| ec   | 24 | 8  | true | WARL | 0x0 | Error code, provides a description of the detected error. Encodings defined in section 2.7 of RISC-V RERI spec                                                                                                |
| le   | 32 | 4  | true | WARL | 0x0 | The encoding that describes how to interpret/process the locator register value                                                                                                                               |
| ie   | 36 | 4  | true | WARL | 0x0 | The encoding that describes how to interpret/process the information register value                                                                                                                           |
| se   | 40 | 4  | true | WARL | 0x0 | The encoding that describes how to interpret/process the supplemental info register value                                                                                                                     |
| te   | 44 | 4  | true | WARL | 0x0 | The encoding that describes how to interpret/process the timestamp register value                                                                                                                             |
| cec  | 48 | 16 | true | WARL | 0x0 | Corrected error counter. Incremented on each corrected error if cece is set to 1.                                                                                                                             |

#### c\_fe\_error\_record\_0\_addr

**Address:** 0x42008050

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Address register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                |
|------------|------------|-----------|----------|--------|-------------|--------------------------------------------|
| address    | 0          | 64        | true     | WARL   | 0x0         | Address associated with the detected error |

#### c\_fe\_error\_record\_0\_info

**Address:** 0x42008058

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Information register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                           |
|------------|------------|-----------|----------|--------|-------------|---------------------------------------|
| info       | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom error information |

#### c\_fe\_error\_record\_0\_suppl\_info

**Address:** 0x42008060

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Supplemental information register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                        |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------------|
| suppl_info | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom supplemental error information |

### c\_fe\_error\_record\_0\_timestamp

**Address:** 0x42008068

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Timestamp register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                   |
|------------|------------|-----------|----------|--------|-------------|-------------------------------|
| timestamp  | 0          | 64        | true     | WARL   | 0x0         | Timestamp of the error record |

### c\_fe\_error\_record\_0\_locator

**Address:** 0x42008070

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Locator register of error record 0

| Field Name  | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                          |
|-------------|------------|-----------|----------|--------|-------------|--------------------------------------------------------------------------------------|
| description | 0          | 32        | true     | WARL   | 0x0         | Contains description of the error event telling what happened [descriptor, instance] |
| location    | 32         | 32        | true     | WARL   | 0x0         | Contains location of the error event telling where it happened [chip, ip, instance]  |

### c\_fe\_error\_record\_1\_control

**Address:** 0x42008080

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000001

**Description:** Control register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                                             |
|------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| else       | 0          | 1         | false    | WARL   | 0x1         | If 1, the hardware unit logs and signals errors in the error record. If 0, the hardware unit will not log RAS errors                                                                                    |
| cece       | 1          | 1         | false    | WARL   | 0x0         | If 1, hardware will count corrected errors in the cec field of the status register. If 0, cec does not count and retains its value.                                                                     |
| ces        | 2          | 2         | false    | WARL   | 0x0         | Used to enable signaling of corrected errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |

|                     |    |    |       |           |     |                                                                                                                                                                                                                    |
|---------------------|----|----|-------|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ueds                | 4  | 2  | false | WARL      | 0x0 | Used to enable signaling of uncorrected deferred errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| uecs                | 6  | 2  | false | WARL      | 0x0 | Used to enable signaling of uncorrected critical errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| wpri_reserved_31_8  | 8  | 24 | false | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| eid                 | 32 | 16 | true  | WARL      | 0x0 | Error injection delay                                                                                                                                                                                              |
| sinv                | 48 | 1  | true  | WARL      | 0x0 | Status register invalidate. When written to 1, causes the valid (v) field of the status register to be cleared.                                                                                                    |
| srdp                | 49 | 1  | true  | WARL      | 0x0 | Set read in progress. When 1, causes read in progress (rdip) to be set                                                                                                                                             |
| wpri_reserved_55_48 | 50 | 10 | false | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| custom_unused       | 60 | 4  | false | Read-Only | 0x0 | Reserved                                                                                                                                                                                                           |

### c\_fe\_error\_record\_1\_status

**Address:** 0x42008088

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Status register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                      |
|------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v          | 0          | 1         | true     | WARL   | 0x0         | If 1, error record holds a valid error                                                                                                                           |
| ce         | 1          | 1         | true     | WARL   | 0x0         | Set to 1 if the detected error was corrected                                                                                                                     |
| ued        | 2          | 1         | true     | WARL   | 0x0         | Set to 1 if the error could not be corrected but was deferred                                                                                                    |
| uec        | 3          | 1         | true     | WARL   | 0x0         | Set to 1 if the error could not be corrected or deferred and needs critical handling by RAS handler                                                              |
| pri        | 4          | 2         | true     | WARL   | 0x0         | Priority of the error. Indicates relative priority among errors of the same class.                                                                               |
| mo         | 6          | 1         | true     | WARL   | 0x0         | Set to 1 if more errors of the same class and severity occur                                                                                                     |
| c          | 7          | 1         | true     | WARL   | 0x0         | Containable bit. If 1, indicates that the error has not propagated beyond the boundaries of the hardware unit                                                    |
| tt         | 8          | 3         | true     | WARL   | 0x0         | Transaction type. 0=Unspecified or NA, 1=UNSPECIFIED, 2-3=Reserved for future standard use, 4=Explicit read, 5=Explicit write, 6=Implicit read, 7=Implicit write |
| iv         | 11         | 1         | true     | WARL   | 0x0         | Information valid, set to 1 if error reports additional information in the info register                                                                         |

|                      |    |    |      |           |     |                                                                                                                                                                                                               |
|----------------------|----|----|------|-----------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| at                   | 12 | 4  | true | WARL      | 0x0 | Address type, indicates the type of address reported in the addr register. 0=None, 1=Supervisor Physical Address, 2=Guest Physical Address, 3=Virtual Address, 4-15=Component Specific                        |
| siv                  | 16 | 1  | true | WARL      | 0x0 | Supplemental information valid, set to 1 if error reports supplemental information in the suppl_info register                                                                                                 |
| tsv                  | 17 | 1  | true | WARL      | 0x0 | Timestamp valid, set to 1 if the error record holds a valid timestamp in the timestamp register                                                                                                               |
| lv                   | 18 | 1  | true | WARL      | 0x0 | Locator valid, set to 1 if the error record holds a valid timestamp in the locator register                                                                                                                   |
| wpri_reserve_d_19_19 | 19 | 1  | true | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                  |
| scrub                | 20 | 1  | true | WARL      | 0x0 | Set to 1 to indicate that the storage location has been updated with the corrected value. Only valid for corrected errors.                                                                                    |
| ceco                 | 21 | 1  | true | WARL      | 0x0 | Corrected error counter overflow                                                                                                                                                                              |
| wpri_reserve_d_22_22 | 22 | 1  | true | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                  |
| rdip                 | 23 | 1  | true | WARL      | 0x0 | Read in progress. Set to 1 when a new error is recorded in an invalid status register. Cleared to 0 when a valid status is overwritten. When control.sinv is set, status.v is cleared to 0 only if rdip is 1. |
| ec                   | 24 | 8  | true | WARL      | 0x0 | Error code, provides a description of the detected error. Encodings defined in section 2.7 of RISC-V RERI spec                                                                                                |
| le                   | 32 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the locator register value                                                                                                                               |
| ie                   | 36 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the information register value                                                                                                                           |
| se                   | 40 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the supplemental info register value                                                                                                                     |
| te                   | 44 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the timestamp register value                                                                                                                             |
| cec                  | 48 | 16 | true | WARL      | 0x0 | Corrected error counter. Incremented on each corrected error if cece is set to 1.                                                                                                                             |

### c\_fe\_error\_record\_1\_addr

**Address:** 0x42008090

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Address register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                |
|------------|------------|-----------|----------|--------|-------------|--------------------------------------------|
| address    | 0          | 64        | true     | WARL   | 0x0         | Address associated with the detected error |

### c\_fe\_error\_record\_1\_info

**Address:** 0x42008098

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Information register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                           |
|------------|------------|-----------|----------|--------|-------------|---------------------------------------|
| info       | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom error information |

### c\_fe\_error\_record\_1\_suppl\_info

**Address:** 0x420080A0

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Supplemental information register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                        |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------------|
| suppl_info | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom supplemental error information |

### c\_fe\_error\_record\_1\_timestamp

**Address:** 0x420080A8

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Timestamp register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                   |
|------------|------------|-----------|----------|--------|-------------|-------------------------------|
| timestamp  | 0          | 64        | true     | WARL   | 0x0         | Timestamp of the error record |

### c\_fe\_error\_record\_1\_locator

**Address:** 0x420080B0

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Locator register of error record 1

| Field Name  | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                          |
|-------------|------------|-----------|----------|--------|-------------|--------------------------------------------------------------------------------------|
| description | 0          | 32        | true     | WARL   | 0x0         | Contains description of the error event telling what happened [descriptor, instance] |
| location    | 32         | 32        | true     | WARL   | 0x0         | Contains location of the error event telling where it happened [chip, ip, instance]  |

### c\_ls\_error\_header\_vendor

**Address:** 0x42008100

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000821

**Description:** Vendor and implementation ID

| Field Name          | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                 |
|---------------------|------------|-----------|----------|-----------|-------------|-------------------------------------------------------------------------------------------------------------|
| vendor_id           | 0          | 32        | false    | Read-Only | 0x821       | Provides the JEDEC manufacturer ID of the provider of the component hosting the error bank                  |
| imp_id              | 32         | 16        | false    | Read-Only | 0x0         | Provides a unique identity, defined by the vendor, to identify the component and revisions of the component |
| wpri_reserved_63_48 | 48         | 16        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                |

### c\_ls\_error\_header\_bank\_info

**Address:** 0x42008108

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0100000000430000

**Description:** Error bank information

| Field Name          | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                 |
|---------------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| inst_id             | 0          | 16        | false    | Read-Only | 0x0         | Identifies a unique instance of an error bank within a package or die, ideally in the whole system; Not being used, tied to 0                                                               |
| n_err_recs          | 16         | 6         | false    | Read-Only | 0x3         | Indicates the number of error records implemented by the error bank                                                                                                                         |
| layout              | 22         | 2         | false    | Read-Only | 0x1         | Indicates the layout of the registers in the error bank and the error records. The layout encoding 0 indicates the registers are arranged and have meaning as defined by the specification. |
| wpri_reserved_55_24 | 24         | 32        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                |
| version             | 56         | 8         | false    | Read-Only | 0x1         | Version of the arch register layout specification implemented by the error bank                                                                                                             |

### c\_ls\_error\_header\_valid\_summary

**Address:** 0x42008110

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000001

**Description:** Summary of valid error records

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|              |   |    |       |           |     |                                                                                                |
|--------------|---|----|-------|-----------|-----|------------------------------------------------------------------------------------------------|
| sv           | 0 | 1  | false | Read-Only | 0x1 | Indicates that the valid_bitmap provides a summary of the valid bits from the status registers |
| valid_bitmap | 1 | 63 | true  | Read-Only | 0x0 | Summary of the valid bits from status registers in the error bank                              |

### c\_ls\_error\_record\_0\_control

**Address:** 0x42008140

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000001

**Description:** Control register of error record 0

| Field Name          | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                        |
|---------------------|------------|-----------|----------|-----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| else                | 0          | 1         | false    | WARL      | 0x1         | If 1, the hardware unit logs and signals errors in the error record. If 0, the hardware unit will not log RAS errors                                                                                               |
| cece                | 1          | 1         | false    | WARL      | 0x0         | If 1, hardware will count corrected errors in the cec field of the status register. If 0, cec does not count and retains its value.                                                                                |
| ces                 | 2          | 2         | false    | WARL      | 0x0         | Used to enable signaling of corrected errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal.            |
| ueds                | 4          | 2         | false    | WARL      | 0x0         | Used to enable signaling of uncorrected deferred errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| uecs                | 6          | 2         | false    | WARL      | 0x0         | Used to enable signaling of uncorrected critical errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| wpri_reserved_31_8  | 8          | 24        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| eid                 | 32         | 16        | true     | WARL      | 0x0         | Error injection delay                                                                                                                                                                                              |
| sinv                | 48         | 1         | true     | WARL      | 0x0         | Status register invalidate. When written to 1, causes the valid (v) field of the status register to be cleared.                                                                                                    |
| srdep               | 49         | 1         | true     | WARL      | 0x0         | Set read in progress. When 1, causes read in progress (rdip) to be set                                                                                                                                             |
| wpri_reserved_55_48 | 50         | 10        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| custom_unused       | 60         | 4         | false    | Read-Only | 0x0         | Reserved                                                                                                                                                                                                           |

### c\_ls\_error\_record\_0\_status

**Address:** 0x42008148

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Status register of error record 0

| Field Name          | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                   |
|---------------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v                   | 0          | 1         | true     | WARL      | 0x0         | If 1, error record holds a valid error                                                                                                                                                                        |
| ce                  | 1          | 1         | true     | WARL      | 0x0         | Set to 1 if the detected error was corrected                                                                                                                                                                  |
| ued                 | 2          | 1         | true     | WARL      | 0x0         | Set to 1 if the error could not be corrected but was deferred                                                                                                                                                 |
| uec                 | 3          | 1         | true     | WARL      | 0x0         | Set to 1 if the error could not be corrected or deferred and needs critical handling by RAS handler                                                                                                           |
| pri                 | 4          | 2         | true     | WARL      | 0x0         | Priority of the error. Indicates relative priority among errors of the same class.                                                                                                                            |
| mo                  | 6          | 1         | true     | WARL      | 0x0         | Set to 1 if more errors of the same class and severity occur                                                                                                                                                  |
| c                   | 7          | 1         | true     | WARL      | 0x0         | Containable bit. If 1, indicates that the error has not propagated beyond the boundaries of the hardware unit                                                                                                 |
| tt                  | 8          | 3         | true     | WARL      | 0x0         | Transaction type. 0=Unspecified or NA, 1=UNSPECIFIED, 2-3=Reserved for future standard use, 4=Explicit read, 5=Explicit write, 6=Implicit read, 7=Implicit write                                              |
| iv                  | 11         | 1         | true     | WARL      | 0x0         | Information valid, set to 1 if error reports additional information in the info register                                                                                                                      |
| at                  | 12         | 4         | true     | WARL      | 0x0         | Address type, indicates the type of address reported in the addr register. 0=None, 1=Supervisor Physical Address, 2=Guest Physical Address, 3=Virtual Address, 4-15=Component Specific                        |
| siv                 | 16         | 1         | true     | WARL      | 0x0         | Supplemental information valid, set to 1 if error reports supplemental information in the suppl_info register                                                                                                 |
| tsv                 | 17         | 1         | true     | WARL      | 0x0         | Timestamp valid, set to 1 if the error record holds a valid timestamp in the timestamp register                                                                                                               |
| lv                  | 18         | 1         | true     | WARL      | 0x0         | Locator valid, set to 1 if the error record holds a valid timestamp in the locator register                                                                                                                   |
| wpri_reserved_19_19 | 19         | 1         | true     | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                                  |
| scrub               | 20         | 1         | true     | WARL      | 0x0         | Set to 1 to indicate that the storage location has been updated with the corrected value. Only valid for corrected errors.                                                                                    |
| ceco                | 21         | 1         | true     | WARL      | 0x0         | Corrected error counter overflow                                                                                                                                                                              |
| wpri_reserved_22_22 | 22         | 1         | true     | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                                  |
| rdip                | 23         | 1         | true     | WARL      | 0x0         | Read in progress. Set to 1 when a new error is recorded in an invalid status register. Cleared to 0 when a valid status is overwritten. When control.sinv is set, status.v is cleared to 0 only if rdip is 1. |
| ec                  | 24         | 8         | true     | WARL      | 0x0         | Error code, provides a description of the detected error. Encodings defined in section 2.7 of RISC-V RERI spec                                                                                                |
| le                  | 32         | 4         | true     | WARL      | 0x0         | The encoding that describes how to interpret/process the locator register value                                                                                                                               |

|     |    |    |      |      |     |                                                                                           |
|-----|----|----|------|------|-----|-------------------------------------------------------------------------------------------|
| ie  | 36 | 4  | true | WARL | 0x0 | The encoding that describes how to interpret/process the information register value       |
| se  | 40 | 4  | true | WARL | 0x0 | The encoding that describes how to interpret/process the supplemental info register value |
| te  | 44 | 4  | true | WARL | 0x0 | The encoding that describes how to interpret/process the timestamp register value         |
| cec | 48 | 16 | true | WARL | 0x0 | Corrected error counter. Incremented on each corrected error if cece is set to 1.         |

### c\_ls\_error\_record\_0\_addr

**Address:** 0x42008150

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Address register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                |
|------------|------------|-----------|----------|--------|-------------|--------------------------------------------|
| address    | 0          | 64        | true     | WARL   | 0x0         | Address associated with the detected error |

### c\_ls\_error\_record\_0\_info

**Address:** 0x42008158

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Information register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                           |
|------------|------------|-----------|----------|--------|-------------|---------------------------------------|
| info       | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom error information |

### c\_ls\_error\_record\_0\_suppl\_info

**Address:** 0x42008160

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Supplemental information register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                        |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------------|
| suppl_info | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom supplemental error information |

### c\_ls\_error\_record\_0\_timestamp

**Address:** 0x42008168

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Timestamp register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                   |
|------------|------------|-----------|----------|--------|-------------|-------------------------------|
| timestamp  | 0          | 64        | true     | WARL   | 0x0         | Timestamp of the error record |

### c\_ls\_error\_record\_0\_locator

**Address:** 0x42008170

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Locator register of error record 0

| Field Name  | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                          |
|-------------|------------|-----------|----------|--------|-------------|--------------------------------------------------------------------------------------|
| description | 0          | 32        | true     | WARL   | 0x0         | Contains description of the error event telling what happened [descriptor, instance] |
| location    | 32         | 32        | true     | WARL   | 0x0         | Contains location of the error event telling where it happened [chip, ip, instance]  |

### c\_ls\_error\_record\_1\_control

**Address:** 0x42008180

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000001

**Description:** Control register of error record 1

| Field Name         | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                        |
|--------------------|------------|-----------|----------|-----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| else               | 0          | 1         | false    | WARL      | 0x1         | If 1, the hardware unit logs and signals errors in the error record. If 0, the hardware unit will not log RAS errors                                                                                               |
| cece               | 1          | 1         | false    | WARL      | 0x0         | If 1, hardware will count corrected errors in the cec field of the status register. If 0, cec does not count and retains its value.                                                                                |
| ces                | 2          | 2         | false    | WARL      | 0x0         | Used to enable signaling of corrected errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal.            |
| ueds               | 4          | 2         | false    | WARL      | 0x0         | Used to enable signaling of uncorrected deferred errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| uecs               | 6          | 2         | false    | WARL      | 0x0         | Used to enable signaling of uncorrected critical errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| wpri_reserved_31_8 | 8          | 24        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| eid                | 32         | 16        | true     | WARL      | 0x0         | Error injection delay                                                                                                                                                                                              |

|                     |    |    |       |           |     |                                                                                                                 |
|---------------------|----|----|-------|-----------|-----|-----------------------------------------------------------------------------------------------------------------|
| sinv                | 48 | 1  | true  | WARL      | 0x0 | Status register invalidate. When written to 1, causes the valid (v) field of the status register to be cleared. |
| srdp                | 49 | 1  | true  | WARL      | 0x0 | Set read in progress. When 1, causes read in progress (rdip) to be set                                          |
| wpri_reserved_55_48 | 50 | 10 | false | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                    |
| custom_unused       | 60 | 4  | false | Read-Only | 0x0 | Reserved                                                                                                        |

### c\_ls\_error\_record\_1\_status

**Address:** 0x42008188

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Status register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                            |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v          | 0          | 1         | true     | WARL   | 0x0         | If 1, error record holds a valid error                                                                                                                                                 |
| ce         | 1          | 1         | true     | WARL   | 0x0         | Set to 1 if the detected error was corrected                                                                                                                                           |
| ued        | 2          | 1         | true     | WARL   | 0x0         | Set to 1 if the error could not be corrected but was deferred                                                                                                                          |
| uec        | 3          | 1         | true     | WARL   | 0x0         | Set to 1 if the error could not be corrected or deferred and needs critical handling by RAS handler                                                                                    |
| pri        | 4          | 2         | true     | WARL   | 0x0         | Priority of the error. Indicates relative priority among errors of the same class.                                                                                                     |
| mo         | 6          | 1         | true     | WARL   | 0x0         | Set to 1 if more errors of the same class and severity occur                                                                                                                           |
| c          | 7          | 1         | true     | WARL   | 0x0         | Containable bit. If 1, indicates that the error has not propagated beyond the boundaries of the hardware unit                                                                          |
| tt         | 8          | 3         | true     | WARL   | 0x0         | Transaction type. 0=Unspecified or NA, 1=UNSPECIFIED, 2-3=Reserved for future standard use, 4=Explicit read, 5=Explicit write, 6=Implicit read, 7=Implicit write                       |
| iv         | 11         | 1         | true     | WARL   | 0x0         | Information valid, set to 1 if error reports additional information in the info register                                                                                               |
| at         | 12         | 4         | true     | WARL   | 0x0         | Address type, indicates the type of address reported in the addr register. 0=None, 1=Supervisor Physical Address, 2=Guest Physical Address, 3=Virtual Address, 4-15=Component Specific |
| siv        | 16         | 1         | true     | WARL   | 0x0         | Supplemental information valid, set to 1 if error reports supplemental information in the suppl_info register                                                                          |
| tsv        | 17         | 1         | true     | WARL   | 0x0         | Timestamp valid, set to 1 if the error record holds a valid timestamp in the timestamp register                                                                                        |
| lv         | 18         | 1         | true     | WARL   | 0x0         | Locator valid, set to 1 if the error record holds a valid timestamp in the locator register                                                                                            |

|                     |    |    |      |           |     |                                                                                                                                                                                                               |
|---------------------|----|----|------|-----------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| wpri_reserved_19_19 | 19 | 1  | true | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                  |
| scrub               | 20 | 1  | true | WARL      | 0x0 | Set to 1 to indicate that the storage location has been updated with the corrected value. Only valid for corrected errors.                                                                                    |
| ceco                | 21 | 1  | true | WARL      | 0x0 | Corrected error counter overflow                                                                                                                                                                              |
| wpri_reserved_22_22 | 22 | 1  | true | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                  |
| rdip                | 23 | 1  | true | WARL      | 0x0 | Read in progress. Set to 1 when a new error is recorded in an invalid status register. Cleared to 0 when a valid status is overwritten. When control.sinv is set, status.v is cleared to 0 only if rdip is 1. |
| ec                  | 24 | 8  | true | WARL      | 0x0 | Error code, provides a description of the detected error. Encodings defined in section 2.7 of RISC-V RERI spec                                                                                                |
| le                  | 32 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the locator register value                                                                                                                               |
| ie                  | 36 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the information register value                                                                                                                           |
| se                  | 40 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the supplemental info register value                                                                                                                     |
| te                  | 44 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the timestamp register value                                                                                                                             |
| cec                 | 48 | 16 | true | WARL      | 0x0 | Corrected error counter. Incremented on each corrected error if cece is set to 1.                                                                                                                             |

### c\_ls\_error\_record\_1\_addr

**Address:** 0x42008190

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Address register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                |
|------------|------------|-----------|----------|--------|-------------|--------------------------------------------|
| address    | 0          | 64        | true     | WARL   | 0x0         | Address associated with the detected error |

### c\_ls\_error\_record\_1\_info

**Address:** 0x42008198

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Information register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                           |
|------------|------------|-----------|----------|--------|-------------|---------------------------------------|
| info       | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom error information |

### c\_ls\_error\_record\_1\_suppl\_info

**Address:** 0x420081A0

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Supplemental information register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                        |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------------|
| suppl_info | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom supplemental error information |

### c\_ls\_error\_record\_1\_timestamp

**Address:** 0x420081A8

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Timestamp register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                   |
|------------|------------|-----------|----------|--------|-------------|-------------------------------|
| timestamp  | 0          | 64        | true     | WARL   | 0x0         | Timestamp of the error record |

### c\_ls\_error\_record\_1\_locator

**Address:** 0x420081B0

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Locator register of error record 1

| Field Name  | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                          |
|-------------|------------|-----------|----------|--------|-------------|--------------------------------------------------------------------------------------|
| description | 0          | 32        | true     | WARL   | 0x0         | Contains description of the error event telling what happened [descriptor, instance] |
| location    | 32         | 32        | true     | WARL   | 0x0         | Contains location of the error event telling where it happened [chip, ip, instance]  |

### c\_ls\_error\_record\_2\_control

**Address:** 0x420081C0

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000001

**Description:** Control register of error record 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                          |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------------------------------------------------------------------------------|
| else       | 0          | 1         | false    | WARL   | 0x1         | If 1, the hardware unit logs and signals errors in the error record. If 0, the hardware unit will not log RAS errors |

|                     |    |    |       |           |     |                                                                                                                                                                                                                    |
|---------------------|----|----|-------|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cece                | 1  | 1  | false | WARL      | 0x0 | If 1, hardware will count corrected errors in the cec field of the status register. If 0, cec does not count and retains its value.                                                                                |
| ces                 | 2  | 2  | false | WARL      | 0x0 | Used to enable signaling of corrected errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal.            |
| ueds                | 4  | 2  | false | WARL      | 0x0 | Used to enable signaling of uncorrected deferred errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| uecs                | 6  | 2  | false | WARL      | 0x0 | Used to enable signaling of uncorrected critical errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| wpri_reserved_31_8  | 8  | 24 | false | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| eid                 | 32 | 16 | true  | WARL      | 0x0 | Error injection delay                                                                                                                                                                                              |
| sinv                | 48 | 1  | true  | WARL      | 0x0 | Status register invalidate. When written to 1, causes the valid (v) field of the status register to be cleared.                                                                                                    |
| srdp                | 49 | 1  | true  | WARL      | 0x0 | Set read in progress. When 1, causes read in progress (rdip) to be set                                                                                                                                             |
| wpri_reserved_55_48 | 50 | 10 | false | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| custom_unused       | 60 | 4  | false | Read-Only | 0x0 | Reserved                                                                                                                                                                                                           |

## c\_ls\_error\_record\_2\_status

**Address:** 0x420081C8

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Status register of error record 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                   |
|------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------|
| v          | 0          | 1         | true     | WARL   | 0x0         | If 1, error record holds a valid error                                                                        |
| ce         | 1          | 1         | true     | WARL   | 0x0         | Set to 1 if the detected error was corrected                                                                  |
| ued        | 2          | 1         | true     | WARL   | 0x0         | Set to 1 if the error could not be corrected but was deferred                                                 |
| uec        | 3          | 1         | true     | WARL   | 0x0         | Set to 1 if the error could not be corrected or deferred and needs critical handling by RAS handler           |
| pri        | 4          | 2         | true     | WARL   | 0x0         | Priority of the error. Indicates relative priority among errors of the same class.                            |
| mo         | 6          | 1         | true     | WARL   | 0x0         | Set to 1 if more errors of the same class and severity occur                                                  |
| c          | 7          | 1         | true     | WARL   | 0x0         | Containable bit. If 1, indicates that the error has not propagated beyond the boundaries of the hardware unit |

|                     |    |    |      |           |     |                                                                                                                                                                                                               |
|---------------------|----|----|------|-----------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tt                  | 8  | 3  | true | WARL      | 0x0 | Transaction type. 0=Unspecified or NA, 1=UNSPECIFIED, 2-3=Reserved for future standard use, 4=Explicit read, 5=Explicit write, 6=Implicit read, 7=Implicit write                                              |
| iv                  | 11 | 1  | true | WARL      | 0x0 | Information valid, set to 1 if error reports additional information in the info register                                                                                                                      |
| at                  | 12 | 4  | true | WARL      | 0x0 | Address type, indicates the type of address reported in the addr register. 0=None, 1=Supervisor Physical Address, 2=Guest Physical Address, 3=Virtual Address, 4-15=Component Specific                        |
| siv                 | 16 | 1  | true | WARL      | 0x0 | Supplemental information valid, set to 1 if error reports supplemental information in the suppl_info register                                                                                                 |
| tsv                 | 17 | 1  | true | WARL      | 0x0 | Timestamp valid, set to 1 if the error record holds a valid timestamp in the timestamp register                                                                                                               |
| lv                  | 18 | 1  | true | WARL      | 0x0 | Locator valid, set to 1 if the error record holds a valid timestamp in the locator register                                                                                                                   |
| wpri_reserved_19_19 | 19 | 1  | true | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                  |
| scrub               | 20 | 1  | true | WARL      | 0x0 | Set to 1 to indicate that the storage location has been updated with the corrected value. Only valid for corrected errors.                                                                                    |
| ceco                | 21 | 1  | true | WARL      | 0x0 | Corrected error counter overflow                                                                                                                                                                              |
| wpri_reserved_22_22 | 22 | 1  | true | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                  |
| rdip                | 23 | 1  | true | WARL      | 0x0 | Read in progress. Set to 1 when a new error is recorded in an invalid status register. Cleared to 0 when a valid status is overwritten. When control.sinv is set, status.v is cleared to 0 only if rdip is 1. |
| ec                  | 24 | 8  | true | WARL      | 0x0 | Error code, provides a description of the detected error. Encodings defined in section 2.7 of RISC-V RERI spec                                                                                                |
| le                  | 32 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the locator register value                                                                                                                               |
| ie                  | 36 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the information register value                                                                                                                           |
| se                  | 40 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the supplemental info register value                                                                                                                     |
| te                  | 44 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the timestamp register value                                                                                                                             |
| cec                 | 48 | 16 | true | WARL      | 0x0 | Corrected error counter. Incremented on each corrected error if cece is set to 1.                                                                                                                             |

## c\_ls\_error\_record\_2\_addr

**Address:** 0x420081D0

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Address register of error record 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                |
|------------|------------|-----------|----------|--------|-------------|--------------------------------------------|
| address    | 0          | 64        | true     | WARL   | 0x0         | Address associated with the detected error |

### **c\_ls\_error\_record\_2\_info**

**Address:** 0x420081D8

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Information register of error record 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                           |
|------------|------------|-----------|----------|--------|-------------|---------------------------------------|
| info       | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom error information |

### **c\_ls\_error\_record\_2\_suppl\_info**

**Address:** 0x420081E0

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Supplemental information register of error record 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                        |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------------|
| suppl_info | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom supplemental error information |

### **c\_ls\_error\_record\_2\_timestamp**

**Address:** 0x420081E8

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Timestamp register of error record 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                   |
|------------|------------|-----------|----------|--------|-------------|-------------------------------|
| timestamp  | 0          | 64        | true     | WARL   | 0x0         | Timestamp of the error record |

### **c\_ls\_error\_record\_2\_locator**

**Address:** 0x420081F0

**cluster\_mask:** 0x01E00000

**core\_mask:** 0x00070000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Locator register of error record 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|             |    |    |      |      |     |                                                                                      |
|-------------|----|----|------|------|-----|--------------------------------------------------------------------------------------|
| description | 0  | 32 | true | WARL | 0x0 | Contains description of the error event telling what happened [descriptor, instance] |
| location    | 32 | 32 | true | WARL | 0x0 | Contains location of the error event telling where it happened [chip, ip, instance]  |

### trRamControl

**Address:** 0x42080000

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000008

**Description:** None

| Field Name       | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------------|------------|-----------|----------|--------|-------------|-------------|
| trramactive      | 0          | 1         | true     | WARL   | 0x0         |             |
| trramenable      | 1          | 1         | true     | WARL   | 0x0         |             |
| rsvd22           | 2          | 1         | -        | WPRI   | -           | Reserved    |
| trramempty       | 3          | 1         | true     | WARL   | 0x1         |             |
| trrammode        | 4          | 1         | true     | WARL   | 0x0         |             |
| rsvd75           | 5          | 3         | -        | WPRI   | -           | Reserved    |
| trramstopon wrap | 8          | 1         | true     | WARL   | 0x0         |             |
| rsvd329          | 9          | 23        | -        | WPRI   | -           | Reserved    |

### trRamStartLow

**Address:** 0x42080010

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** None

| Field Name    | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|---------------|------------|-----------|----------|--------|-------------|-------------|
| rsvd_1_0      | 0          | 2         | false    | WARL   | 0x0         |             |
| trramstartlow | 2          | 30        | true     | WARL   | 0x0         |             |

### trRamStartHigh

**Address:** 0x42080014

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** None

| Field Name     | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|----------------|------------|-----------|----------|--------|-------------|-------------|
| trramstarthigh | 0          | 32        | true     | WARL   | 0x0         |             |

### trRamLimitLow

**Address:** 0x42080018

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00008000

**Description:** None

| Field Name    | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|---------------|------------|-----------|----------|--------|-------------|-------------|
| rsvd_1_0      | 0          | 2         | false    | WARL   | 0x0         |             |
| trramlimitlow | 2          | 30        | true     | WARL   | 0x2000      |             |

### trRamLimitHigh

**Address:** 0x4208001C

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** None

| Field Name     | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|----------------|------------|-----------|----------|--------|-------------|-------------|
| trramlimithigh | 0          | 32        | true     | WARL   | 0x0         |             |

### trRamWPLow

**Address:** 0x42080020

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** None

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| trramwrap  | 0          | 1         | true     | WARL   | 0x0         |             |
| rsvd11     | 1          | 1         | -        | WPRI   | -           | Reserved    |
| trramwplow | 2          | 30        | true     | WARL   | 0x0         |             |

### trRamWPHigh

**Address:** 0x42080024

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** None

| Field Name  | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|-------------|------------|-----------|----------|--------|-------------|-------------|
| trramwphigh | 0          | 32        | true     | WARL   | 0x0         |             |

### trRamRPLow

**Address:** 0x42080028

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** None

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| rsvd_1_0   | 0          | 2         | false    | WARL   | 0x0         |             |
| trramrplow | 2          | 30        | true     | WARL   | 0x0         |             |

### trRamRPHigh

**Address:** 0x4208002C

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** None

| Field Name  | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|-------------|------------|-----------|----------|--------|-------------|-------------|
| trramrphigh | 0          | 32        | true     | WARL   | 0x0         |             |

### trRamData

**Address:** 0x42080040

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** None

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description |
|------------|------------|-----------|----------|-----------|-------------|-------------|
| trramdata  | 0          | 32        | true     | Read Only | 0x0         |             |

### trFunnelControl

**Address:** 0x42081000

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000008

**Description:** None

| Field Name     | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|----------------|------------|-----------|----------|--------|-------------|-------------|
| trfunnelactive | 0          | 1         | true     | WARL   | 0x0         |             |
| trfunnelenable | 1          | 1         | true     | WARL   | 0x0         |             |
| rsvd22         | 2          | 1         | -        | WPRI   | -           | Reserved    |
| trfunnelempy   | 3          | 1         | true     | WARL   | 0x1         |             |
| rsvd324        | 4          | 28        | -        | WPRI   | -           | Reserved    |

### trFunnelImpl

**Address:** 0x42081004

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000801

**Description:** None

| Field Name       | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------------|------------|-----------|----------|--------|-------------|-------------|
| trfunnelvermajor | 0          | 4         | false    | WARL   | 0x1         |             |
| trfunnelverminor | 4          | 4         | false    | WARL   | 0x0         | e           |
| trfunnelcomptype | 8          | 4         | false    | WARL   | 0x8         |             |
| rsvd3212         | 12         | 20        | -        | WPRI   | -           | Reserved    |

### trFunnelDisInput

**Address:** 0x42081008

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** None

| Field Name       | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                     |
|------------------|------------|-----------|----------|--------|-------------|-----------------------------------------------------------------|
| trfunneldisinput | 0          | 16        | true     | WARL   | 0x0         | Bits 7:0 are reserved for N-trace srcs and 15:8 for Dst sources |
| rsvd3216         | 16         | 16        | -        | WPRI   | -           | Reserved                                                        |

### rstctl\_ip\_ver\_reg

**Address:** 0x42102000

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00200000

**Description:** CPL Reset Controller IP Version Reg

| Field Name    | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                 |
|---------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| step_version  | 0          | 10        | false    | Read Only | 0x0         | Represents step version of the IP. Incremented for bug fixes of the IP_x000B_Reset to '0' if either MAJOR_VERSION or MINOR_VERSION updates. |
| minor_version | 10         | 11        | false    | Read Only | 0x0         | Represents minor version of the IP. Incremented for feature changes of the IP_x000B_Reset to '0' if MAJOR_VERSION update.                   |
| major_version | 21         | 11        | false    | Read Only | 0x1         | Represents major version of the IP. Incremented for major architecture changes of the IP                                                    |

### rstctl\_cpl\_cl\_cold\_reset\_reg

**Address:** 0x42102004

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** CPL RSTCTL CPL Cluster Cold Reset Reg

| Field Name               | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                             |
|--------------------------|------------|-----------|----------|--------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| cpl_cluster_cold_reset_n | 0          | 1         | false    | WARL   | 0x0         | CPL initiated Cluster cold reset. Resets entire cluster except CPL, PLL Wrapper_x000B_0x1: Reset de-asserted_x000B_0x0: Reset asserted. |
| rsvd                     | 1          | 31        | false    | WARL   | 0x0         | Reserved                                                                                                                                |

### **rstctl\_cpl\_cl\_warm\_reset\_reg**

**Address:** 0x42102008

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** CPL RSTCTL CPL Cluster Warm Reset Reg

| Field Name                  | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                                      |
|-----------------------------|------------|-----------|----------|--------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cpl_cluster_warm_reset_n    | 0          | 1         | true     | WARL   | 0x0         | cl_warm and rf_warm reset control.0x0: Reset asserted.0x1: Reset de-asserted..                                                                                                                   |
| cpl_force_ss_to_ref_clock_n | 1          | 1         | true     | WARL   | 0x0         | Control to switch cluster clock to ref clock. 0x0: Cluster clock source switched to Reference clock independent of PLL clock mux selection. 0x1: Cluster clock source is based PLL mux selection |
| rsvd                        | 2          | 30        | true     | WARL   | 0x0         | Reserved                                                                                                                                                                                         |

### **rstctl\_cpl\_cl\_no\_fetch\_reg**

**Address:** 0x4210200C

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x000000FF

**Description:** CPL RSTCTL CPL Cluster No Fetch Reg

| Field Name           | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                                                 |
|----------------------|------------|-----------|----------|--------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cpl_cluster_no_fetch | 0          | 8         | true     | WARL   | 0xFF        | Controls CPU execution. One bit per core. Bitwise decoding:_x000B_0x0: CPU execution is not blocked_x000B_0x1: CPU is in no Fetch state.                                                                    |
| rsvd                 | 8          | 16        | true     | WARL   | 0x0         | Reserved                                                                                                                                                                                                    |
| spare_smc2cpl_req    | 24         | 3         | true     | WARL   | 0x0         | SMC-CPL Spare handshake request. Each bit indicate. 0x0: Request is not set; 0x1: Request is set                                                                                                            |
| cluster_core_go      | 27         | 1         | true     | WARL   | 0x0         | CPL sets this bit to indicate SMC once it completes boot config. CPL has to write '0' to clear the status. 0x0: CPL has not completed boot config or SMC cleared status; 0x1: CPL has completed boot config |
| spare_cpl2smc_ack    | 28         | 3         | true     | WARL   | 0x0         | CPL-SMC Spare handshake acknowledgement. Each bit indicate 0x0: Ack is not set; 0x1: Ack is set                                                                                                             |
| config_done          | 31         | 1         | true     | WARL   | 0x0         | CPL sets this bit to indicate SMC that it completed boot config. SMC has to write '0' to clear status. 0x0: CPL has not completed boot config or SMC cleared status. 0x1: CPL has completed boot config     |

### rstctl\_wait\_event\_ctrl\_reg

**Address:** 0x42102010

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x0000000F

**Description:** CPL RSTCTL Wait event Control Register

| Field Name           | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                             |
|----------------------|------------|-----------|----------|--------|-------------|-------------------------------------------------------------------------------------------------------------------------|
| fence_counter_preset | 0          | 8         | false    | WARL   | 0xF         | Fence sequencer counter preset value. Used for controlling fence_disable assertion and de-assertion timing during reset |
| spare                | 8          | 24        | false    | WARL   | 0x0         | Spare Bits                                                                                                              |

### rc\_scratchpad32

**Address:** 0x42102014

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0xF4F4F4F4

**Description:** Scratchpad register for DV

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| data       | 0          | 32        | false    | WARL   | 0xF4F4F4F4  |             |

### pll\_ip\_ver\_reg

**Address:** 0x42103000

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00200000

**Description:** CPL PLL IP Version Reg

| Field Name    | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                 |
|---------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| step_version  | 0          | 10        | false    | Read Only | 0x0         | Represents step version of the IP. Incremented for bug fixes of the IP_x000B_Reset to '0' if either MAJOR_VERSION or MINOR_VERSION updates. |
| minor_version | 10         | 11        | false    | Read Only | 0x0         | Represents minor version of the IP. Incremented for feature changes of the IP_x000B_Reset to '0' if MAJOR_VERSION update.                   |
| major_version | 21         | 11        | false    | Read Only | 0x1         | Represents major version of the IP. Incremented for major architecture changes of the IP                                                    |

### pll\_control\_reg

**Address:** 0x42103004

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000020

**Description:** CPL PLL Control Register

| Field Name                | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                        |
|---------------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pll_dfs_req               | 0          | 1         | false    | WARL   | 0x0         | Trigger to change PLL frequency. SW to clear this bit and set to update new DFS settings                                                                                           |
| pll_shutdown_req          | 1          | 1         | false    | WARL   | 0x0         | SW initiated PLL shutdown request_x000B_0x0: No shutdown request_x000B_0x1: shutdown request from SW. Shutdown status indicated using shutdown_done interrupt to SW.               |
| pll_wakeup_req            | 2          | 1         | false    | WARL   | 0x0         | SW initiated PLL wakeup request_x000B_0x0: No wakeup request_x000B_0x1: wakeup request from SW. Wakeup status indicated using wakeup_done interrupt to SW.                         |
| pll_bypass                | 3          | 1         | false    | WARL   | 0x0         | Bypass mode control_x000B_0x1: Bypass mode is enabled (FOUT = FIN)_x000B_0x0: PLL operates normally                                                                                |
| pll_scalar_mode           | 4          | 1         | false    | WARL   | 0x0         | Glitch free scalar mode frequency switch. When this bit is set only scalar_div_value is sampled, all other parameters are masked._x000B_0x0: Normal DFS_x000B_0x1: Scalar mode DFS |
| pll_resetb                | 5          | 1         | false    | WARL   | 0x1         | PLL reset override0x0: PLL kept in reset state0x1: PLL reset de-asserted                                                                                                           |
| pll_resetb_override       | 6          | 1         | false    | WARL   | 0x0         | PLL reset override control0x0: PLL reset driven from FSM0x1: PLL reset driven from MMR control(pll_resetb)                                                                         |
| dis_inactive_pll_shutdown | 7          | 1         | false    | WARL   | 0x0         | Inactive PLL shutdown override0x0: PLL not driving cl_clk shutdown after PRESET2 timeout0x1: PLL not driving cl_clk continue in active state                                       |
| pll_sel                   | 8          | 4         | false    | WARL   | 0x0         | PLL mux sel override0x0: RF_CLK0x1: SC_CLK0x2: FB_CLK0x3: PLL0_CLK0x4: PLL1_CLK0xF - 0x5: Reserved                                                                                 |
| pll_sel_override          | 12         | 1         | false    | WARL   | 0x0         | PLL mux sel override control0x0: PLL mux sel driven from FSM0x1: PLL mux sel driven from MMR control(pll_sel)                                                                      |
| ext_dfs_ongoing           | 13         | 1         | false    | WARL   | 0x0         | Indicates CPL initiated DFS request based on EXT DFS request. This is used to control ext_dfs_done interrupt. 0x0: No ext DFS ongoing. 0x1: Ext DFS ongoing                        |
| ext_dfs_abort             | 14         | 1         | false    | WARL   | 0x0         | Indicates CPL aborted EXT DFS request. This is used to control ext_dfs_abort interrupt. 0x0: Ext DFS abort de-asserted by CPL. 0x1: Ext DFS abort set by CPL                       |
| spare                     | 15         | 17        | false    | WARL   | 0x0         | Spare Bits                                                                                                                                                                         |

## pll\_status\_reg

**Address:** 0x42103008

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** CPL PLL status Register

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|                   |    |    |      |           |     |                                                                                         |
|-------------------|----|----|------|-----------|-----|-----------------------------------------------------------------------------------------|
| pll0_active       | 0  | 1  | true | Read Only | 0x0 | Indicates PLL0 active status_x000B_0x0: PLL0 is shutdown_x000B_0x1: PLL0 is active      |
| pll0_locked       | 1  | 1  | true | Read Only | 0x0 | Indicates PLL0 lock status_x000B_0x0: PLL0 is not locked_x000B_0x1: PLL0 is locked      |
| pll0_reset        | 2  | 1  | true | Read Only | 0x0 | Indicates PLL0 Reset Status_x000B_0x0: PLL0 is not in Reset_x000B_0x1: PLL0 is in Reset |
| pll0_spare_status | 3  | 13 | true | Read Only | 0x0 | PLL0 spare status                                                                       |
| pll1_active       | 16 | 1  | true | Read Only | 0x0 | Indicates PLL1 active status_x000B_0x0: PLL1 is shutdown_x000B_0x1: PLL1 is active      |
| pll1_locked       | 17 | 1  | true | Read Only | 0x0 | Indicates PLL1 lock status_x000B_0x0: PLL1 is not locked_x000B_0x1: PLL1 is locked      |
| pll1_reset        | 18 | 1  | true | Read Only | 0x0 | Indicates PLL1 Reset Status_x000B_0x0: PLL1 is not in Reset_x000B_0x1: PLL1 is in Reset |
| pll1_spare_status | 19 | 13 | true | Read Only | 0x0 | PLL1 spare status                                                                       |

## pll\_interrupts\_reg

**Address:** 0x4210300C

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** CPL PLL interrupts

| Field Name        | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                              |
|-------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------------------------------------------------------------------------------------|
| dfs_done          | 0          | 1         | true     | W1C    | 0x0         | Parameter update done interrupt to uC on on DFS done_x000B_SW will write this bit to clear the interrupt                                 |
| shutdown_done     | 1          | 1         | true     | W1C    | 0x0         | shutdown status interrupt to uC on thermalSW will write this bit to clear the interrupt                                                  |
| wakeup_done       | 2          | 1         | true     | W1C    | 0x0         | Wake interrupt to uC on C4 exit_x000B_SW will write this bit to clear the interrupt                                                      |
| pll_lock_lost     | 3          | 1         | true     | W1C    | 0x0         | Indicates if active PLL lost its lock                                                                                                    |
| cold_powerup_done | 4          | 1         | true     | W1C    | 0x0         | Cold powerup done interrupt to SMC SW will write this bit to clear the interrupt                                                         |
| ext_dfs_req       | 5          | 1         | true     | W1C    | 0x0         | HW to set this bit when pll_control_reg[ext_pll_dfs_req] is set to trigger CPL interrupt. SW will write this bit to clear the interrupt. |
| ext_dfs_done      | 6          | 1         | true     | W1C    | 0x0         | External initiated DFS completion indication to SMC. SW will write this bit to clear the interrupt                                       |
| ext_dfs_abort     | 7          | 1         | true     | W1C    | 0x0         | External initiated DFS abort indication to SMC. SW will write this bit to clear the interrupt                                            |
| spare             | 8          | 24        | true     | W1C    | 0x0         | Spare Bits                                                                                                                               |

### pll\_wait\_event\_ctrl\_reg

**Address:** 0x42103010

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x000F500F

**Description:** CPL PLL Control Register

| Field Name          | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                             |
|---------------------|------------|-----------|----------|--------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pll_counter_preset0 | 0          | 8         | false    | WARL   | 0xF         | PLL counter preset value. Used for controlling RESETB assertion timing T2, when PLL parameters changed                                                                                  |
| pll_counter_preset1 | 8          | 8         | false    | WARL   | 0x50        | PLL counter preset value. Used for controlling LOCK assertion timing T3, when PLL parameters changed                                                                                    |
| pll_counter_preset2 | 16         | 8         | false    | WARL   | 0xF         | PLL counter preset value. Used for controlling inactive PLL reset assertion timing. Inactive PLL is in reset after switching to configured PLL indicated by pll_status_reg:PLL*_Active. |
| spare               | 24         | 8         | false    | WARL   | 0x0         | Spare Bits                                                                                                                                                                              |

### pll\_parameters0\_reg

**Address:** 0x42103014

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** CPL PLL parameters0 Register

| Field Name          | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description              |
|---------------------|------------|-----------|----------|--------|-------------|--------------------------|
| spare_control_param | 0          | 32        | true     | WARL   | 0x0         | Spare Control Parameters |

### pll\_parameters1\_reg

**Address:** 0x42103018

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** CPL PLL parameters1 Register

| Field Name          | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description              |
|---------------------|------------|-----------|----------|--------|-------------|--------------------------|
| spare_control_param | 0          | 32        | true     | WARL   | 0x0         | Spare Control Parameters |

### pll\_parameters2\_reg

**Address:** 0x4210301C

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** CPL PLL parameters2 Register

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|                     |   |    |      |      |     |                          |
|---------------------|---|----|------|------|-----|--------------------------|
| spare_control_param | 0 | 32 | true | WARL | 0x0 | Spare Control Parameters |
|---------------------|---|----|------|------|-----|--------------------------|

### pll\_parameters3\_reg

**Address:** 0x42103020

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** CPL PLL parameters3 Register

| Field Name          | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description              |
|---------------------|------------|-----------|----------|--------|-------------|--------------------------|
| spare_control_param | 0          | 32        | true     | WARL   | 0x0         | Spare Control Parameters |

### pll\_ext\_control\_reg

**Address:** 0x42103024

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** CPL PLL EXT Control Register

| Field Name      | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                                                                                                                                                             |
|-----------------|------------|-----------|----------|--------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ext_pll_dfs_req | 0          | 1         | false    | WARL   | 0x0         | DFS request from External agent. SMC configures pll_ctrl_reg[p_state] and set this bit. SMC to clear this bit once pll_interrupt_reg[ext_dfs_done] set. 0x0 : No DFS request from SMC or SMC cleared it. 0x1 : DFS request from SMC, generates interrupt to CPL; 0x1 : DFS request from SMC, generates interrupt to CPL |
| p_state         | 1          | 4         | false    | WARL   | 0x0         | P-State Request from SMC. 0x0 : Unused; 0x1 : P1 State; 0x2 : P2 State; 0x3 : P3 State; 0x4 : P4 State; 0x5 to 0x15 : Reserved                                                                                                                                                                                          |
| spare           | 5          | 27        | false    | WARL   | 0x0         | Spare Bits                                                                                                                                                                                                                                                                                                              |

### cc\_scratchpad32

**Address:** 0x42103028

**cluster\_mask:** 0x01E00000

**Size:** 32 bits

**Reset Value:** 0xF5F5F5F5

**Description:** Scratchpad register for DV

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| data       | 0          | 32        | false    | WARL   | 0xF5F5F5F5  |             |

### mbx\_doorbell\_capability\_reg

**Address:** 0x42170000

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000103

**Description:** Mailbox Capability Register

| Field Name              | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                     |
|-------------------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------|
| types_avail             | 0          | 8         | false    | WARL   | 0x3         | If Bit[0] = 1, Type0 is Available, If Bit[1] = 1, Type1 is Available, If Bit[2] = 1, Type2 is Available, and so on till bit[7]. |
| mbox_intr_s<br>upported | 8          | 1         | false    | WARL   | 0x1         | Mailbox Interrupt Capability. At least one type of Interrupts is supported on Mailbox Status Change.                            |
| rsvd_63_to_9            | 9          | 55        | false    | WARL   | 0x0         |                                                                                                                                 |

## msg\_reg

**Address:** 0x42170008

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Message Register

| Field Name    | Bit Offset | Bit Width | Volatile | Access     | Reset Value | Description                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|------------|-----------|----------|------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data0         | 0          | 16        | false    | read-write | 0x0         | Data is used to define the message fields as per Type protocol. Use Extended Data1-3 if needed for a given message Type.                                                                                                                                                                                                                                                                       |
| mbox_num      | 16         | 4         | false    | read-write | 0x0         | Mbox# to track the type message. With Type and Mbox# values, FW can track the life of a message.                                                                                                                                                                                                                                                                                               |
| extended_data | 20         | 3         | false    | read-write | 0x0         | If 0x1, use Extended Data1, if 0x2 use Extended Data1 & Extended Data0, and so on                                                                                                                                                                                                                                                                                                              |
| type          | 23         | 3         | false    | read-write | 0x0         | For Ascalon Cluster, see table "type" for encoding and number of mbox-es per Type                                                                                                                                                                                                                                                                                                              |
| enable_reload | 26         | 1         | false    | read-write | 0x0         | Reload mailbox. Repeat Mailbox commands "N" times with an interval of "M" cycles, before marking Mailbox as complete. N is "repeat_count and M is reload_delay_count                                                                                                                                                                                                                           |
| rsvd_63_to_26 | 27         | 37        | false    | read-only  | 0x0         | <tr><th>Type Name [2:0]</th><th>Number of mbox-es</th><th>Message Fields</th><th>Response Type </th><tr><tr><td>0x0: pmnetwork</td>15</td> Command, Response </td>Response from pmnetwork (including Data)</tr><tr><td>0x1: Temp Hub</td>8</td> Command, Response </td>Response from Temp Hub, Dedicated registers for Temp Snapshot Data)</tr><tr><td>0x2 to 0x3: Reserved</td><td><td> </tr> |

## msg\_reg\_data

**Address:** 0x42170010

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Extended Data Registers for Message

| Field Name     | Bit Offset | Bit Width | Volatile | Access     | Reset Value | Description                                                    |
|----------------|------------|-----------|----------|------------|-------------|----------------------------------------------------------------|
| extended_data1 | 0          | 16        | false    | read-write | 0x0         | Data is used to define the message fields as per Type protocol |

|                |    |    |       |            |     |                                                                |
|----------------|----|----|-------|------------|-----|----------------------------------------------------------------|
| extended_data2 | 16 | 16 | false | read-write | 0x0 | Data is used to define the message fields as per Type protocol |
| extended_data3 | 32 | 16 | false | read-write | 0x0 | Data is used to define the message fields as per Type protocol |
| extended_data4 | 48 | 16 | false | read-write | 0x0 | Data is used to define the message fields as per Type protocol |

### msg\_type\_sts\_reg

**Address:** 0x42170018

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Status of Mailbox across all types.

| Field Name    | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|------------|-----------|----------|-----------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type0_sts     | 0          | 4         | true     | read-only | 0x0         | Each Status field is a 4-bit field<br><br>Group0: If Bit[3] is set to 0, decode the rest of bits as follows: <br>Bit0: Complete: At least one Mailbox is in COMPLETE state.<br>Bit 1: N_Complete: # of Mailbox in COMPLETE state equals N (threshold value defined in Config register). N should be non-zero for this bit to be set. <br>Bit 2: Filter_Complete: At least one Mailbox selected by Filters (in Type Configuration) is in COMPLETE STATE.<br>Group1: If Bit[3] is set to 1, the status belongs to Group 1. Group 1 is reserved for future use. <br> |
| type1_sts     | 4          | 4         | true     | read-only | 0x0         | See Type0Sts Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| type2_sts     | 8          | 4         | false    | read-only | 0x0         | See Type0Sts Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| type3_sts     | 12         | 4         | false    | read-only | 0x0         | See Type0Sts Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| type4_sts     | 16         | 4         | false    | read-only | 0x0         | See Type0Sts Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| type5_sts     | 20         | 4         | false    | read-only | 0x0         | See Type0Sts Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| type6_sts     | 24         | 4         | false    | read-only | 0x0         | See Type0Sts Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| type7_sts     | 28         | 4         | false    | read-only | 0x0         | See Type0Sts Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| rsvd_63_to_32 | 32         | 32        | false    | read-only | 0x0         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

### mbox\_sts\_type1\_type0\_reg

**Address:** 0x42170020

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x5555000000000000

**Description:** PM Network Mailbox Status (Used for communication w/ satellites for PM transition (type1) and Temp Hubs (type0))

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|                |    |   |      |           |     |                                                                                                                                                                                                                                                                                                                                                                        |
|----------------|----|---|------|-----------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type0_mb0_sts  | 0  | 2 | true | read-only | 0x0 | Bit[1]: Empty, Bit[0]: Done <br>2'b00: EMPTY: Mailbox Empty w/ all previous Completion cleared by FW <br>2'b01: NOT SUPPORTED: This mailbox# is not supported. The mailbox is not considered for determining Type Status. <br>2'b10: PROGRESS: Mailbox Not Empty. Mailbox Message in Progress. <br>2'b11: COMPLETE: Mailbox Not Empty. Mailbox Message Completed. <br> |
| type0_mb1_sts  | 2  | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type0_mb2_sts  | 4  | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type0_mb3_sts  | 6  | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type0_mb4_sts  | 8  | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type0_mb5_sts  | 10 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type0_mb6_sts  | 12 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type0_mb7_sts  | 14 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type0_mb8_sts  | 16 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type0_mb9_sts  | 18 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type0_mb10_sts | 20 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type0_mb11_sts | 22 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type0_mb12_sts | 24 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type0_mb13_sts | 26 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type0_mb14_sts | 28 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type0_mb15_sts | 30 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type1_mb0_sts  | 32 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type1_mb1_sts  | 34 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type1_mb2_sts  | 36 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type1_mb3_sts  | 38 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type1_mb4_sts  | 40 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type1_mb5_sts  | 42 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type1_mb6_sts  | 44 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |
| type1_mb7_sts  | 46 | 2 | true | read-only | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description                                                                                                                                                                                                                                                                                                                            |

|                |    |   |      |           |     |                                             |
|----------------|----|---|------|-----------|-----|---------------------------------------------|
| type1_mb8_sts  | 48 | 2 | true | read-only | 0x1 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb9_sts  | 50 | 2 | true | read-only | 0x1 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb10_sts | 52 | 2 | true | read-only | 0x1 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb11_sts | 54 | 2 | true | read-only | 0x1 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb12_sts | 56 | 2 | true | read-only | 0x1 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb13_sts | 58 | 2 | true | read-only | 0x1 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb14_sts | 60 | 2 | true | read-only | 0x1 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb15_sts | 62 | 2 | true | read-only | 0x1 | See mbox_sts_type0_reg[mb0_sts] Description |

### mbox\_sts\_type3\_type2\_reg

**Address:** 0x42170028

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x5555555555555555

**Description:** Reserved for Future use

| Field Name           | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description |
|----------------------|------------|-----------|----------|-----------|-------------|-------------|
| type2_mb_15_to_0_sts | 0          | 32        | false    | read-only | 0x55555555  |             |
| type3_mb_15_to_0_sts | 32         | 32        | false    | read-only | 0x55555555  |             |

### mbox\_sts\_type1\_type0\_reg\_w2c

**Address:** 0x42170030

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Reserved for Future use

| Field Name        | Bit Offset | Bit Width | Volatile | Access     | Reset Value | Description                                                                         |
|-------------------|------------|-----------|----------|------------|-------------|-------------------------------------------------------------------------------------|
| type0_mb0_sts_w2c | 0          | 2         | false    | read-write | 0x0         | If [1:0] = COMPLETE and Mailbox is in COMPLETE, then the mailbox is reset to EMPTY. |
| type0_mb1_sts_w2c | 2          | 2         | false    | read-write | 0x0         | See mbox_sts_type0_reg[mb0_sts] Description                                         |
| type0_mb2_sts_w2c | 4          | 2         | false    | read-write | 0x0         | See mbox_sts_type0_reg[mb0_sts] Description                                         |
| type0_mb3_sts_w2c | 6          | 2         | false    | read-write | 0x0         | See mbox_sts_type0_reg[mb0_sts] Description                                         |
| type0_mb4_sts_w2c | 8          | 2         | false    | read-write | 0x0         | See mbox_sts_type0_reg[mb0_sts] Description                                         |
| type0_mb5_sts_w2c | 10         | 2         | false    | read-write | 0x0         | See mbox_sts_type0_reg[mb0_sts] Description                                         |

|                    |    |   |       |            |     |                                             |
|--------------------|----|---|-------|------------|-----|---------------------------------------------|
| type0_mb6_sts_w2c  | 12 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type0_mb7_sts_w2c  | 14 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type0_mb8_sts_w2c  | 16 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type0_mb9_sts_w2c  | 18 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type0_mb10_sts_w2c | 20 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type0_mb11_sts_w2c | 22 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type0_mb12_sts_w2c | 24 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type0_mb13_sts_w2c | 26 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type0_mb14_sts_w2c | 28 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type0_mb15_sts_w2c | 30 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb0_sts_w2c  | 32 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb1_sts_w2c  | 34 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb2_sts_w2c  | 36 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb3_sts_w2c  | 38 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb4_sts_w2c  | 40 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb5_sts_w2c  | 42 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb6_sts_w2c  | 44 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb7_sts_w2c  | 46 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb8_sts_w2c  | 48 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb9_sts_w2c  | 50 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb10_sts_w2c | 52 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb11_sts_w2c | 54 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb12_sts_w2c | 56 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb13_sts_w2c | 58 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb14_sts_w2c | 60 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |
| type1_mb15_sts_w2c | 62 | 2 | false | read-write | 0x0 | See mbox_sts_type0_reg[mb0_sts] Description |

## mbx\_sts\_type3\_type2\_reg\_w2c

**Address:** 0x42170038

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x5555555555555555

**Description:** Reserved for Future use

| Field Name               | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description |
|--------------------------|------------|-----------|----------|-----------|-------------|-------------|
| type2_mb_15_to_0_sts_w2c | 0          | 32        | false    | read-only | 0x55555555  |             |
| type3_mb_15_to_0_sts_w2c | 32         | 32        | false    | read-only | 0x55555555  |             |

## mbx\_cfg\_type1\_type0\_reg

**Address:** 0x42170040

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x000000070000000F

**Description:** Mailbox Configuration Register for Type 0 & Type 1

| Field Name         | Bit Offset | Bit Width | Volatile | Access     | Reset Value | Description                                                                                                                                                                                                                                                                                                                |
|--------------------|------------|-----------|----------|------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type0_mbox_id_max  | 0          | 4         | false    | read-only  | 0xF         | The maximum id value allowed for a mbox# in the message register for the type specified by this cfg register.                                                                                                                                                                                                              |
| type0_mbox_ie      | 4          | 4         | false    | read-write | 0x0         | Interrupt Enables <br>If Bit[0] = 1, Enable Interrupt if Status = COMPLETE <br>If Bit[1] = 1, Enable Interrupt if Status = N_COMPLETE <br>If Bit[2] = 1, Enable Interrupt if Status = FILTER_COMPLETE <br>Bit[3]: Reserved. for Future <br>Interrupts are cleared when the corresponding mbox# status bit is cleared. <br> |
| type0_mbox_fltr    | 8          | 16        | false    | read-write | 0x0         | Mailbox Filter <br>If Mailbox Filter[0] = 1 and mbox0 = COMPLETE, mark FILTER_COMPLETE as 1. <br>If Mailbox Filter[1] = 1 and mbox1 = COMPLETE, mark FILTER_COMPLETE as 1. <br>If Mailbox Filter[2] = 1 and mbox2 = COMPLETE, mark FILTER_COMPLETE as 1. <br>and so on. <br>                                               |
| type0_thresh_old_n | 24         | 5         | false    | read-write | 0x0         | N-Complete Threshold <br>If "threshold_n" number of mailboxes reach COMPLETE state, mark N_COMPLETE Type Status as "1".                                                                                                                                                                                                    |
| rsvd_30_to_29      | 29         | 2         | false    | read-only  | 0x0         | Rserved                                                                                                                                                                                                                                                                                                                    |
| type0_enable       | 31         | 1         | false    | read-write | 0x0         | Mailbox message is enabled once this bit is set by FW.                                                                                                                                                                                                                                                                     |
| type1_mbox_id_max  | 32         | 4         | false    | read-only  | 0x7         | The maximum value allowed for a mbox# in the message register for the type specified by this cfg register.                                                                                                                                                                                                                 |

|                    |    |    |       |            |     |                                                                                                                                                                                                                                                                                                                            |
|--------------------|----|----|-------|------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type1_mbox_ie      | 36 | 4  | false | read-write | 0x0 | Interrupt Enables <br>If Bit[0] = 1, Enable Interrupt if Status = COMPLETE <br>If Bit[1] = 1, Enable Interrupt if Status = N_COMPLETE <br>If Bit[2] = 1, Enable Interrupt if Status = FILTER_COMPLETE <br>Bit[3]: Reserved. for Future <br>Interrupts are cleared when the corresponding mbox# status bit is cleared. <br> |
| type1_mbox_filtr   | 40 | 16 | false | read-write | 0x0 | Mailbox Filter <br>If Mailbox Filter[0] = 1 and mbox0 = COMPLETE, mark FILTER_COMPLETE as 1. <br>If Mailbox Filter[1] = 1 and mbox1 = COMPLETE, mark FILTER_COMPLETE as 1. <br>If Mailbox Filter[2] = 1 and mbox2 = COMPLETE, mark FILTER_COMPLETE as 1. <br>and so on. <br>                                               |
| type1_thresh_old_n | 56 | 5  | false | read-write | 0x0 | N-Complete Threshold <br>If "threshold_n" number of mailboxes reach COMPLETE state, mark N_COMPLETE Type Status as "1".                                                                                                                                                                                                    |
| rsvd_61_to_60      | 61 | 2  | false | read-only  | 0x0 | Reserved                                                                                                                                                                                                                                                                                                                   |
| type1_enable       | 63 | 1  | false | read-write | 0x0 | Mailbox message is enabled once this bit is set by FW. Virtual Channel intended for Temp Hub traffic.                                                                                                                                                                                                                      |

### mbx\_cfg\_type3\_type2\_reg

**Address:** 0x42170048

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Mailbox Configuration Register for Type 3 & Type 2 (Rsvd for future use)

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description |
|------------|------------|-----------|----------|-----------|-------------|-------------|
| rsvd       | 0          | 64        | false    | read-only | 0x0         | Reserved    |

### mbx\_query

**Address:** 0x42170060

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** FW register to query the mailbox status

| Field Name   | Bit Offset | Bit Width | Volatile | Access     | Reset Value | Description                                                                            |
|--------------|------------|-----------|----------|------------|-------------|----------------------------------------------------------------------------------------|
| mbox_num     | 0          | 4         | false    | read-write | 0x0         | Mbox# of message type. With Type and Mbox# values, FW can track the life of a message. |
| type         | 4          | 3         | false    | read-write | 0x0         | For Ascalon Cluster, see table "type" for number of mbox-es per Type                   |
| rsvd_63_to_7 | 7          | 57        | false    | read-only  | 0x0         | Reserved                                                                               |

### mbx\_query\_reply

**Address:** 0x42170068

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Reply to Mailbox Query

| Field Name     | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                 |
|----------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------|
| immediate_data | 0          | 32        | true     | read-only | 0x0         | Reply data. Data fields determined by Type.                                                 |
| memory_ptr_vld | 32         | 1         | true     | read-only | 0x0         | If set to 1, additional data in memory. Use mbx_query_reply_1[memory_ptr] for address       |
| mbx_num        | 33         | 4         | true     | read-only | 0x0         | Reflects the mailbox number requested in the last mbx_query register                        |
| type           | 37         | 3         | true     | read-only | 0x0         | Reflects the type requested in the last mbx_query register                                  |
| mbx_status     | 40         | 2         | true     | read-only | 0x0         | Status of Mbox                                                                              |
| mbx_error      | 42         | 2         | true     | read-only | 0x0         | 0: No Errors; 1: Message completed w/ unexpected satellite response, 2-3: Rsvd. For future. |
| rsvd_62_to_44  | 44         | 19        | false    | read-only | 0x0         | Additional data in memory for mbox query.                                                   |
| valid          | 63         | 1         | true     | read-only | 0x0         | Qualifer for memory_ptr_vld, immediate_data, and mbox_error                                 |

## mbx\_query\_reply\_1

**Address:** 0x42170070

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Additional data for mbox query.

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                               |
|------------|------------|-----------|----------|-----------|-------------|-------------------------------------------|
| memory_ptr | 0          | 64        | true     | read-only | 0x0         | Additional data in memory for mbox query. |

## db\_cfg\_reg

**Address:** 0x42170078

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Doorbell Cfg Register

| Field Name     | Bit Offset | Bit Width | Volatile | Access     | Reset Value | Description                                                                                                                                        |
|----------------|------------|-----------|----------|------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| priotiy0_db_ie | 0          | 1         | false    | read-write | 0x0         | Priority0 Doorbells Interrupt Enable1 bit for each typeSingle bit to enable interrupts in case the corresponding status bit transitions from 1 → 0 |
| priotiy1_db_ie | 1          | 1         | false    | read-write | 0x0         | Priority1 Doorbells Interrupt Enable1 bit for each typeSingle bit to enable interrupts in case the corresponding status bit transitions from 1 → 0 |
| rsvd_7_2       | 2          | 6         | false    | read-only  | 0x0         |                                                                                                                                                    |

|                     |    |    |       |            |     |                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------|----|----|-------|------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| priority0_db_events | 8  | 8  | false | read-write | 0x0 | Use this field to generate priority 0 interrupt on HW events. Set the corresponding bit to generate priority 0 interrupt for the events. For now, only support bit-6 and bit-7 events. bit 0: CACHE_FLUSH_STATUS, bit 1: TEMP_SENSOR_DATA_RSP, bit 2: TEMP_HUB_DATA_RSP, bit 3: ZKR_SEED_DEMAND, bit 4: REG_RD_DATA, bit 5: REG_WR_ACK, bit 6: PM_TRANSITION_REQ, bit 7: PKT_CORE_C_STATUS |
| priority1_db_events | 16 | 8  | false | read-write | 0x0 | Use this field to generate priority 1 interrupt on HW events. Use the same bit definition as priority0_db_events.                                                                                                                                                                                                                                                                          |
| rsvd_63_24          | 24 | 40 | false | read-only  | 0x0 |                                                                                                                                                                                                                                                                                                                                                                                            |

### db\_sts\_reg

**Address:** 0x42170080

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Doorbell Status

| Field Name       | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                             |
|------------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| priority0_db_sts | 0          | 8         | true     | read-only | 0x0         | Priority0 Doorbells Status. Each bit represents the status of event. The event definition of each bit is same as listed in "priority0_db_events" field. |
| priority1_db_sts | 8          | 8         | true     | read-only | 0x0         | Priority1 Doorbells Status. Each bit represents the status of event. The event definition of each bit is same as listed in "priority0_db_events" field. |
| rsvd_63_16       | 16         | 48        | false    | read-only | 0x0         |                                                                                                                                                         |

### db\_value\_event0\_reg

**Address:** 0x42170088

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Doorbell Related Data for Event 0

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description |
|------------|------------|-----------|----------|-----------|-------------|-------------|
| data       | 0          | 64        | true     | read-only | 0x0         |             |

### db\_value\_event1\_reg

**Address:** 0x42170090

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Doorbell Related Data for Event 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|      |   |    |      |           |     |  |
|------|---|----|------|-----------|-----|--|
| data | 0 | 64 | true | read-only | 0x0 |  |
|------|---|----|------|-----------|-----|--|

### db\_value\_event2\_reg

**Address:** 0x42170098

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Doorbell Related Data for Event 2

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description |
|------------|------------|-----------|----------|-----------|-------------|-------------|
| data       | 0          | 64        | true     | read-only | 0x0         |             |

### db\_value\_event3\_reg

**Address:** 0x421700A0

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Doorbell Related Data for Event 3 (ZKR Seed Demand)

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                       |
|------------|------------|-----------|----------|-----------|-------------|-------------------------------------------------------------------|
| core0_data | 0          | 8         | true     | read-only | 0x0         | Request from Core-0, core0 data[6:4] ZKR Seed Demand From Core 0) |
| core1_data | 8          | 8         | true     | read-only | 0x0         | Request from Core-1, core1 data[6:4] ZKR Seed Demand From Core 1) |
| core2_data | 16         | 8         | true     | read-only | 0x0         | Request from Core-2, core2 data[6:4] ZKR Seed Demand From Core 2) |
| core3_data | 24         | 8         | true     | read-only | 0x0         | Request from Core-3, core3 data[6:4] ZKR Seed Demand From Core 3) |
| core4_data | 32         | 8         | true     | read-only | 0x0         | Request from Core-4, core4 data[6:4] ZKR Seed Demand From Core 4) |
| core5_data | 40         | 8         | true     | read-only | 0x0         | Request from Core-5, core5 data[6:4] ZKR Seed Demand From Core 5) |
| core6_data | 48         | 8         | true     | read-only | 0x0         | Request from Core-6, core6 data[6:4] ZKR Seed Demand From Core 6) |
| core7_data | 56         | 8         | true     | read-only | 0x0         | Request from Core-7, core7 data[6:4] ZKR Seed Demand From Core 7) |

### db\_value\_event4\_reg

**Address:** 0x421700B0

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Doorbell Related Data for Event 4

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description |
|------------|------------|-----------|----------|-----------|-------------|-------------|
| data       | 0          | 64        | true     | read-only | 0x0         |             |

### db\_value\_event5\_reg

**Address:** 0x421700B8

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Doorbell Related Data for Event 5

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description |
|------------|------------|-----------|----------|-----------|-------------|-------------|
| data       | 0          | 64        | true     | read-only | 0x0         |             |

## db\_value\_event6\_reg

**Address:** 0x421700C0

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Doorbell Related Data for Event 6

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                         |
|------------|------------|-----------|----------|-----------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| core0_data | 0          | 8         | true     | read-only | 0x0         | Request from Core-0, core0 data[6:4] Pointer in the DFS frequency table, bit[3:0]: Core C-state Requested. If 4, requested C-state is 4. If 3 requestet C-state is C3. (other Values not supported) |
| core1_data | 8          | 8         | true     | read-only | 0x0         | Request from Core-1, core1 data[6:4] Pointer in the DFS frequency table, bit[3:0]: Core C-state Requested. If 4, requested C-state is 4. If 3 requestet C-state is C3. (other Values not supported) |
| core2_data | 16         | 8         | true     | read-only | 0x0         | Request from Core-2, core2 data[6:4] Pointer in the DFS frequency table, bit[3:0]: Core C-state Requested. If 4, requested C-state is 4. If 3 requestet C-state is C3. (other Values not supported) |
| core3_data | 24         | 8         | true     | read-only | 0x0         | Request from Core-3, core3 data[6:4] Pointer in the DFS frequency table, bit[3:0]: Core C-state Requested. If 4, requested C-state is 4. If 3 requestet C-state is C3. (other Values not supported) |
| core4_data | 32         | 8         | true     | read-only | 0x0         | Request from Core-4, core4 data[6:4] Pointer in the DFS frequency table, bit[3:0]: Core C-state Requested. If 4, requested C-state is 4. If 3 requestet C-state is C3. (other Values not supported) |
| core5_data | 40         | 8         | true     | read-only | 0x0         | Request from Core-5, core5 data[6:4] Pointer in the DFS frequency table, bit[3:0]: Core C-state Requested. If 4, requested C-state is 4. If 3 requestet C-state is C3. (other Values not supported) |
| core6_data | 48         | 8         | true     | read-only | 0x0         | Request from Core-6, core6 data[6:4] Pointer in the DFS frequency table, bit[3:0]: Core C-state Requested. If 4, requested C-state is 4. If 3 requestet C-state is C3. (other Values not supported) |
| core7_data | 56         | 8         | true     | read-only | 0x0         | Request from Core-7, core7 data[6:4] Pointer in the DFS frequency table, bit[3:0]: Core C-state Requested. If 4, requested C-state is 4. If 3 requestet C-state is C3. (other Values not supported) |

## db\_value\_event7\_reg

**Address:** 0x421700C8

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Doorbell Related Data for Event 7

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                              |
|------------|------------|-----------|----------|-----------|-------------|----------------------------------------------------------------------------------------------------------|
| core0_data | 0          | 8         | true     | read-only | 0x0         | Core-0 C-state Status Update. If 2, Core in C2, If 3 Core in C3, 4 Core in C4, If 0 in Core in C0 (Wake) |
| core1_data | 8          | 8         | true     | read-only | 0x0         | Core-1 C-state Status Update. If 2, Core in C2, If 3 Core in C3, 4 Core in C4, If 0 in Core in C0 (Wake) |
| core2_data | 16         | 8         | true     | read-only | 0x0         | Core-2 C-state Status Update. If 2, Core in C2, If 3 Core in C3, 4 Core in C4, If 0 in Core in C0 (Wake) |
| core3_data | 24         | 8         | true     | read-only | 0x0         | Core-3 C-state Status Update. If 2, Core in C2, If 3 Core in C3, 4 Core in C4, If 0 in Core in C0 (Wake) |
| core4_data | 32         | 8         | true     | read-only | 0x0         | Core-4 C-state Status Update. If 2, Core in C2, If 3 Core in C3, 4 Core in C4, If 0 in Core in C0 (Wake) |
| core5_data | 40         | 8         | true     | read-only | 0x0         | Core-5 C-state Status Update. If 2, Core in C2, If 3 Core in C3, 4 Core in C4, If 0 in Core in C0 (Wake) |
| core6_data | 48         | 8         | true     | read-only | 0x0         | Core-6 C-state Status Update. If 2, Core in C2, If 3 Core in C3, 4 Core in C4, If 0 in Core in C0 (Wake) |
| core7_data | 56         | 8         | true     | read-only | 0x0         | Core-7 C-state Status Update. If 2, Core in C2, If 3 Core in C3, 4 Core in C4, If 0 in Core in C0 (Wake) |

## db\_sts\_reg\_w2c

**Address:** 0x421700D0

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Doorbell Status

| Field Name       | Bit Offset | Bit Width | Volatile | Access     | Reset Value | Description                                                                                                                       |
|------------------|------------|-----------|----------|------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------|
| priority0_db_sts | 0          | 8         | false    | read-write | 0x0         | Write 1 to clear the the status of pri0 event. The event definition of each bit is same as listed in "priority0_db_events" field. |
| priority1_db_sts | 8          | 8         | false    | read-write | 0x0         | Write 1 to clear the the status of pri1 event. The event definition of each bit is same as listed in "priority0_db_events" field. |
| rsvd_63_16       | 16         | 48        | false    | read-only  | 0x0         |                                                                                                                                   |

## temperature\_data\_base\_address

**Address:** 0x421700D8

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Address to load Temp Hub Snapshot

| Field Name   | Bit Offset | Bit Width | Volatile | Access     | Reset Value | Description                                                                                                          |
|--------------|------------|-----------|----------|------------|-------------|----------------------------------------------------------------------------------------------------------------------|
| address      | 0          | 19        | false    | read-write | 0x0         |                                                                                                                      |
| max_sample_s | 19         | 13        | false    | read-write | 0x0         | Maximum number of temperature hub responses to be recorded before rollover in the memory to start from base address. |
| rsvd_63_32   | 32         | 32        | false    | read-only  | 0x0         |                                                                                                                      |

## **mbx\_reload\_parameters**

**Address:** 0x421700E0

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** None

| Field Name         | Bit Offset | Bit Width | Volatile | Access     | Reset Value | Description                                                                                                                                     |
|--------------------|------------|-----------|----------|------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| reload_delay_count | 0          | 32        | false    | read-write | 0x0         | if mailbox has "enable_reload" set, wait "reload_delay_count" before repeating Mailbox "repeat_count" times before marking Mailbox as COMPLETE. |
| repeat_count       | 32         | 32        | false    | read-write | 0x0         | if mailbox has "enable_reload" set, Repeat Mailbox "repeat_count" times before marking Mailbox as COMPLETE.                                     |

## **mb\_scratchpad64**

**Address:** 0x421700E8

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x8F8F8F8F8F8F8F8F

**Description:** Scratchpad register for DV

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value        | Description |
|------------|------------|-----------|----------|--------|--------------------|-------------|
| data       | 0          | 64        | false    | WARL   | 0x8F8F8F8F8F8F8F8F |             |

## **pmnw\_cfg**

**Address:** 0x421700F0

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000049

**Description:** None

| Field Name    | Bit Offset | Bit Width | Volatile | Access     | Reset Value | Description                                                                             |
|---------------|------------|-----------|----------|------------|-------------|-----------------------------------------------------------------------------------------|
| num_st_l_rnds | 0          | 3         | false    | read-write | 0x1         | Sets the number of standard and large round sets in the PMNW arbiter before an XL round |
| num_l_rnds    | 3          | 3         | false    | read-write | 0x1         | Sets the number of large rounds in the PMNW arbiter (during large grants)               |

|             |   |    |       |            |     |                                                                                 |
|-------------|---|----|-------|------------|-----|---------------------------------------------------------------------------------|
| num_st_rnds | 6 | 3  | false | read-write | 0x1 | Sets the number of standard rounds in the PMNW arbiter (during standard grants) |
| rsvd_63_9   | 9 | 55 | false | read-only  | 0x0 |                                                                                 |

## mtime

**Address:** 0x42180000

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Machine-level time counter

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mtime      | 0          | 64        | true     | WARL   | 0x0         |             |

## mtimecmp0

**Address:** 0x42188000

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x00000000FFFFFFFF

**Description:** Machine-level time compare register for Core 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mtimecmp   | 0          | 64        | false    | WARL   | 0xFFFFFFFF  |             |

## mtimecmp1

**Address:** 0x42188008

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x00000000FFFFFFFF

**Description:** Machine-level time compare register for Core 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mtimecmp   | 0          | 64        | false    | WARL   | 0xFFFFFFFF  |             |

## mtimecmp2

**Address:** 0x42188010

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x00000000FFFFFFFF

**Description:** Machine-level time compare register for Core 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mtimecmp   | 0          | 64        | false    | WARL   | 0xFFFFFFFF  |             |

## mtimecmp3

**Address:** 0x42188018

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x00000000FFFFFFFF

**Description:** Machine-level time compare register for Core 3

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mtimecmp   | 0          | 64        | false    | WARL   | 0xFFFFFFFF  |             |

#### mtimecmp4

**Address:** 0x42188020

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x00000000FFFFFFFF

**Description:** Machine-level time compare register for Core 4

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mtimecmp   | 0          | 64        | false    | WARL   | 0xFFFFFFFF  |             |

#### mtimecmp5

**Address:** 0x42188028

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x00000000FFFFFFFF

**Description:** Machine-level time compare register for Core 5

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mtimecmp   | 0          | 64        | false    | WARL   | 0xFFFFFFFF  |             |

#### mtimecmp6

**Address:** 0x42188030

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x00000000FFFFFFFF

**Description:** Machine-level time compare register for Core 6

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mtimecmp   | 0          | 64        | false    | WARL   | 0xFFFFFFFF  |             |

#### mtimecmp7

**Address:** 0x42188038

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x00000000FFFFFFFF

**Description:** Machine-level time compare register for Core 7

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mtimecmp   | 0          | 64        | false    | WARL   | 0xFFFFFFFF  |             |

## mtimecmp8

**Address:** 0x42188040

**cluster\_mask:** 0x01E00000

**Size:** 64 bits

**Reset Value:** 0x00000000FFFFFFFF

**Description:** Machine-level time compare register for Cluster CPL

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mtimecmp   | 0          | 64        | false    | WARL   | 0xFFFFFFFF  |             |

## sc\_ctrl

**Address:** 0x421A0000

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000B4D20

**Description:** Configure slice, control flush, configure PMU counters

| Field Name                 | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                          |
|----------------------------|------------|-----------|----------|-----------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| force_init_done            | 0          | 1         | false    | WARL      | 0x0         | If 1, will force the hardware init state machine to stop running, and force SC to believe array init was completed.                                  |
| flush_enable               | 1          | 1         | true     | WARL      | 0x0         | If 1, will enable the SC Flush State Machine.                                                                                                        |
| flush_abort                | 2          | 1         | true     | WARL      | 0x0         | If 1, will abort the in progress SC Flush.                                                                                                           |
| pmu_counter_en             | 3          | 1         | false    | WARL      | 0x0         | Enables the PMU Counters                                                                                                                             |
| pmu_counter_reset          | 4          | 1         | false    | WARL      | 0x0         | Resets all the PMU Counters                                                                                                                          |
| wrevict_disable            | 5          | 1         | false    | WARL      | 0x1         | Disable WriteEvictFull generation from SC to HN                                                                                                      |
| aegis_cmo_enable           | 6          | 1         | false    | WARL      | 0x0         | Aegis Custom CMO Flow enablement                                                                                                                     |
| creq_mshr_threshold        | 7          | 6         | false    | WARL      | 0x1A        | MSHR Occupancy Threshold for Core Requests                                                                                                           |
| cpreq_mshr_threshold       | 13         | 6         | false    | WARL      | 0x1A        | MSHR Occupancy Threshold for Completer Port Requests                                                                                                 |
| hwa_write_policy           | 19         | 2         | false    | WARL      | 0x1         | Determines how new hardware assertions are logged in sc_hardware_assertion. 0=disabled, 1=log first, 2=log latest, 3=log worst (highest hwatype val) |
| cla_debug_mshr_id_write_en | 21         | 1         | true     | WARL      | 0x0         | Set to 1 to force the CLA MSHR debug id to be equal to the value in the cla_debug_mshr_id field. Hardware will reset this bit to 0 upon completion.  |
| cla_debug_mshr_id          | 22         | 5         | false    | WARL      | 0x0         | CLA MSHR debug id overwrite value                                                                                                                    |
| unused                     | 27         | 37        | false    | Read-Only | 0x0         | Reserved                                                                                                                                             |

## sc\_status

**Address:** 0x421A0008

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Status of the slice

| Field Name       | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                 |
|------------------|------------|-----------|----------|-----------|-------------|-------------------------------------------------------------|
| array_init_done  | 0          | 1         | true     | Read-Only | 0x0         | Hardware sets to 1 when array init is complete              |
| flush_done       | 1          | 1         | true     | Read-Only | 0x0         | Hardware sets to 1 when SC Flush successfully done.         |
| flush_abort_done | 2          | 1         | true     | Read-Only | 0x0         | Hardware sets to 1 when SC Flush Abort successfully done.   |
| sp_reconfig_done | 3          | 1         | true     | Read-Only | 0x0         | Hardware sets to 1 when Dynamic SP reconfiguration is done. |
| unused           | 4          | 60        | true     | Read-Only | 0x0         | Reserved                                                    |

### sc\_sp

**Address:** 0x421A0010

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** ScratchPad configuration

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                      |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| size       | 0          | 6         | false    | WARL   | 0x0         | Ways reserved for ScratchPad in SharedCache; Programming needs to be in a legal range inclusive of sc_harvest_strap programming; Effectively, this field is write and read legal |
| unused     | 6          | 22        | false    | WARL   | 0x0         | Reserved                                                                                                                                                                         |
| base       | 28         | 36        | false    | WARL   | 0x0         | Base Address for ScratchPad Access, specifies address bits                                                                                                                       |

### sc\_dft\_scratch

**Address:** 0x421A0038

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x6F6F6F6F6F6F6F6F

**Description:** Scratch register used for testing. Has no HW functionality.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value        | Description |
|------------|------------|-----------|----------|--------|--------------------|-------------|
| data       | 0          | 64        | false    | WARL   | 0x6F6F6F6F6F6F6F6F |             |

### sc\_chicken\_bits

**Address:** 0x421A0040

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0080070000000000

**Description:** None

| Field Name                       | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                               |
|----------------------------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------------------------------------|
| mshr_disable                     | 0          | 32        | false    | WARL      | 0x0         | Per MSHR disable vector                                                                                                   |
| mshr_pipelined_disable           | 32         | 1         | false    | WARL      | 0x0         | If 1, disables pipelining. Only one MSHR can access the pipe at a time                                                    |
| qos_disable                      | 33         | 1         | false    | WARL      | 0x0         | If 1, disables QOS way partitioning. All RCIDs will have access to all ways                                               |
| ras_signal_disable               | 34         | 3         | false    | WARL      | 0x0         | If 1, disables RAS interrupt signaling {platform,high,low}.                                                               |
| no_allocate                      | 37         | 1         | false    | WARL      | 0x0         | If 1, no lines will be allocated in the cache                                                                             |
| snp_fwd_disable                  | 38         | 1         | false    | WARL      | 0x0         | If 1, disables fwd snoops                                                                                                 |
| snp_upgrade_invalidate           | 39         | 1         | false    | WARL      | 0x0         | If 1, all snoops will invalidate                                                                                          |
| num_same_addr_queue_enabled      | 40         | 3         | false    | WARL      | 0x7         | Size of the same address queue, min value =1 required.                                                                    |
| core_snp_no_upg                  | 43         | 1         | false    | WARL      | 0x0         | Disable Upgrading SnpOnce to SnpClean - added for RVDE-18926                                                              |
| rsvd44                           | 44         | 1         | false    | WARL      | 0x0         | Reserved for ECO use                                                                                                      |
| sp_reconfig_chnlblock_disable    | 45         | 1         | false    | WARL      | 0x0         | Disable Request Channel blocking when SP reconfig is happening                                                            |
| sp_reconfig_cp_chnlblock_disable | 46         | 1         | false    | WARL      | 0x0         | Disable Completer Port Request Channel blocking when SP reconfig is happening                                             |
| slice_cg_disable                 | 47         | 1         | false    | WARL      | 0x0         | Disable the SC slice level clock gater.                                                                                   |
| intra_slice_clock_gate_disable   | 48         | 1         | false    | WARL      | 0x0         | Disable mid level clock gating inside the slice.                                                                          |
| livelock_replay_cnt_max          | 49         | 2         | false    | WARL      | 0x0         | Programmable Value of Replay Counter to include in Livelock detection.                                                    |
| livelock_dealloc_timer_max       | 51         | 2         | false    | WARL      | 0x0         | Programmable Value of Livelock detection based on Dealloc timer.                                                          |
| mshr_bypass_disable              | 53         | 1         | false    | WARL      | 0x0         | If 1, disables MSHR bypassing                                                                                             |
| sc_clock_halt_disable            | 54         | 1         | false    | WARL      | 0x0         | If 1, disables any clock halt to SC                                                                                       |
| ras_victim_chi_err_uec           | 55         | 1         | false    | WARL      | 0x1         | If 1, SC will log a RAS_CAT_UEC error if an error is seen for the response to a victim write. If 0, will log RAS_CAT_NONE |
| rsvd                             | 56         | 5         | false    | WARL      | 0x0         | Reserved                                                                                                                  |
| unused                           | 61         | 3         | false    | Read-Only | 0x0         | Reserved                                                                                                                  |

## sc\_cc\_capabilities

**Address:** 0x421A00C0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000001800

**Description:** None

| Field Name              | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                         |
|-------------------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------------------------------|
| ver                     | 0          | 8         | false    | Read-Only | 0x0         | Version of the specification implemented by the capacity controller                                                 |
| nblks                   | 8          | 16        | true     | Read-Only | 0x18        | Total number of allocatable capacity blocks in the capacity controller                                              |
| frcid                   | 24         | 1         | false    | Read-Only | 0x0         | If 1, the controller supports an operation to flush and deallocate the capacity blocks occupied by an RCID          |
| cunits                  | 25         | 1         | false    | Read-Only | 0x0         | If 1, the controller supports a limit on the capacity units that can be occupied by an RCID                         |
| rpfx                    | 26         | 1         | false    | Read-Only | 0x0         | If 1, the controller uses RCID along with P least significant bits of the MCID to compute an effective MCID         |
| p                       | 27         | 4         | false    | Read-Only | 0x0         | If RPFx is 1, the controller uses RCID along with P least significant bits of the MCID to compute an effective MCID |
| wpri_reserve<br>d_63_31 | 31         | 33        | false    | Read-Only | 0x0         | Reserved for future RISCv CBQRI standard use                                                                        |

### sc\_cc\_mon\_ctl

**Address:** 0x421A00C8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Used to control monitoring of capacity usage by a MCID. When the controller does not support monitoring, this register is read only zero

| Field Name              | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                                              |
|-------------------------|------------|-----------|----------|-----------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| op                      | 0          | 5         | false    | Read-Only | 0x0         | Used to instruct the controller to perform an operation. Op codes defined in CBQRI spec 3.2                                                                                                                                              |
| at                      | 5          | 3         | false    | Read-Only | 0x0         | Used to program the access type to count, and its validity is indicated by the ATV field                                                                                                                                                 |
| mcid                    | 8          | 12        | false    | Read-Only | 0x0         | Monitoring counter ID                                                                                                                                                                                                                    |
| evt_id                  | 20         | 8         | false    | Read-Only | 0x0         | Used to program the identifier of the event to count in the monitoring counter selected by MCID                                                                                                                                          |
| atv                     | 28         | 1         | false    | Read-Only | 0x0         | If 0, the counter counts requests with all access types, and the AT is ignored                                                                                                                                                           |
| wpri_reserve<br>d_31_29 | 29         | 3         | false    | Read-Only | 0x0         | Reserved for future RISCv CBQRI standard use                                                                                                                                                                                             |
| status                  | 32         | 7         | false    | Read-Only | 0x0         | Status of the requested operation. Status codes defined in CBQRI spec 3.2                                                                                                                                                                |
| busy                    | 39         | 1         | false    | Read-Only | 0x0         | A write to the cc_mon_ctl sets the read-only BUSY bit to 1 indicating the controller is performing the requested operation. When the BUSY bit reads 0, the operation is complete, and the read-only STATUS field provides a status value |
| wpri_reserve<br>d_63_40 | 40         | 24        | false    | Read-Only | 0x0         | Status of the requested operation                                                                                                                                                                                                        |

## sc\_cc\_mon\_ctr\_val

**Address:** 0x421A00D0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Holds a snapshot of the counter selected by the READ\_COUNTER operation. When controller does not support usage monitoring, this register always reads 0

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                  |
|------------|------------|-----------|----------|-----------|-------------|------------------------------|
| ctr        | 0          | 63        | false    | Read-Only | 0x0         | Counter value                |
| inv        | 63         | 1         | false    | Read-Only | 0x0         | If 0, counter value is valid |

## sc\_cc\_alloc\_ctl

**Address:** 0x421A00D8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Used to configure allocation of capacity to an RCID per access-type

| Field Name              | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                                                |
|-------------------------|------------|-----------|----------|-----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| op                      | 0          | 5         | false    | WARL      | 0x0         | Used to instruct the controller to perform an operation. Op codes defined in CBQRI spec 3.4                                                                                                                                                |
| at                      | 5          | 3         | false    | WARL      | 0x0         | Used to program the access type to allocate. If the controller does not support capacity allocation per access-type, this field is read-only zero                                                                                          |
| rcid                    | 8          | 12        | false    | WARL      | 0x0         | Resource control ID                                                                                                                                                                                                                        |
| wpri_reserve<br>d_31_20 | 20         | 12        | false    | Read-Only | 0x0         | Reserved for future RISCv CBQRI standard use                                                                                                                                                                                               |
| status                  | 32         | 7         | true     | Read-Only | 0x0         | Status of the requested operation. Status codes defined in CBQRI spec 3.4                                                                                                                                                                  |
| busy                    | 39         | 1         | true     | Read-Only | 0x0         | A write to the cc_alloc_ctl sets the read-only BUSY bit to 1 indicating the controller is performing the requested operation. When the BUSY bit reads 0, the operation is complete, and the read-only STATUS field provides a status value |
| wpri_reserve<br>d_63_40 | 40         | 24        | false    | Read-Only | 0x0         | Status of the requested operation                                                                                                                                                                                                          |

## sc\_cc\_block\_mask

**Address:** 0x421A00E0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** This register has NBLCKs bits each corresponding to one allocatable capacity block in the controller

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|              |    |    |       |           |     |                       |
|--------------|----|----|-------|-----------|-----|-----------------------|
| way_bit_mask | 0  | 32 | true  | WARL      | 0x0 | Way allocation bitmap |
| unused       | 32 | 32 | false | Read-Only | 0x0 |                       |

### sc\_cc\_cunits

**Address:** 0x421A00E8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** If the controller supports configuring limits on capacity units that may be occupied in the allocated capacity blocks then this register sets an upper limit on the number of capacity units that can be occupied by an RCID in the capacity blocks allocated for an AT. A value of zero specified in the cc\_cunits register indicates that no limit is configured.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| unused     | 0          | 64        | false    | WARL   | 0x0         |             |

### sc\_pmu\_select\_0

**Address:** 0x421A0140

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Select PMU event for counter 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| select     | 0          | 5         | false    | WARL   | 0x0         |             |
| rsvd645    | 5          | 59        | -        | WPRI   | -           | Reserved    |

### sc\_pmu\_select\_1

**Address:** 0x421A0148

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Select PMU event for counter 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| select     | 0          | 5         | false    | WARL   | 0x0         |             |
| rsvd645    | 5          | 59        | -        | WPRI   | -           | Reserved    |

### sc\_pmu\_select\_2

**Address:** 0x421A0150

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Select PMU event for counter 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| select     | 0          | 5         | false    | WARL   | 0x0         |             |
| rsvd645    | 5          | 59        | -        | WPRI   | -           | Reserved    |

### sc\_pmu\_select\_3

**Address:** 0x421A0158

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Select PMU event for counter 3

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| select     | 0          | 5         | false    | WARL   | 0x0         |             |
| rsvd645    | 5          | 59        | -        | WPRI   | -           | Reserved    |

### sc\_pmu\_select\_4

**Address:** 0x421A0160

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Select PMU event for counter 4

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| select     | 0          | 5         | false    | WARL   | 0x0         |             |
| rsvd645    | 5          | 59        | -        | WPRI   | -           | Reserved    |

### sc\_pmu\_select\_5

**Address:** 0x421A0168

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Select PMU event for counter 5

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| select     | 0          | 5         | false    | WARL   | 0x0         |             |
| rsvd645    | 5          | 59        | -        | WPRI   | -           | Reserved    |

### sc\_pmu\_select\_6

**Address:** 0x421A0170

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Select PMU event for counter 6

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|         |   |    |       |      |     |          |
|---------|---|----|-------|------|-----|----------|
| select  | 0 | 5  | false | WARL | 0x0 |          |
| rsvd645 | 5 | 59 | -     | WPRI | -   | Reserved |

### sc\_pmu\_select\_7

**Address:** 0x421A0178

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Select PMU event for counter 7

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| select     | 0          | 5         | false    | WARL   | 0x0         |             |
| rsvd645    | 5          | 59        | -        | WPRI   | -           | Reserved    |

### sc\_pmu\_counter\_0

**Address:** 0x421A01C0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** None

| Field Name  | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description    |
|-------------|------------|-----------|----------|-----------|-------------|----------------|
| counter_val | 0          | 64        | true     | Read-Only | 0x0         | Perf counter 0 |

### sc\_pmu\_counter\_1

**Address:** 0x421A01C8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** None

| Field Name  | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description    |
|-------------|------------|-----------|----------|-----------|-------------|----------------|
| counter_val | 0          | 64        | true     | Read-Only | 0x0         | Perf counter 1 |

### sc\_pmu\_counter\_2

**Address:** 0x421A01D0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** None

| Field Name  | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description    |
|-------------|------------|-----------|----------|-----------|-------------|----------------|
| counter_val | 0          | 64        | true     | Read-Only | 0x0         | Perf counter 2 |

### sc\_pmu\_counter\_3

**Address:** 0x421A01D8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** None

| Field Name  | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description    |
|-------------|------------|-----------|----------|-----------|-------------|----------------|
| counter_val | 0          | 64        | true     | Read-Only | 0x0         | Perf counter 3 |

### sc\_pmu\_counter\_4

**Address:** 0x421A01E0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** None

| Field Name  | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description    |
|-------------|------------|-----------|----------|-----------|-------------|----------------|
| counter_val | 0          | 64        | true     | Read-Only | 0x0         | Perf counter 4 |

### sc\_pmu\_counter\_5

**Address:** 0x421A01E8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** None

| Field Name  | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description    |
|-------------|------------|-----------|----------|-----------|-------------|----------------|
| counter_val | 0          | 64        | true     | Read-Only | 0x0         | Perf counter 5 |

### sc\_pmu\_counter\_6

**Address:** 0x421A01F0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** None

| Field Name  | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description    |
|-------------|------------|-----------|----------|-----------|-------------|----------------|
| counter_val | 0          | 64        | true     | Read-Only | 0x0         | Perf counter 6 |

### sc\_pmu\_counter\_7

**Address:** 0x421A01F8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** None

| Field Name  | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description    |
|-------------|------------|-----------|----------|-----------|-------------|----------------|
| counter_val | 0          | 64        | true     | Read-Only | 0x0         | Perf counter 7 |

### sc\_dbg\_cla\_counter0\_cfg

**Address:** 0x421A0280

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configure CLA counter0. One of the 4 counters used for counting cycles after a event match, and trigger a action on match.

| Field Name      | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                                                                            |
|-----------------|------------|-----------|----------|--------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| counter         | 0          | 16        | true     | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| target          | 16         | 16        | false    | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| reset_on_target | 32         | 1         | false    | WARL   | 0x0         | When Counter = target, reset to 0. With this bit set, counter will provide a periodic tick w/ frequency of (target +1) if the action is set to Auto Incr. The periodic tick will continue till Stop Auto Incr or Clear Counter action. |
| upper_counter   | 33         | 15        | true     | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| upper_target    | 48         | 15        | false    | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| rsvd            | 63         | 1         | false    | WARL   | 0x0         |                                                                                                                                                                                                                                        |

### sc\_dbg\_cla\_counter1\_cfg

**Address:** 0x421A0288

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configure CLA counter1. One of the 4 counters used for counting cycles after a event match, and trigger a action on match.

| Field Name      | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                                                                            |
|-----------------|------------|-----------|----------|--------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| counter         | 0          | 16        | true     | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| target          | 16         | 16        | false    | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| reset_on_target | 32         | 1         | false    | WARL   | 0x0         | When Counter = target, reset to 0. With this bit set, counter will provide a periodic tick w/ frequency of (target +1) if the action is set to Auto Incr. The periodic tick will continue till Stop Auto Incr or Clear Counter action. |
| upper_counter   | 33         | 15        | true     | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| upper_target    | 48         | 15        | false    | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| rsvd            | 63         | 1         | false    | WARL   | 0x0         |                                                                                                                                                                                                                                        |

### sc\_dbg\_cla\_counter2\_cfg

**Address:** 0x421A0290

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configure CLA counter2. One of the 4 counters used for counting cycles after a event match, and trigger a action on match.

| Field Name      | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                                                                            |
|-----------------|------------|-----------|----------|--------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| counter         | 0          | 16        | true     | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| target          | 16         | 16        | false    | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| reset_on_target | 32         | 1         | false    | WARL   | 0x0         | When Counter = target, reset to 0. With this bit set, counter will provide a periodic tick w/ frequency of (target +1) if the action is set to Auto Incr. The periodic tick will continue till Stop Auto Incr or Clear Counter action. |
| upper_counter   | 33         | 15        | true     | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| upper_target    | 48         | 15        | false    | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| rsvd            | 63         | 1         | false    | WARL   | 0x0         |                                                                                                                                                                                                                                        |

### sc\_dbg\_cla\_counter3\_cfg

**Address:** 0x421A0298

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configure CLA counter3. One of the 4 counters used for counting cycles after a event match, and trigger a action on match.

| Field Name      | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                                                                            |
|-----------------|------------|-----------|----------|--------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| counter         | 0          | 16        | true     | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| target          | 16         | 16        | false    | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| reset_on_target | 32         | 1         | false    | WARL   | 0x0         | When Counter = target, reset to 0. With this bit set, counter will provide a periodic tick w/ frequency of (target +1) if the action is set to Auto Incr. The periodic tick will continue till Stop Auto Incr or Clear Counter action. |
| upper_counter   | 33         | 15        | true     | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| upper_target    | 48         | 15        | false    | WARL   | 0x0         |                                                                                                                                                                                                                                        |
| rsvd            | 63         | 1         | false    | WARL   | 0x0         |                                                                                                                                                                                                                                        |

### sc\_dbg\_node0\_eap0

**Address:** 0x421A02A0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name             | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                   |
|------------------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node              | 0          | 2         | false    | WARL   | 0x0         | Select Destination Node                                                                                                                                       |
| action0                | 2          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action1                | 8          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| logical_op             | 14         | 2         | false    | WARL   | 0x0         | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0            | 16         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| event_type1            | 22         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| custom_action_0        | 28         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_1        | 32         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_0_enable | 36         | 1         | false    | WARL   | 0x0         | Enable custom_action_0                                                                                                                                        |
| custom_action_1_enable | 37         | 1         | false    | WARL   | 0x0         | Enable custom_action_1                                                                                                                                        |
| event_type2            | 38         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| udf                    | 44         | 8         | false    | WARL   | 0x0         | User Defined Function with event type 0, 1, and 2 as arguments                                                                                                |
| action2                | 52         | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action3                | 58         | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |

### sc\_dbg\_node0\_eap1

**Address:** 0x421A02A8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name      | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                   |
|-----------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node       | 0          | 2         | false    | WARL   | 0x0         | Select Destination Node                                                                                                                                       |
| action0         | 2          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action1         | 8          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| logical_op      | 14         | 2         | false    | WARL   | 0x0         | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0     | 16         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| event_type1     | 22         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| custom_action_0 | 28         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_1 | 32         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |

|                       |    |   |       |      |     |                                                                |
|-----------------------|----|---|-------|------|-----|----------------------------------------------------------------|
| custom_action0_enable | 36 | 1 | false | WARL | 0x0 | Enable custom_action_0                                         |
| custom_action1_enable | 37 | 1 | false | WARL | 0x0 | Enable custom_action_1                                         |
| event_type2           | 38 | 6 | false | WARL | 0x0 | Select a trigger event                                         |
| udf                   | 44 | 8 | false | WARL | 0x0 | User Defined Function with event type 0, 1, and 2 as arguments |
| action2               | 52 | 6 | false | WARL | 0x0 | Select an Action                                               |
| action3               | 58 | 6 | false | WARL | 0x0 | Select an Action                                               |

### sc\_dbg\_node0\_eap2

**Address:** 0x421A02B0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name            | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                   |
|-----------------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node             | 0          | 2         | false    | WARL   | 0x0         | Select Destination Node                                                                                                                                       |
| action0               | 2          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action1               | 8          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| logical_op            | 14         | 2         | false    | WARL   | 0x0         | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0           | 16         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| event_type1           | 22         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| custom_action0        | 28         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action1        | 32         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action0_enable | 36         | 1         | false    | WARL   | 0x0         | Enable custom_action_0                                                                                                                                        |
| custom_action1_enable | 37         | 1         | false    | WARL   | 0x0         | Enable custom_action_1                                                                                                                                        |
| event_type2           | 38         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| udf                   | 44         | 8         | false    | WARL   | 0x0         | User Defined Function with event type 0, 1, and 2 as arguments                                                                                                |
| action2               | 52         | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action3               | 58         | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |

### sc\_dbg\_node0\_eap3

**Address:** 0x421A02B8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name             | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                   |
|------------------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node              | 0          | 2         | false    | WARL   | 0x0         | Select Destination Node                                                                                                                                       |
| action0                | 2          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action1                | 8          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| logical_op             | 14         | 2         | false    | WARL   | 0x0         | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0            | 16         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| event_type1            | 22         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| custom_action_0        | 28         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_1        | 32         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_0_enable | 36         | 1         | false    | WARL   | 0x0         | Enable custom_action_0                                                                                                                                        |
| custom_action_1_enable | 37         | 1         | false    | WARL   | 0x0         | Enable custom_action_1                                                                                                                                        |
| event_type2            | 38         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| udf                    | 44         | 8         | false    | WARL   | 0x0         | User Defined Function with event type 0, 1, and 2 as arguments                                                                                                |
| action2                | 52         | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action3                | 58         | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |

### sc\_dbg\_node1\_eap0

**Address:** 0x421A02C0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name             | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                   |
|------------------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node              | 0          | 2         | false    | WARL   | 0x0         | Select Destination Node                                                                                                                                       |
| action0                | 2          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action1                | 8          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| logical_op             | 14         | 2         | false    | WARL   | 0x0         | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0            | 16         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| event_type1            | 22         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| custom_action_0        | 28         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_1        | 32         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_0_enable | 36         | 1         | false    | WARL   | 0x0         | Enable custom_action_0                                                                                                                                        |

|                       |    |   |       |      |     |                                                                |
|-----------------------|----|---|-------|------|-----|----------------------------------------------------------------|
| custom_action1_enable | 37 | 1 | false | WARL | 0x0 | Enable custom_action_1                                         |
| event_type2           | 38 | 6 | false | WARL | 0x0 | Select a trigger event                                         |
| udf                   | 44 | 8 | false | WARL | 0x0 | User Defined Function with event type 0, 1, and 2 as arguments |
| action2               | 52 | 6 | false | WARL | 0x0 | Select an Action                                               |
| action3               | 58 | 6 | false | WARL | 0x0 | Select an Action                                               |

### sc\_dbg\_node1\_eap1

**Address:** 0x421A02C8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name            | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                   |
|-----------------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node             | 0          | 2         | false    | WARL   | 0x0         | Select Destination Node                                                                                                                                       |
| action0               | 2          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action1               | 8          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| logical_op            | 14         | 2         | false    | WARL   | 0x0         | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0           | 16         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| event_type1           | 22         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| custom_action0        | 28         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action1        | 32         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action0_enable | 36         | 1         | false    | WARL   | 0x0         | Enable custom_action_0                                                                                                                                        |
| custom_action1_enable | 37         | 1         | false    | WARL   | 0x0         | Enable custom_action_1                                                                                                                                        |
| event_type2           | 38         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| udf                   | 44         | 8         | false    | WARL   | 0x0         | User Defined Function with event type 0, 1, and 2 as arguments                                                                                                |
| action2               | 52         | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action3               | 58         | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |

### sc\_dbg\_node1\_eap2

**Address:** 0x421A02D0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|                        |    |   |       |      |     |                                                                                                                                                               |
|------------------------|----|---|-------|------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node              | 0  | 2 | false | WARL | 0x0 | Select Destination Node                                                                                                                                       |
| action0                | 2  | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |
| action1                | 8  | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |
| logical_op             | 14 | 2 | false | WARL | 0x0 | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0            | 16 | 6 | false | WARL | 0x0 | Select a trigger event                                                                                                                                        |
| event_type1            | 22 | 6 | false | WARL | 0x0 | Select a trigger event                                                                                                                                        |
| custom_action_0        | 28 | 4 | false | WARL | 0x0 | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_1        | 32 | 4 | false | WARL | 0x0 | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_0_enable | 36 | 1 | false | WARL | 0x0 | Enable custom_action_0                                                                                                                                        |
| custom_action_1_enable | 37 | 1 | false | WARL | 0x0 | Enable custom_action_1                                                                                                                                        |
| event_type2            | 38 | 6 | false | WARL | 0x0 | Select a trigger event                                                                                                                                        |
| udf                    | 44 | 8 | false | WARL | 0x0 | User Defined Function with event type 0, 1, and 2 as arguments                                                                                                |
| action2                | 52 | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |
| action3                | 58 | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |

### sc\_dbg\_node1\_eap3

**Address:** 0x421A02D8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name             | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                   |
|------------------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node              | 0          | 2         | false    | WARL   | 0x0         | Select Destination Node                                                                                                                                       |
| action0                | 2          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action1                | 8          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| logical_op             | 14         | 2         | false    | WARL   | 0x0         | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0            | 16         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| event_type1            | 22         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| custom_action_0        | 28         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_1        | 32         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_0_enable | 36         | 1         | false    | WARL   | 0x0         | Enable custom_action_0                                                                                                                                        |

|                       |    |   |       |      |     |                                                                |
|-----------------------|----|---|-------|------|-----|----------------------------------------------------------------|
| custom_action1_enable | 37 | 1 | false | WARL | 0x0 | Enable custom_action_1                                         |
| event_type2           | 38 | 6 | false | WARL | 0x0 | Select a trigger event                                         |
| udf                   | 44 | 8 | false | WARL | 0x0 | User Defined Function with event type 0, 1, and 2 as arguments |
| action2               | 52 | 6 | false | WARL | 0x0 | Select an Action                                               |
| action3               | 58 | 6 | false | WARL | 0x0 | Select an Action                                               |

### sc\_dbg\_node2\_eap0

**Address:** 0x421A02E0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name              | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                   |
|-------------------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node               | 0          | 2         | false    | WARL   | 0x0         | Select Destination Node                                                                                                                                       |
| action0                 | 2          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action1                 | 8          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| logical_op              | 14         | 2         | false    | WARL   | 0x0         | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0             | 16         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| event_type1             | 22         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| custom_action_n_0       | 28         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_n_1       | 32         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_n0_enable | 36         | 1         | false    | WARL   | 0x0         | Enable custom_action_0                                                                                                                                        |
| custom_action_n1_enable | 37         | 1         | false    | WARL   | 0x0         | Enable custom_action_1                                                                                                                                        |
| event_type2             | 38         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| udf                     | 44         | 8         | false    | WARL   | 0x0         | User Defined Function with event type 0, 1, and 2 as arguments                                                                                                |
| action2                 | 52         | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action3                 | 58         | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |

### sc\_dbg\_node2\_eap1

**Address:** 0x421A02E8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|                        |    |   |       |      |     |                                                                                                                                                               |
|------------------------|----|---|-------|------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node              | 0  | 2 | false | WARL | 0x0 | Select Destination Node                                                                                                                                       |
| action0                | 2  | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |
| action1                | 8  | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |
| logical_op             | 14 | 2 | false | WARL | 0x0 | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0            | 16 | 6 | false | WARL | 0x0 | Select a trigger event                                                                                                                                        |
| event_type1            | 22 | 6 | false | WARL | 0x0 | Select a trigger event                                                                                                                                        |
| custom_action_0        | 28 | 4 | false | WARL | 0x0 | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_1        | 32 | 4 | false | WARL | 0x0 | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_0_enable | 36 | 1 | false | WARL | 0x0 | Enable custom_action_0                                                                                                                                        |
| custom_action_1_enable | 37 | 1 | false | WARL | 0x0 | Enable custom_action_1                                                                                                                                        |
| event_type2            | 38 | 6 | false | WARL | 0x0 | Select a trigger event                                                                                                                                        |
| udf                    | 44 | 8 | false | WARL | 0x0 | User Defined Function with event type 0, 1, and 2 as arguments                                                                                                |
| action2                | 52 | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |
| action3                | 58 | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |

## sc\_dbg\_node2\_eap2

**Address:** 0x421A02F0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name             | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                   |
|------------------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node              | 0          | 2         | false    | WARL   | 0x0         | Select Destination Node                                                                                                                                       |
| action0                | 2          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action1                | 8          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| logical_op             | 14         | 2         | false    | WARL   | 0x0         | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0            | 16         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| event_type1            | 22         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| custom_action_0        | 28         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_1        | 32         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_0_enable | 36         | 1         | false    | WARL   | 0x0         | Enable custom_action_0                                                                                                                                        |

|                       |    |   |       |      |     |                                                                |
|-----------------------|----|---|-------|------|-----|----------------------------------------------------------------|
| custom_action1_enable | 37 | 1 | false | WARL | 0x0 | Enable custom_action_1                                         |
| event_type2           | 38 | 6 | false | WARL | 0x0 | Select a trigger event                                         |
| udf                   | 44 | 8 | false | WARL | 0x0 | User Defined Function with event type 0, 1, and 2 as arguments |
| action2               | 52 | 6 | false | WARL | 0x0 | Select an Action                                               |
| action3               | 58 | 6 | false | WARL | 0x0 | Select an Action                                               |

### sc\_dbg\_node2\_eap3

**Address:** 0x421A02F8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name              | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                   |
|-------------------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node               | 0          | 2         | false    | WARL   | 0x0         | Select Destination Node                                                                                                                                       |
| action0                 | 2          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action1                 | 8          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| logical_op              | 14         | 2         | false    | WARL   | 0x0         | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0             | 16         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| event_type1             | 22         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| custom_action_n_0       | 28         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_n_1       | 32         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_n0_enable | 36         | 1         | false    | WARL   | 0x0         | Enable custom_action_0                                                                                                                                        |
| custom_action_n1_enable | 37         | 1         | false    | WARL   | 0x0         | Enable custom_action_1                                                                                                                                        |
| event_type2             | 38         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| udf                     | 44         | 8         | false    | WARL   | 0x0         | User Defined Function with event type 0, 1, and 2 as arguments                                                                                                |
| action2                 | 52         | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action3                 | 58         | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |

### sc\_dbg\_node3\_eap0

**Address:** 0x421A0300

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|                        |    |   |       |      |     |                                                                                                                                                               |
|------------------------|----|---|-------|------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node              | 0  | 2 | false | WARL | 0x0 | Select Destination Node                                                                                                                                       |
| action0                | 2  | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |
| action1                | 8  | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |
| logical_op             | 14 | 2 | false | WARL | 0x0 | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0            | 16 | 6 | false | WARL | 0x0 | Select a trigger event                                                                                                                                        |
| event_type1            | 22 | 6 | false | WARL | 0x0 | Select a trigger event                                                                                                                                        |
| custom_action_0        | 28 | 4 | false | WARL | 0x0 | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_1        | 32 | 4 | false | WARL | 0x0 | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_0_enable | 36 | 1 | false | WARL | 0x0 | Enable custom_action_0                                                                                                                                        |
| custom_action_1_enable | 37 | 1 | false | WARL | 0x0 | Enable custom_action_1                                                                                                                                        |
| event_type2            | 38 | 6 | false | WARL | 0x0 | Select a trigger event                                                                                                                                        |
| udf                    | 44 | 8 | false | WARL | 0x0 | User Defined Function with event type 0, 1, and 2 as arguments                                                                                                |
| action2                | 52 | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |
| action3                | 58 | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |

### sc\_dbg\_node3\_eap1

**Address:** 0x421A0308

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name             | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                   |
|------------------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node              | 0          | 2         | false    | WARL   | 0x0         | Select Destination Node                                                                                                                                       |
| action0                | 2          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action1                | 8          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| logical_op             | 14         | 2         | false    | WARL   | 0x0         | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0            | 16         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| event_type1            | 22         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| custom_action_0        | 28         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_1        | 32         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_0_enable | 36         | 1         | false    | WARL   | 0x0         | Enable custom_action_0                                                                                                                                        |

|                       |    |   |       |      |     |                                                                |
|-----------------------|----|---|-------|------|-----|----------------------------------------------------------------|
| custom_action1_enable | 37 | 1 | false | WARL | 0x0 | Enable custom_action_1                                         |
| event_type2           | 38 | 6 | false | WARL | 0x0 | Select a trigger event                                         |
| udf                   | 44 | 8 | false | WARL | 0x0 | User Defined Function with event type 0, 1, and 2 as arguments |
| action2               | 52 | 6 | false | WARL | 0x0 | Select an Action                                               |
| action3               | 58 | 6 | false | WARL | 0x0 | Select an Action                                               |

### sc\_dbg\_node3\_eap2

**Address:** 0x421A0310

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name              | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                   |
|-------------------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node               | 0          | 2         | false    | WARL   | 0x0         | Select Destination Node                                                                                                                                       |
| action0                 | 2          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action1                 | 8          | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| logical_op              | 14         | 2         | false    | WARL   | 0x0         | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0             | 16         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| event_type1             | 22         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| custom_action_n_0       | 28         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_n_1       | 32         | 4         | false    | WARL   | 0x0         | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_n0_enable | 36         | 1         | false    | WARL   | 0x0         | Enable custom_action_0                                                                                                                                        |
| custom_action_n1_enable | 37         | 1         | false    | WARL   | 0x0         | Enable custom_action_1                                                                                                                                        |
| event_type2             | 38         | 6         | false    | WARL   | 0x0         | Select a trigger event                                                                                                                                        |
| udf                     | 44         | 8         | false    | WARL   | 0x0         | User Defined Function with event type 0, 1, and 2 as arguments                                                                                                |
| action2                 | 52         | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |
| action3                 | 58         | 6         | false    | WARL   | 0x0         | Select an Action                                                                                                                                              |

### sc\_dbg\_node3\_eap3

**Address:** 0x421A0318

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** CLA Action(s) are tied to events(s) using event-action pairing registers. One of the 8 EAPs (Two EAPs per Node).

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|                        |    |   |       |      |     |                                                                                                                                                               |
|------------------------|----|---|-------|------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dest_node              | 0  | 2 | false | WARL | 0x0 | Select Destination Node                                                                                                                                       |
| action0                | 2  | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |
| action1                | 8  | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |
| logical_op             | 14 | 2 | false | WARL | 0x0 | Relation to be satisfied among events to activate the actions                                                                                                 |
| event_type0            | 16 | 6 | false | WARL | 0x0 | Select a trigger event                                                                                                                                        |
| event_type1            | 22 | 6 | false | WARL | 0x0 | Select a trigger event                                                                                                                                        |
| custom_action_0        | 28 | 4 | false | WARL | 0x0 | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_1        | 32 | 4 | false | WARL | 0x0 | Select the bit position of custom action bus to be set when EAP trigger is met. External blocks can define and implement the specific action for a given bit. |
| custom_action_0_enable | 36 | 1 | false | WARL | 0x0 | Enable custom_action_0                                                                                                                                        |
| custom_action_1_enable | 37 | 1 | false | WARL | 0x0 | Enable custom_action_1                                                                                                                                        |
| event_type2            | 38 | 6 | false | WARL | 0x0 | Select a trigger event                                                                                                                                        |
| udf                    | 44 | 8 | false | WARL | 0x0 | User Defined Function with event type 0, 1, and 2 as arguments                                                                                                |
| action2                | 52 | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |
| action3                | 58 | 6 | false | WARL | 0x0 | Select an Action                                                                                                                                              |

### sc\_dbg\_signal\_mask0

**Address:** 0x421A0320

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Used to define debug bus match event. Match event (0x2) is triggered when debug\_bus & mask = match.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                  |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------|
| value      | 0          | 64        | false    | WARL   | 0x0         | Mask to be applied to debug bus before match |

### sc\_dbg\_signal\_match0

**Address:** 0x421A0328

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Used to define debug bus match event. Match event (0x2) is triggered when debug\_bus & mask = match.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                     |
|------------|------------|-----------|----------|--------|-------------|---------------------------------|
| value      | 0          | 64        | false    | WARL   | 0x0         | Value used for debug bus match. |

### sc\_dbg\_signal\_mask1

**Address:** 0x421A0330

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Used to define debug bus match event. Match event (0x4) is triggered when debug\_bus & mask = match.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                  |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------|
| value      | 0          | 64        | false    | WARL   | 0x0         | Mask to be applied to debug bus before match |

### sc\_dbg\_signal\_match1

**Address:** 0x421A0338

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Used to define debug bus match event. Match event (0x4) is triggered when debug\_bus & mask = match.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                     |
|------------|------------|-----------|----------|--------|-------------|---------------------------------|
| value      | 0          | 64        | false    | WARL   | 0x0         | Value used for debug bus match. |

### sc\_dbg\_signal\_edge\_detect\_cfg

**Address:** 0x421A0340

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Register to configure debug bus for edge triggers.

| Field Name       | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                           |
|------------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------|
| signal0_select   | 0          | 6         | false    | WARL   | 0x0         | Define which signal to select for edge detection.                                     |
| pos_edge_signal0 | 6          | 1         | false    | WARL   | 0x0         | Defines which edge to detect for signal from signal0_select. 1: Pos edge, 0: Neg Edge |
| signal1_select   | 7          | 6         | false    | WARL   | 0x0         | Define which signal to select for edge detection.                                     |
| pos_edge_signal1 | 13         | 1         | false    | WARL   | 0x0         | Defines which edge to detect for signal from signal1_select. 1: Pos edge, 0: Neg Edge |
| rsvd6414         | 14         | 50        | -        | WPRI   | -           | Reserved                                                                              |

### sc\_dbg\_eap\_status

**Address:** 0x421A0348

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Register to indicate if EAP Pair was activated. Use the corresponding w2c register to reset the status. There are 2 bits

for each EAP. The bit corresponds to action-0 and action-1 of the EAP. The register is used by SW to know if an action was taken (ex: NMI ISR needs to know which of the EAP triggered the NMI)

| Field Name     | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|----------------|------------|-----------|----------|--------|-------------|-------------|
| node0_eap0     | 0          | 1         | true     | WARL   | 0x0         |             |
| node0_eap1     | 1          | 1         | true     | WARL   | 0x0         |             |
| node0_eap2     | 2          | 1         | true     | WARL   | 0x0         |             |
| node0_eap3     | 3          | 1         | true     | WARL   | 0x0         |             |
| node1_eap0     | 4          | 1         | true     | WARL   | 0x0         |             |
| node1_eap1     | 5          | 1         | true     | WARL   | 0x0         |             |
| node1_eap2     | 6          | 1         | true     | WARL   | 0x0         |             |
| node1_eap3     | 7          | 1         | true     | WARL   | 0x0         |             |
| node2_eap0     | 8          | 1         | true     | WARL   | 0x0         |             |
| node2_eap1     | 9          | 1         | true     | WARL   | 0x0         |             |
| node2_eap2     | 10         | 1         | true     | WARL   | 0x0         |             |
| node2_eap3     | 11         | 1         | true     | WARL   | 0x0         |             |
| node3_eap0     | 12         | 1         | true     | WARL   | 0x0         |             |
| node3_eap1     | 13         | 1         | true     | WARL   | 0x0         |             |
| node3_eap2     | 14         | 1         | true     | WARL   | 0x0         |             |
| node3_eap3     | 15         | 1         | true     | WARL   | 0x0         |             |
| rsvd_31_16     | 16         | 16        | true     | WARL   | 0x0         |             |
| node0_eap0_w2c | 32         | 1         | true     | WARL   | 0x0         |             |
| node0_eap1_w2c | 33         | 1         | true     | WARL   | 0x0         |             |
| node0_eap2_w2c | 34         | 1         | true     | WARL   | 0x0         |             |
| node0_eap3_w2c | 35         | 1         | true     | WARL   | 0x0         |             |
| node1_eap0_w2c | 36         | 1         | true     | WARL   | 0x0         |             |
| node1_eap1_w2c | 37         | 1         | true     | WARL   | 0x0         |             |
| node1_eap2_w2c | 38         | 1         | true     | WARL   | 0x0         |             |
| node1_eap3_w2c | 39         | 1         | true     | WARL   | 0x0         |             |
| node2_eap0_w2c | 40         | 1         | true     | WARL   | 0x0         |             |
| node2_eap1_w2c | 41         | 1         | true     | WARL   | 0x0         |             |
| node2_eap2_w2c | 42         | 1         | true     | WARL   | 0x0         |             |
| node2_eap3_w2c | 43         | 1         | true     | WARL   | 0x0         |             |
| node3_eap0_w2c | 44         | 1         | true     | WARL   | 0x0         |             |
| node3_eap1_w2c | 45         | 1         | true     | WARL   | 0x0         |             |

|                |    |    |      |      |     |          |
|----------------|----|----|------|------|-----|----------|
| node3_eap2_w2c | 46 | 1  | true | WARL | 0x0 |          |
| node3_eap3_w2c | 47 | 1  | true | WARL | 0x0 |          |
| rsvd6448       | 48 | 16 | -    | WPRI | -   | Reserved |

### sc\_dbg\_cla\_ctrl\_status

**Address:** 0x421A0350

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000001B00

**Description:** Ctrl/Status register to Enable EAP after EAP programming is complete, and read the current node.

| Field Name                | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                            |
|---------------------------|------------|-----------|----------|-----------|-------------|----------------------------------------------------------------------------------------|
| current_node              | 0          | 2         | true     | WARL      | 0x0         | Read the current node.                                                                 |
| rsvd42                    | 2          | 3         | -        | WPRI      | -           | Reserved                                                                               |
| enable_eap                | 5          | 1         | false    | WARL      | 0x0         | Set Enable EAP to 1 after EAP programming is complete                                  |
| enable_cla                | 6          | 1         | false    | WARL      | 0x0         | Set Enable CLA                                                                         |
| cla_chain_loop_delay      | 7          | 7         | false    | WARL      | 0x36        | Creates a window in which we don't forward the incoming Xtrigger and Clock_Halt signal |
| disable_global_clock_halt | 14         | 1         | false    | WARL      | 0x0         | If this bit is set, clock halt action is not applied to global clocks                  |
| disable_local_clock_halt  | 15         | 1         | false    | WARL      | 0x0         | If this bit is set, clock halt action is not applied to local clocks                   |
| rsvd_62_16                | 16         | 47        | false    | Read-Only | 0x0         |                                                                                        |
| cla_lock                  | 63         | 1         | false    | WARL      | 0x0         | If this bit is set, then CLA is locked and the enable_cla is latched forever           |

### sc\_dbg\_transition\_mask

**Address:** 0x421A0358

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** 3 registers to configure "transition event". This register is to select the signals of interest using a mask.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| value      | 0          | 64        | false    | WARL   | 0x0         |             |

### sc\_dbg\_transition\_from\_value

**Address:** 0x421A0360

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** 3 registers to configure "transition event". The event is triggered on transition from Value A to Value B. This register specifies Value A.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| value      | 0          | 64        | false    | WARL   | 0x0         |             |

### sc\_dbg\_transition\_to\_value

**Address:** 0x421A0368

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** 3 registers to configure "transition event". The event is triggered on transition from Value A to Value B. This register specifies Value B.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| value      | 0          | 64        | false    | WARL   | 0x0         |             |

### sc\_dbg\_ones\_count\_mask

**Address:** 0x421A0370

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** 2 registers to configure "ones count" event. This register is to select the signals of interest using a mask.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| value      | 0          | 64        | false    | WARL   | 0x0         |             |

### sc\_dbg\_ones\_count\_value

**Address:** 0x421A0378

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** 2 registers to configure "ones count" event. Event triggered when the sum of the signals match the value specified in this register.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| value      | 0          | 64        | false    | WARL   | 0x0         |             |

### sc\_dbg\_any\_change

**Address:** 0x421A0380

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Event triggered when a subset of debug signals change. The mask is used to select the subset of the debug signals.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | false    | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node0\_eap0

**Address:** 0x421A0388

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node0 Eap0 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node0\_eap1

**Address:** 0x421A0390

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node0 Eap1 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node0\_eap2

**Address:** 0x421A0398

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node0 Eap2 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node0\_eap3

**Address:** 0x421A03A0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node0 Eap3 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node1\_eap0

**Address:** 0x421A03A8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node1 Eap0 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node1\_eap1

**Address:** 0x421A03B0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node1 Eap0 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node1\_eap2

**Address:** 0x421A03B8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node1 Eap2 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node1\_eap3

**Address:** 0x421A03C0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node1 Eap3 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node2\_eap0

**Address:** 0x421A03C8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node2 Eap0 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node2\_eap1

**Address:** 0x421A03D0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node2 Eap1 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node2\_eap2

**Address:** 0x421A03D8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node2 Eap2 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node2\_eap3

**Address:** 0x421A03E0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node2 Eap3 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node3\_eap0

**Address:** 0x421A03E8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node3 Eap0 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node3\_eap1

**Address:** 0x421A03F0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node3 Eap0 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node3\_eap2

**Address:** 0x421A03F8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node3 Eap2 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_signal\_snapshot\_node3\_eap3

**Address:** 0x421A0400

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** debug Bus Value when Node3 Eap3 Trigger is met.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | true     | WARL   | 0x0         |             |

### sc\_dbg\_mux\_control\_A

**Address:** 0x421A0408

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Debug mux select A (default)

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                     |
|------------|------------|-----------|----------|-----------|-------------|---------------------------------|
| dbm_mode   | 0          | 2         | false    | WARL      | 0x0         | Mode of the debug mux           |
| dbm_id     | 2          | 6         | false    | WARL      | 0x0         | Id for the targeted debug mux   |
| rsvd       | 8          | 8         | false    | Read-Only | 0x0         |                                 |
| muxselseg0 | 16         | 6         | false    | WARL      | 0x0         | Select Set 0 for Debug Mux X0Y0 |
| muxselseg1 | 22         | 6         | false    | WARL      | 0x0         | Select Set 1 for Debug Mux X0Y0 |
| muxselseg2 | 28         | 6         | false    | WARL      | 0x0         | Select Set 2 for Debug Mux X0Y0 |
| muxselseg3 | 34         | 6         | false    | WARL      | 0x0         | Select Set 3 for Debug Mux X0Y0 |
| unused     | 40         | 24        | false    | Read-Only | 0x0         |                                 |

### sc\_dbg\_mux\_control\_B

**Address:** 0x421A0410

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Debug mux select B

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                     |
|------------|------------|-----------|----------|-----------|-------------|---------------------------------|
| dbm_mode   | 0          | 2         | false    | WARL      | 0x0         | Mode of the debug mux           |
| dbm_id     | 2          | 6         | false    | WARL      | 0x0         | Id for the targeted debug mux   |
| rsvd       | 8          | 8         | false    | Read-Only | 0x0         |                                 |
| muxselseg0 | 16         | 6         | false    | WARL      | 0x0         | Select Set 0 for Debug Mux X0Y0 |
| muxselseg1 | 22         | 6         | false    | WARL      | 0x0         | Select Set 1 for Debug Mux X0Y0 |
| muxselseg2 | 28         | 6         | false    | WARL      | 0x0         | Select Set 2 for Debug Mux X0Y0 |
| muxselseg3 | 34         | 6         | false    | WARL      | 0x0         | Select Set 3 for Debug Mux X0Y0 |
| unused     | 40         | 24        | false    | Read-Only | 0x0         |                                 |

### sc\_dbg\_signal\_mask2

**Address:** 0x421A0418

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Used to define debug bus match event. Match event (0x1C) is triggered when debug\_bus & mask = match.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                  |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------|
| value      | 0          | 64        | false    | WARL   | 0x0         | Mask to be applied to debug bus before match |

### sc\_dbg\_signal\_match2

**Address:** 0x421A0420

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Used to define debug bus match event. Match event (0x1C) is triggered when debug\_bus & mask = match.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                     |
|------------|------------|-----------|----------|--------|-------------|---------------------------------|
| value      | 0          | 64        | false    | WARL   | 0x0         | Value used for debug bus match. |

### sc\_dbg\_signal\_mask3

**Address:** 0x421A0428

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Used to define debug bus match event. Match event (0x1D) is triggered when debug\_bus & mask = match.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                  |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------|
| value      | 0          | 64        | false    | WARL   | 0x0         | Mask to be applied to debug bus before match |

### sc\_dbg\_signal\_match3

**Address:** 0x421A0430

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Used to define debug bus match event. Match event (0x1D) is triggered when debug\_bus & mask = match.

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                     |
|------------|------------|-----------|----------|--------|-------------|---------------------------------|
| value      | 0          | 64        | false    | WARL   | 0x0         | Value used for debug bus match. |

### sc\_dbg\_signal\_delay\_mux\_sel

**Address:** 0x421A0438

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Debug signal shift mux select

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                             |
|------------|------------|-----------|----------|-----------|-------------|-----------------------------------------|
| muxselseg0 | 0          | 2         | false    | WARL      | 0x0         | Signal shift Select Set 0 for Debug Mux |
| muxselseg1 | 2          | 2         | false    | WARL      | 0x0         | Signal shift Select Set 1 for Debug Mux |
| muxselseg2 | 4          | 2         | false    | WARL      | 0x0         | Signal shift Select Set 2 for Debug Mux |
| muxselseg3 | 6          | 2         | false    | WARL      | 0x0         | Signal shift Select Set 3 for Debug Mux |
| muxselseg4 | 8          | 2         | false    | WARL      | 0x0         | Signal shift Select Set 4 for Debug Mux |
| muxselseg5 | 10         | 2         | false    | WARL      | 0x0         | Signal shift Select Set 5 for Debug Mux |
| muxselseg6 | 12         | 2         | false    | WARL      | 0x0         | Signal shift Select Set 6 for Debug Mux |
| muxselseg7 | 14         | 2         | false    | WARL      | 0x0         | Signal shift Select Set 7 for Debug Mux |
| rsvd       | 16         | 48        | false    | Read-Only | 0x0         |                                         |

### sc\_dbg\_cla\_timestampsync

**Address:** 0x421A0440

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** This value is loaded to the finegrain timestamp counter based on a cross trigger to sync the time value across the multiple cluster's CLA blocks

| Field Name         | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|--------------------|------------|-----------|----------|--------|-------------|-------------|
| timestamp          | 0          | 63        | false    | WARL   | 0x0         |             |
| timesyncmodeenable | 63         | 1         | false    | WARL   | 0x0         |             |

### sc\_error\_header\_vendor

**Address:** 0x421A0480

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000821

**Description:** Vendor and implementation ID

| Field Name          | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                 |
|---------------------|------------|-----------|----------|-----------|-------------|-------------------------------------------------------------------------------------------------------------|
| vendor_id           | 0          | 32        | false    | Read-Only | 0x821       | Provides the JEDEC manufacturer ID of the provider of the component hosting the error bank                  |
| imp_id              | 32         | 16        | false    | Read-Only | 0x0         | Provides a unique identity, defined by the vendor, to identify the component and revisions of the component |
| wpri_reserved_63_48 | 48         | 16        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                |

### sc\_error\_header\_bank\_info

**Address:** 0x421A0488

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0100000000430000

**Description:** Error bank information

| Field Name          | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                 |
|---------------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| inst_id             | 0          | 16        | false    | Read-Only | 0x0         | Identifies a unique instance of an error bank within a package or die, ideally in the whole system. Not being used, tied to 0                                                               |
| n_err_recs          | 16         | 6         | false    | Read-Only | 0x3         | Indicates the number of error records implemented by the error bank                                                                                                                         |
| layout              | 22         | 2         | false    | Read-Only | 0x1         | Indicates the layout of the registers in the error bank and the error records. The layout encoding 0 indicates the registers are arranged and have meaning as defined by the specification. |
| wpri_reserved_55_24 | 24         | 32        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                |
| version             | 56         | 8         | false    | Read-Only | 0x1         | Version of the arch register layout specification implemented by the error bank                                                                                                             |

### sc\_error\_header\_valid\_summary

**Address:** 0x421A0490

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000001

**Description:** Summary of valid error records

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|              |   |    |       |           |     |                                                                                                |
|--------------|---|----|-------|-----------|-----|------------------------------------------------------------------------------------------------|
| sv           | 0 | 1  | false | Read-Only | 0x1 | Indicates that the valid_bitmap provides a summary of the valid bits from the status registers |
| valid_bitmap | 1 | 63 | true  | Read-Only | 0x0 | Summary of the valid bits from status registers in the error bank                              |

### sc\_error\_header\_reserved\_0

**Address:** 0x421A0498

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Reserved for future RISC-V RERI standard use

| Field Name    | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                  |
|---------------|------------|-----------|----------|-----------|-------------|----------------------------------------------|
| reserved_63_0 | 0          | 64        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use |

### sc\_error\_header\_reserved\_1

**Address:** 0x421A04A0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Reserved for future RISC-V RERI standard use

| Field Name    | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                  |
|---------------|------------|-----------|----------|-----------|-------------|----------------------------------------------|
| reserved_63_0 | 0          | 64        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use |

### sc\_error\_header\_reserved\_2

**Address:** 0x421A04A8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Reserved for future RISC-V RERI standard use

| Field Name    | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                  |
|---------------|------------|-----------|----------|-----------|-------------|----------------------------------------------|
| reserved_63_0 | 0          | 64        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use |

### sc\_error\_header\_reserved\_3

**Address:** 0x421A04B0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Reserved for future RISC-V RERI standard use

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|               |   |    |       |           |     |                                              |
|---------------|---|----|-------|-----------|-----|----------------------------------------------|
| reserved_63_0 | 0 | 64 | false | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use |
|---------------|---|----|-------|-----------|-----|----------------------------------------------|

### sc\_error\_header\_custom\_0

**Address:** 0x421A04B8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Reserved for custom use

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description             |
|------------|------------|-----------|----------|-----------|-------------|-------------------------|
| unused     | 0          | 64        | false    | Read-Only | 0x0         | Reserved for custom use |

### sc\_error\_record\_0\_control

**Address:** 0x421A04C0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000001

**Description:** Control register of error record 0

| Field Name          | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                        |
|---------------------|------------|-----------|----------|-----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| else                | 0          | 1         | false    | WARL      | 0x1         | If 1, the hardware unit logs and signals errors in the error record. If 0, the hardware unit will not log RAS errors                                                                                               |
| cece                | 1          | 1         | false    | WARL      | 0x0         | If 1, hardware will count corrected errors in the cec field of the status register. If 0, cec does not count and retains its value.                                                                                |
| ces                 | 2          | 2         | false    | WARL      | 0x0         | Used to enable signaling of corrected errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal.            |
| ueds                | 4          | 2         | false    | WARL      | 0x0         | Used to enable signaling of uncorrected deferred errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| uecs                | 6          | 2         | false    | WARL      | 0x0         | Used to enable signaling of uncorrected critical errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| wpri_reserved_31_8  | 8          | 24        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| eid                 | 32         | 16        | true     | WARL      | 0x0         | Error injection delay                                                                                                                                                                                              |
| sinv                | 48         | 1         | true     | WARL      | 0x0         | Status register invalidate. When written to 1, causes the valid (v) field of the status register to be cleared.                                                                                                    |
| srdep               | 49         | 1         | true     | WARL      | 0x0         | Set read in progress. When 1, causes read in progress (rdip) to be set                                                                                                                                             |
| wpri_reserved_55_48 | 50         | 10        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |

|               |    |   |       |           |     |          |
|---------------|----|---|-------|-----------|-----|----------|
| custom_unused | 60 | 4 | false | Read-Only | 0x0 | Reserved |
|---------------|----|---|-------|-----------|-----|----------|

## sc\_error\_record\_0\_status

**Address:** 0x421A04C8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Status register of error record 0

| Field Name          | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                            |
|---------------------|------------|-----------|----------|-----------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v                   | 0          | 1         | true     | WARL      | 0x0         | If 1, error record holds a valid error                                                                                                                                                 |
| ce                  | 1          | 1         | true     | WARL      | 0x0         | Set to 1 if the detected error was corrected                                                                                                                                           |
| ued                 | 2          | 1         | true     | WARL      | 0x0         | Set to 1 if the error could not be corrected but was deferred                                                                                                                          |
| uec                 | 3          | 1         | true     | WARL      | 0x0         | Set to 1 if the error could not be corrected or deferred and needs critical handling by RAS handler                                                                                    |
| pri                 | 4          | 2         | true     | WARL      | 0x0         | Priority of the error. Indicates relative priority among errors of the same class.                                                                                                     |
| mo                  | 6          | 1         | true     | WARL      | 0x0         | Set to 1 if more errors of the same class and severity occur                                                                                                                           |
| c                   | 7          | 1         | true     | WARL      | 0x0         | Containable bit. If 1, indicates that the error has not propagated beyond the boundaries of the hardware unit                                                                          |
| tt                  | 8          | 3         | true     | WARL      | 0x0         | Transaction type. 0=Unspecified or NA, 1=UNSPECIFIED, 2-3=Reserved for future standard use, 4=Explicit read, 5=Explicit write, 6=Implicit read, 7=Implicit write                       |
| iv                  | 11         | 1         | true     | WARL      | 0x0         | Information valid, set to 1 if error reports additional information in the info register                                                                                               |
| at                  | 12         | 4         | true     | WARL      | 0x0         | Address type, indicates the type of address reported in the addr register. 0=None, 1=Supervisor Physical Address, 2=Guest Physical Address, 3=Virtual Address, 4-15=Component Specific |
| siv                 | 16         | 1         | true     | WARL      | 0x0         | Supplemental information valid, set to 1 if error reports supplemental information in the suppl_info register                                                                          |
| tsv                 | 17         | 1         | true     | WARL      | 0x0         | Timestamp valid, set to 1 if the error record holds a valid timestamp in the timestamp register                                                                                        |
| lv                  | 18         | 1         | true     | WARL      | 0x0         | Locator valid, set to 1 if the error record holds a valid timestamp in the locator register                                                                                            |
| wpri_reserved_19_19 | 19         | 1         | true     | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                           |
| scrub               | 20         | 1         | true     | WARL      | 0x0         | Set to 1 to indicate that the storage location has been updated with the corrected value. Only valid for corrected errors.                                                             |
| ceco                | 21         | 1         | true     | WARL      | 0x0         | Corrected error counter overflow                                                                                                                                                       |
| wpri_reserved_22_22 | 22         | 1         | true     | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                           |

|      |    |    |      |      |     |                                                                                                                                                                                                               |
|------|----|----|------|------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rdip | 23 | 1  | true | WARL | 0x0 | Read in progress. Set to 1 when a new error is recorded in an invalid status register. Cleared to 0 when a valid status is overwritten. When control.sinv is set, status.v is cleared to 0 only if rdip is 1. |
| ec   | 24 | 8  | true | WARL | 0x0 | Error code, provides a description of the detected error. Encodings defined in section 2.7 of RISC-V RERI spec                                                                                                |
| le   | 32 | 4  | true | WARL | 0x0 | The encoding that describes how to interpret/process the locator register value                                                                                                                               |
| ie   | 36 | 4  | true | WARL | 0x0 | The encoding that describes how to interpret/process the information register value                                                                                                                           |
| se   | 40 | 4  | true | WARL | 0x0 | The encoding that describes how to interpret/process the supplemental info register value                                                                                                                     |
| te   | 44 | 4  | true | WARL | 0x0 | The encoding that describes how to interpret/process the timestamp register value                                                                                                                             |
| cec  | 48 | 16 | true | WARL | 0x0 | Corrected error counter. Incremented on each corrected error if cece is set to 1.                                                                                                                             |

#### sc\_error\_record\_0\_addr

**Address:** 0x421A04D0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Address register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                |
|------------|------------|-----------|----------|--------|-------------|--------------------------------------------|
| address    | 0          | 64        | true     | WARL   | 0x0         | Address associated with the detected error |

#### sc\_error\_record\_0\_info

**Address:** 0x421A04D8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Information register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                           |
|------------|------------|-----------|----------|--------|-------------|---------------------------------------|
| info       | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom error information |

#### sc\_error\_record\_0\_suppl\_info

**Address:** 0x421A04E0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Supplemental information register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                        |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------------|
| suppl_info | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom supplemental error information |

### sc\_error\_record\_0\_timestamp

**Address:** 0x421A04E8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Timestamp register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                   |
|------------|------------|-----------|----------|--------|-------------|-------------------------------|
| timestamp  | 0          | 64        | true     | WARL   | 0x0         | Timestamp of the error record |

### sc\_error\_record\_0\_locator

**Address:** 0x421A04F0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Locator register of error record 0

| Field Name  | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                          |
|-------------|------------|-----------|----------|--------|-------------|--------------------------------------------------------------------------------------|
| description | 0          | 32        | true     | WARL   | 0x0         | Contains description of the error event telling what happened [descriptor, instance] |
| location    | 32         | 32        | true     | WARL   | 0x0         | Contains location of the error event telling where it happened [chip, ip, instance]  |

### sc\_error\_record\_1\_control

**Address:** 0x421A0500

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000001

**Description:** Control register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                                             |
|------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| else       | 0          | 1         | false    | WARL   | 0x1         | If 1, the hardware unit logs and signals errors in the error record. If 0, the hardware unit will not log RAS errors                                                                                    |
| cece       | 1          | 1         | false    | WARL   | 0x0         | If 1, hardware will count corrected errors in the cec field of the status register. If 0, cec does not count and retains its value.                                                                     |
| ces        | 2          | 2         | false    | WARL   | 0x0         | Used to enable signaling of corrected errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |

|                     |    |    |       |           |     |                                                                                                                                                                                                                    |
|---------------------|----|----|-------|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ueds                | 4  | 2  | false | WARL      | 0x0 | Used to enable signaling of uncorrected deferred errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| uecs                | 6  | 2  | false | WARL      | 0x0 | Used to enable signaling of uncorrected critical errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| wpri_reserved_31_8  | 8  | 24 | false | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| eid                 | 32 | 16 | true  | WARL      | 0x0 | Error injection delay                                                                                                                                                                                              |
| sinv                | 48 | 1  | true  | WARL      | 0x0 | Status register invalidate. When written to 1, causes the valid (v) field of the status register to be cleared.                                                                                                    |
| srdep               | 49 | 1  | true  | WARL      | 0x0 | Set read in progress. When 1, causes read in progress (rdip) to be set                                                                                                                                             |
| wpri_reserved_55_48 | 50 | 10 | false | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| custom_unused       | 60 | 4  | false | Read-Only | 0x0 | Reserved                                                                                                                                                                                                           |

## sc\_error\_record\_1\_status

**Address:** 0x421A0508

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Status register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                      |
|------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v          | 0          | 1         | true     | WARL   | 0x0         | If 1, error record holds a valid error                                                                                                                           |
| ce         | 1          | 1         | true     | WARL   | 0x0         | Set to 1 if the detected error was corrected                                                                                                                     |
| ued        | 2          | 1         | true     | WARL   | 0x0         | Set to 1 if the error could not be corrected but was deferred                                                                                                    |
| uec        | 3          | 1         | true     | WARL   | 0x0         | Set to 1 if the error could not be corrected or deferred and needs critical handling by RAS handler                                                              |
| pri        | 4          | 2         | true     | WARL   | 0x0         | Priority of the error. Indicates relative priority among errors of the same class.                                                                               |
| mo         | 6          | 1         | true     | WARL   | 0x0         | Set to 1 if more errors of the same class and severity occur                                                                                                     |
| c          | 7          | 1         | true     | WARL   | 0x0         | Containable bit. If 1, indicates that the error has not propagated beyond the boundaries of the hardware unit                                                    |
| tt         | 8          | 3         | true     | WARL   | 0x0         | Transaction type. 0=Unspecified or NA, 1=UNSPECIFIED, 2-3=Reserved for future standard use, 4=Explicit read, 5=Explicit write, 6=Implicit read, 7=Implicit write |
| iv         | 11         | 1         | true     | WARL   | 0x0         | Information valid, set to 1 if error reports additional information in the info register                                                                         |

|                      |    |    |      |           |     |                                                                                                                                                                                                               |
|----------------------|----|----|------|-----------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| at                   | 12 | 4  | true | WARL      | 0x0 | Address type, indicates the type of address reported in the addr register. 0=None, 1=Supervisor Physical Address, 2=Guest Physical Address, 3=Virtual Address, 4-15=Component Specific                        |
| siv                  | 16 | 1  | true | WARL      | 0x0 | Supplemental information valid, set to 1 if error reports supplemental information in the suppl_info register                                                                                                 |
| tsv                  | 17 | 1  | true | WARL      | 0x0 | Timestamp valid, set to 1 if the error record holds a valid timestamp in the timestamp register                                                                                                               |
| lv                   | 18 | 1  | true | WARL      | 0x0 | Locator valid, set to 1 if the error record holds a valid timestamp in the locator register                                                                                                                   |
| wpri_reserve_d_19_19 | 19 | 1  | true | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                  |
| scrub                | 20 | 1  | true | WARL      | 0x0 | Set to 1 to indicate that the storage location has been updated with the corrected value. Only valid for corrected errors.                                                                                    |
| ceco                 | 21 | 1  | true | WARL      | 0x0 | Corrected error counter overflow                                                                                                                                                                              |
| wpri_reserve_d_22_22 | 22 | 1  | true | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                  |
| rdip                 | 23 | 1  | true | WARL      | 0x0 | Read in progress. Set to 1 when a new error is recorded in an invalid status register. Cleared to 0 when a valid status is overwritten. When control.sinv is set, status.v is cleared to 0 only if rdip is 1. |
| ec                   | 24 | 8  | true | WARL      | 0x0 | Error code, provides a description of the detected error. Encodings defined in section 2.7 of RISC-V RERI spec. For memc, what values to use?                                                                 |
| le                   | 32 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the locator register value                                                                                                                               |
| ie                   | 36 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the information register value                                                                                                                           |
| se                   | 40 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the supplemental info register value                                                                                                                     |
| te                   | 44 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the timestamp register value                                                                                                                             |
| cec                  | 48 | 16 | true | WARL      | 0x0 | Corrected error counter. Incremented on each corrected error if cece is set to 1.                                                                                                                             |

### sc\_error\_record\_1\_addr

**Address:** 0x421A0510

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Address register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                |
|------------|------------|-----------|----------|--------|-------------|--------------------------------------------|
| address    | 0          | 64        | true     | WARL   | 0x0         | Address associated with the detected error |

### sc\_error\_record\_1\_info

**Address:** 0x421A0518

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Information register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                           |
|------------|------------|-----------|----------|--------|-------------|---------------------------------------|
| info       | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom error information |

### sc\_error\_record\_1\_suppl\_info

**Address:** 0x421A0520

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Supplemental information register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                        |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------------|
| suppl_info | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom supplemental error information |

### sc\_error\_record\_1\_timestamp

**Address:** 0x421A0528

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Timestamp register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                   |
|------------|------------|-----------|----------|--------|-------------|-------------------------------|
| timestamp  | 0          | 64        | true     | WARL   | 0x0         | Timestamp of the error record |

### sc\_error\_record\_1\_locator

**Address:** 0x421A0530

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Locator register of error record 1

| Field Name  | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                          |
|-------------|------------|-----------|----------|--------|-------------|--------------------------------------------------------------------------------------|
| description | 0          | 32        | true     | WARL   | 0x0         | Contains description of the error event telling what happened [descriptor, instance] |
| location    | 32         | 32        | true     | WARL   | 0x0         | Contains location of the error event telling where it happened [chip, ip, instance]  |

## sc\_error\_record\_2\_control

**Address:** 0x421A0540

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000001

**Description:** Control register of error record 2

| Field Name          | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                        |
|---------------------|------------|-----------|----------|-----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| else                | 0          | 1         | false    | WARL      | 0x1         | If 1, the hardware unit logs and signals errors in the error record. If 0, the hardware unit will not log RAS errors                                                                                               |
| cece                | 1          | 1         | false    | WARL      | 0x0         | If 1, hardware will count corrected errors in the cec field of the status register. If 0, cec does not count and retains its value.                                                                                |
| ces                 | 2          | 2         | false    | WARL      | 0x0         | Used to enable signaling of corrected errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal.            |
| ueds                | 4          | 2         | false    | WARL      | 0x0         | Used to enable signaling of uncorrected deferred errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| uecs                | 6          | 2         | false    | WARL      | 0x0         | Used to enable signaling of uncorrected critical errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| wpri_reserved_31_8  | 8          | 24        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| eid                 | 32         | 16        | true     | WARL      | 0x0         | Error injection delay                                                                                                                                                                                              |
| sinv                | 48         | 1         | true     | WARL      | 0x0         | Status register invalidate. When written to 1, causes the valid (v) field of the status register to be cleared.                                                                                                    |
| srdp                | 49         | 1         | true     | WARL      | 0x0         | Set read in progress. When 1, causes read in progress (rdip) to be set                                                                                                                                             |
| wpri_reserved_55_48 | 50         | 10        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| custom_unused       | 60         | 4         | false    | Read-Only | 0x0         | Reserved                                                                                                                                                                                                           |

## sc\_error\_record\_2\_status

**Address:** 0x421A0548

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Status register of error record 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                  |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------|
| v          | 0          | 1         | true     | WARL   | 0x0         | If 1, error record holds a valid error       |
| ce         | 1          | 1         | true     | WARL   | 0x0         | Set to 1 if the detected error was corrected |

|                     |    |   |      |           |     |                                                                                                                                                                                                               |
|---------------------|----|---|------|-----------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ued                 | 2  | 1 | true | WARL      | 0x0 | Set to 1 if the error could not be corrected but was deferred                                                                                                                                                 |
| uec                 | 3  | 1 | true | WARL      | 0x0 | Set to 1 if the error could not be corrected or deferred and needs critical handling by RAS handler                                                                                                           |
| pri                 | 4  | 2 | true | WARL      | 0x0 | Priority of the error. Indicates relative priority among errors of the same class.                                                                                                                            |
| mo                  | 6  | 1 | true | WARL      | 0x0 | Set to 1 if more errors of the same class and severity occur                                                                                                                                                  |
| c                   | 7  | 1 | true | WARL      | 0x0 | Containable bit. If 1, indicates that the error has not propagated beyond the boundaries of the hardware unit                                                                                                 |
| tt                  | 8  | 3 | true | WARL      | 0x0 | Transaction type. 0=Unspecified or NA, 1=UNSPECIFIED, 2-3=Reserved for future standard use, 4=Explicit read, 5=Explicit write, 6=Implicit read, 7=Implicit write                                              |
| iv                  | 11 | 1 | true | WARL      | 0x0 | Information valid, set to 1 if error reports additional information in the info register                                                                                                                      |
| at                  | 12 | 4 | true | WARL      | 0x0 | Address type, indicates the type of address reported in the addr register. 0=None, 1=Supervisor Physical Address, 2=Guest Physical Address, 3=Virtual Address, 4-15=Component Specific                        |
| siv                 | 16 | 1 | true | WARL      | 0x0 | Supplemental information valid, set to 1 if error reports supplemental information in the suppl_info register                                                                                                 |
| tsv                 | 17 | 1 | true | WARL      | 0x0 | Timestamp valid, set to 1 if the error record holds a valid timestamp in the timestamp register                                                                                                               |
| lv                  | 18 | 1 | true | WARL      | 0x0 | Locator valid, set to 1 if the error record holds a valid timestamp in the locator register                                                                                                                   |
| wpri_reserved_19_19 | 19 | 1 | true | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                  |
| scrub               | 20 | 1 | true | WARL      | 0x0 | Set to 1 to indicate that the storage location has been updated with the corrected value. Only valid for corrected errors.                                                                                    |
| ceco                | 21 | 1 | true | WARL      | 0x0 | Corrected error counter overflow                                                                                                                                                                              |
| wpri_reserved_22_22 | 22 | 1 | true | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                  |
| rdip                | 23 | 1 | true | WARL      | 0x0 | Read in progress. Set to 1 when a new error is recorded in an invalid status register. Cleared to 0 when a valid status is overwritten. When control.sinv is set, status.v is cleared to 0 only if rdip is 1. |
| ec                  | 24 | 8 | true | WARL      | 0x0 | Error code, provides a description of the detected error. Encodings defined in section 2.7 of RISC-V RERI spec. For memc, what values to use?                                                                 |
| le                  | 32 | 4 | true | WARL      | 0x0 | The encoding that describes how to interpret/process the locator register value                                                                                                                               |
| ie                  | 36 | 4 | true | WARL      | 0x0 | The encoding that describes how to interpret/process the information register value                                                                                                                           |
| se                  | 40 | 4 | true | WARL      | 0x0 | The encoding that describes how to interpret/process the supplemental info register value                                                                                                                     |

|     |    |    |      |      |     |                                                                                   |
|-----|----|----|------|------|-----|-----------------------------------------------------------------------------------|
| te  | 44 | 4  | true | WARL | 0x0 | The encoding that describes how to interpret/process the timestamp register value |
| cec | 48 | 16 | true | WARL | 0x0 | Corrected error counter. Incremented on each corrected error if cece is set to 1. |

### sc\_error\_record\_2\_addr

**Address:** 0x421A0550

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Address register of error record 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                |
|------------|------------|-----------|----------|--------|-------------|--------------------------------------------|
| address    | 0          | 64        | true     | WARL   | 0x0         | Address associated with the detected error |

### sc\_error\_record\_2\_info

**Address:** 0x421A0558

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Information register of error record 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                           |
|------------|------------|-----------|----------|--------|-------------|---------------------------------------|
| info       | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom error information |

### sc\_error\_record\_2\_suppl\_info

**Address:** 0x421A0560

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Supplemental information register of error record 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                        |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------------|
| suppl_info | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom supplemental error information |

### sc\_error\_record\_2\_timestamp

**Address:** 0x421A0568

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Timestamp register of error record 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                   |
|------------|------------|-----------|----------|--------|-------------|-------------------------------|
| timestamp  | 0          | 64        | true     | WARL   | 0x0         | Timestamp of the error record |

## sc\_error\_record\_2\_locator

**Address:** 0x421A0570

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Locator register of error record 2

| Field Name  | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                          |
|-------------|------------|-----------|----------|--------|-------------|--------------------------------------------------------------------------------------|
| description | 0          | 32        | true     | WARL   | 0x0         | Contains description of the error event telling what happened [descriptor, instance] |
| location    | 32         | 32        | true     | WARL   | 0x0         | Contains location of the error event telling where it happened [chip, ip, instance]  |

## sc\_hardware\_assertion

**Address:** 0x421A0580

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Used to log SC hardware assertion fails

| Field Name  | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                |
|-------------|------------|-----------|----------|-----------|-------------|----------------------------------------------------------------------------------------------------------------------------|
| hwatype     | 0          | 10        | true     | Read-Only | 0x0         | HWA identifier. 0 indicates no failure                                                                                     |
| timestamp_v | 10         | 1         | true     | Read-Only | 0x0         | If 1, indicates that timestamp field is valid                                                                              |
| timestamp   | 11         | 27        | true     | Read-Only | 0x0         | RAS timestamp stub of when assertion failure occurred. To directly compare to logged RERI timestamps, left shift by 5 bits |
| rsvd        | 38         | 26        | true     | Read-Only | 0x0         |                                                                                                                            |

## sc\_ras\_inj\_hdr

**Address:** 0x421A05C0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Fault Injection Header

| Field Name      | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                   |
|-----------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------|
| revision_min or | 0          | 4         | false    | WARL   | 0x0         | Indicates that the meaning or behavior of some registers have changed                                         |
| revision_major  | 4          | 4         | false    | WARL   | 0x0         | Indicates that the organization and/or meaning of later fields and/or registers have changed                  |
| data_w          | 8          | 4         | false    | WARL   | 0x0         | Indicates the number of bits, as a power of 2, of the corruption vectors associated with the injection module |
| check_w         | 12         | 4         | false    | WARL   | 0x0         | Indicates the number of bits, as a power of 2, of the corruption vectors associated with the injection module |

|                 |    |    |       |      |     |                                                                                                                                                                                         |
|-----------------|----|----|-------|------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| presentations   | 16 | 16 | false | WARL | 0x0 | Bit vector of which data and check corruption vector presentations are available                                                                                                        |
| pre_processors  | 32 | 4  | false | WARL | 0x0 | Encodes the number of pre-processor instances (and hence total available trigger count)                                                                                                 |
| main_processors | 36 | 4  | false | WARL | 0x0 | Encodes the number of main-processors                                                                                                                                                   |
| post_processors | 40 | 4  | false | WARL | 0x0 | Encodes the number of post-processors (and hence the total available trigger outputs)                                                                                                   |
| rsvd_processors | 44 | 4  | false | WARL | 0x0 | Reserved encoding for processors                                                                                                                                                        |
| injector_enable | 48 | 1  | false | WARL | 0x0 | The injector is enabled and will inject faults when the trigger is asserted and processed according to the provided configurations and setup. This is an overall gate for the injector. |
| rsvd63_49       | 49 | 15 | false | WARL | 0x0 |                                                                                                                                                                                         |

### sc\_ras\_inj\_init

**Address:** 0x421A05C8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Initialisation: write 1 to initialize behavior for the associated Trigger Processor Module and any injector internal state

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| pre        | 0          | 16        | false    | WARL   | 0x0         |             |
| main       | 16         | 16        | false    | WARL   | 0x0         |             |
| post       | 32         | 16        | false    | WARL   | 0x0         |             |
| injector   | 48         | 16        | false    | WARL   | 0x0         |             |

### sc\_ras\_inj\_data\_match\_0

**Address:** 0x421A05D0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Trigger Data Match Value 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| match      | 0          | 64        | false    | WARL   | 0x0         |             |

### sc\_ras\_inj\_data\_mask\_0

**Address:** 0x421A05D8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Trigger Data Match Mask 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | false    | WARL   | 0x0         |             |

### sc\_ras\_inj\_data\_match\_1

**Address:** 0x421A05E0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Trigger Data Match Value 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| match      | 0          | 64        | false    | WARL   | 0x0         |             |

### sc\_ras\_inj\_data\_mask\_1

**Address:** 0x421A05E8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Trigger Data Match Mask 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | false    | WARL   | 0x0         |             |

### sc\_ras\_inj\_data\_match\_2

**Address:** 0x421A05F0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Trigger Data Match Value 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| match      | 0          | 64        | false    | WARL   | 0x0         |             |

### sc\_ras\_inj\_data\_mask\_2

**Address:** 0x421A05F8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Trigger Data Match Mask 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | false    | WARL   | 0x0         |             |

## sc\_ras\_inj\_0\_1\_cfg

**Address:** 0x421A0600

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Configuration register 0-1. Injector 0 if for array writes (encode), and Injector 1 is for array reads (decode)

| Field Name               | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                           |
|--------------------------|------------|-----------|----------|--------|-------------|-----------------------------------------------------------------------------------------------------------------------|
| inj0_tag_wr_select       | 0          | 1         | false    | WARL   | 0x0         | Enables tag array write (encode) to receive the injection when the trigger occurs.                                    |
| inj0_state_wr_select     | 1          | 1         | false    | WARL   | 0x0         | Enables state array write (encode) to receive the injection when the trigger occurs.                                  |
| inj0_replace_wr_select   | 2          | 1         | false    | WARL   | 0x0         | Enables replace array write (encode) to receive the injection when the trigger occurs.                                |
| inj0_data_wr_select      | 3          | 1         | false    | WARL   | 0x0         | Enables data array write (encode) to receive the injection when the trigger occurs.                                   |
| inj0_snf_tag_wr_select   | 4          | 1         | false    | WARL   | 0x0         | Enables snoop filter tag array write (encode) to receive the injection when the trigger occurs.                       |
| inj0_snf_state_wr_select | 5          | 1         | false    | WARL   | 0x0         | Enables snoop filter state array write (encode) to receive the injection when the trigger occurs.                     |
| inj0_structure_wr_rsvd   | 6          | 2         | false    | WARL   | 0x0         | Identifies which of the structures associated with the injector are to receive the injection when the trigger occurs. |
| inj0_rsvd                | 8          | 8         | false    | WARL   | 0x0         |                                                                                                                       |
| inj0_c_data_pld          | 16         | 4         | false    | WARL   | 0x0         | Identifies what is to be presented to the encode or decode block selected by structure                                |
| inj0_c_data_pt           | 20         | 4         | false    | WARL   | 0x0         | Determines how the corrupt data pld field is presented to the encode or decode block selected by structure.           |
| inj0_c_check_pld         | 24         | 4         | false    | WARL   | 0x0         | Identifies what is to be presented to the encode or decode block selected by structure                                |
| inj0_c_check_pt          | 28         | 4         | false    | WARL   | 0x0         | Determines how the corrupt check pld field is presented to the encode or decode block selected by structure.          |
| inj1_tag_rd_select       | 32         | 1         | false    | WARL   | 0x0         | Enables tag array write (decode) to receive the injection when the trigger occurs.                                    |
| inj1_state_rd_select     | 33         | 1         | false    | WARL   | 0x0         | Enables state array write (decode) to receive the injection when the trigger occurs.                                  |
| inj1_replace_rd_select   | 34         | 1         | false    | WARL   | 0x0         | Enables replace array write (decode) to receive the injection when the trigger occurs.                                |
| inj1_data_rd_select      | 35         | 1         | false    | WARL   | 0x0         | Enables data array write (decode) to receive the injection when the trigger occurs.                                   |
| inj1_snf_tag_rd_select   | 36         | 1         | false    | WARL   | 0x0         | Enables snoop filter tag array write (decode) to receive the injection when the trigger occurs.                       |

|                              |    |   |       |      |     |                                                                                                                       |
|------------------------------|----|---|-------|------|-----|-----------------------------------------------------------------------------------------------------------------------|
| inj1_snf_stat<br>e_rd_select | 37 | 1 | false | WARL | 0x0 | Enables snoop filter state array write (decode) to receive the injection when the trigger occurs.                     |
| inj1_structur<br>e_rsvd      | 38 | 2 | false | WARL | 0x0 | Identifies which of the structures associated with the injector are to receive the injection when the trigger occurs. |
| inj1_rsvd                    | 40 | 8 | false | WARL | 0x0 |                                                                                                                       |
| inj1_c_data_<br>pld          | 48 | 4 | false | WARL | 0x0 | Identifies what is to be presented to the encode or decode block selected by structure                                |
| inj1_c_data_<br>pt           | 52 | 4 | false | WARL | 0x0 | Determines how the corrupt data pld field is presented to the encode or decode block selected by structure.           |
| inj1_c_check_<br>pld         | 56 | 4 | false | WARL | 0x0 | Identifies what is to be presented to the encode or decode block selected by structure                                |
| inj1_c_check_<br>pt          | 60 | 4 | false | WARL | 0x0 | Determines how the corrupt check pld field is presented to the encode or decode block selected by structure.          |

### sc\_ras\_tpm\_pre\_proc\_cfg\_0

**Address:** 0x421A0640

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Pre processor 0-7

| Field Name         | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                             |
|--------------------|------------|-----------|----------|--------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc0_functi<br>on | 0          | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc0_meta         | 2          | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc0_rsvd         | 4          | 4         | false    | WARL   | 0x0         |                                                                                                                                                                         |
| proc1_functi<br>on | 8          | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc1_meta         | 10         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc1_rsvd         | 12         | 4         | false    | WARL   | 0x0         |                                                                                                                                                                         |
| proc2_functi<br>on | 16         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc2_meta         | 18         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc2_rsvd         | 20         | 4         | false    | WARL   | 0x0         |                                                                                                                                                                         |
| proc3_functi<br>on | 24         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |

|                |    |   |       |      |     |                                                                                                                                                                         |
|----------------|----|---|-------|------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc3_meta     | 26 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc3_rsvd     | 28 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc4_function | 32 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor: 0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                    |
| proc4_meta     | 34 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc4_rsvd     | 36 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc5_function | 40 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor: 0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                    |
| proc5_meta     | 42 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc5_rsvd     | 44 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc6_function | 48 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor: 0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                    |
| proc6_meta     | 50 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc6_rsvd     | 52 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc7_function | 56 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor: 0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                    |
| proc7_meta     | 58 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc7_rsvd     | 60 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |

### sc\_ras\_tpm\_pre\_proc\_cfg\_1

**Address:** 0x421A0648

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Pre processor 8-15

| Field Name     | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                             |
|----------------|------------|-----------|----------|--------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc8_function | 0          | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor: 0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                    |
| proc8_meta     | 2          | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc8_rsvd     | 4          | 4         | false    | WARL   | 0x0         |                                                                                                                                                                         |

|                 |    |   |       |      |     |                                                                                                                                                                         |
|-----------------|----|---|-------|------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc9_function  | 8  | 2 | false | WARL | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc9_meta      | 10 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc9_rsvd      | 12 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc10_function | 16 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc10_meta     | 18 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc10_rsvd     | 20 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc11_function | 24 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc11_meta     | 26 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc11_rsvd     | 28 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc12_function | 32 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc12_meta     | 34 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc12_rsvd     | 36 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc13_function | 40 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc13_meta     | 42 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc13_rsvd     | 44 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc14_function | 48 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc14_meta     | 50 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc14_rsvd     | 52 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc15_function | 56 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc15_meta     | 58 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc15_rsvd     | 60 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |

## sc\_ras\_tpm\_main\_proc\_cfg\_0

**Address:** 0x421A0650

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Main processor 0-3

| Field Name      | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|------------|-----------|----------|--------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc0_trigger_a | 0          | 4         | false    | WARL   | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc0_trigger_b | 4          | 4         | false    | WARL   | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc0_function  | 8          | 4         | false    | WARL   | 0x0         | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc0_meta      | 12         | 4         | false    | WARL   | 0x0         | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc1_trigger_a | 16         | 4         | false    | WARL   | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc1_trigger_b | 20         | 4         | false    | WARL   | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc1_function  | 24         | 4         | false    | WARL   | 0x0         | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc1_meta      | 28         | 4         | false    | WARL   | 0x0         | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc2_trigger_a | 32         | 4         | false    | WARL   | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |

|                 |    |   |       |      |     |                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|----|---|-------|------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc2_trigger_b | 36 | 4 | false | WARL | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc2_function  | 40 | 4 | false | WARL | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc2_meta      | 44 | 4 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc3_trigger_a | 48 | 4 | false | WARL | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc3_trigger_b | 52 | 4 | false | WARL | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc3_function  | 56 | 4 | false | WARL | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc3_meta      | 60 | 4 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |

### sc\_ras\_tpm\_main\_proc\_cfg\_1

**Address:** 0x421A0658

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Main processor 4-7

| Field Name      | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                            |
|-----------------|------------|-----------|----------|-----------|-------------|----------------------------------------------------------------------------------------------------------------------------------------|
| proc4_trigger_a | 0          | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires. |
| proc4_trigger_b | 4          | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires. |

|                 |    |   |       |           |     |                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|----|---|-------|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc4_function  | 8  | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc4_meta      | 12 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc5_trigger_a | 16 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc5_trigger_b | 20 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc5_function  | 24 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc5_meta      | 28 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc6_trigger_a | 32 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc6_trigger_b | 36 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc6_function  | 40 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc6_meta      | 44 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |

|                 |    |   |       |           |     |                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|----|---|-------|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc7_trigger_a | 48 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc7_trigger_b | 52 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc7_function  | 56 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc7_meta      | 60 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |

## sc\_ras\_tpm\_main\_proc\_cfg\_2

**Address:** 0x421A0660

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Main processor 8-11

| Field Name      | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|------------|-----------|----------|-----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc8_trigger_a | 0          | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc8_trigger_b | 4          | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc8_function  | 8          | 4         | false    | Read-Only | 0x0         | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc8_meta      | 12         | 4         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc9_trigger_a | 16         | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |

|                  |    |   |       |           |     |                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|----|---|-------|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc9_trigger_b  | 20 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc9_function   | 24 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc9_meta       | 28 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc10_trigger_a | 32 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc10_trigger_b | 36 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc10_function  | 40 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc10_meta      | 44 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc11_trigger_a | 48 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc11_trigger_b | 52 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc11_function  | 56 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |

|             |    |   |       |           |     |                                                                                                                                                                                                                                           |
|-------------|----|---|-------|-----------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc11_meta | 60 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD |
|-------------|----|---|-------|-----------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### sc\_ras\_tpm\_main\_proc\_cfg\_3

**Address:** 0x421A0668

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Main processor 12-15

| Field Name       | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|------------|-----------|----------|-----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc12_trigger_a | 0          | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc12_trigger_b | 4          | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc12_function  | 8          | 4         | false    | Read-Only | 0x0         | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc12_meta      | 12         | 4         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc13_trigger_a | 16         | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc13_trigger_b | 20         | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc13_function  | 24         | 4         | false    | Read-Only | 0x0         | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |

|                  |    |   |       |           |     |                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|----|---|-------|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc13_meta      | 28 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc14_trigger_a | 32 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc14_trigger_b | 36 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc14_function  | 40 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc14_meta      | 44 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc15_trigger_a | 48 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc15_trigger_b | 52 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc15_function  | 56 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc15_meta      | 60 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |

### sc\_ras\_tpm\_post\_proc\_cfg\_0

**Address:** 0x421A0670

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Post processor 0-7

| Field Name     | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                        |
|----------------|------------|-----------|----------|--------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| proc0_function | 0          | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc0_meta     | 2          | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc0_rsvd     | 4          | 4         | false    | WARL   | 0x0         |                                                                                                                                                    |
| proc1_function | 8          | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc1_meta     | 10         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc1_rsvd     | 12         | 4         | false    | WARL   | 0x0         |                                                                                                                                                    |
| proc2_function | 16         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc2_meta     | 18         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc2_rsvd     | 20         | 4         | false    | WARL   | 0x0         |                                                                                                                                                    |
| proc3_function | 24         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc3_meta     | 26         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc3_rsvd     | 28         | 4         | false    | WARL   | 0x0         |                                                                                                                                                    |
| proc4_function | 32         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc4_meta     | 34         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc4_rsvd     | 36         | 4         | false    | WARL   | 0x0         |                                                                                                                                                    |
| proc5_function | 40         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc5_meta     | 42         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc5_rsvd     | 44         | 4         | false    | WARL   | 0x0         |                                                                                                                                                    |
| proc6_function | 48         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc6_meta     | 50         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc6_rsvd     | 52         | 4         | false    | WARL   | 0x0         |                                                                                                                                                    |
| proc7_function | 56         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc7_meta     | 58         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |

|            |    |   |       |      |     |  |
|------------|----|---|-------|------|-----|--|
| proc7_rsvd | 60 | 4 | false | WARL | 0x0 |  |
|------------|----|---|-------|------|-----|--|

## sc\_ras\_tpm\_post\_proc\_cfg\_1

**Address:** 0x421A0678

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Post processor 8-15

| Field Name      | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                        |
|-----------------|------------|-----------|----------|-----------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| proc8_function  | 0          | 2         | false    | Read-Only | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc8_meta      | 2          | 2         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc8_rsvd      | 4          | 4         | false    | Read-Only | 0x0         |                                                                                                                                                    |
| proc9_function  | 8          | 2         | false    | Read-Only | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc9_meta      | 10         | 2         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc9_rsvd      | 12         | 4         | false    | Read-Only | 0x0         |                                                                                                                                                    |
| proc10_function | 16         | 2         | false    | Read-Only | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc10_meta     | 18         | 2         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc10_rsvd     | 20         | 4         | false    | Read-Only | 0x0         |                                                                                                                                                    |
| proc11_function | 24         | 2         | false    | Read-Only | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc11_meta     | 26         | 2         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc11_rsvd     | 28         | 4         | false    | Read-Only | 0x0         |                                                                                                                                                    |
| proc12_function | 32         | 2         | false    | Read-Only | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc12_meta     | 34         | 2         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc12_rsvd     | 36         | 4         | false    | Read-Only | 0x0         |                                                                                                                                                    |
| proc13_function | 40         | 2         | false    | Read-Only | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc13_meta     | 42         | 2         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc13_rsvd     | 44         | 4         | false    | Read-Only | 0x0         |                                                                                                                                                    |

|                 |    |   |       |           |     |                                                                                                                                                    |
|-----------------|----|---|-------|-----------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------|
| proc14_function | 48 | 2 | false | Read-Only | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc14_meta     | 50 | 2 | false | Read-Only | 0x0 | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc14_rsvd     | 52 | 4 | false | Read-Only | 0x0 |                                                                                                                                                    |
| proc15_function | 56 | 2 | false | Read-Only | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc15_meta     | 58 | 2 | false | Read-Only | 0x0 | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc15_rsvd     | 60 | 4 | false | Read-Only | 0x0 |                                                                                                                                                    |

### sc\_ras\_tpm\_main\_proc\_capability\_0

**Address:** 0x421A0680

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Capabilities possible with each of the main processors: 0-7

| Field Name     | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                               |
|----------------|------------|-----------|----------|--------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| proc0_pass     | 0          | 1         | true     | WARL   | 0x0         | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc0_leaky    | 1          | 1         | true     | WARL   | 0x0         | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc0_table    | 2          | 1         | true     | WARL   | 0x0         | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc0_lfsr     | 3          | 1         | true     | WARL   | 0x0         | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc0_selftest | 4          | 1         | true     | WARL   | 0x0         | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc0_rsvd     | 5          | 3         | true     | WARL   | 0x0         |                                                                                                                                           |
| proc1_pass     | 8          | 1         | true     | WARL   | 0x0         | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc1_leaky    | 9          | 1         | true     | WARL   | 0x0         | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc1_table    | 10         | 1         | true     | WARL   | 0x0         | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc1_lfsr     | 11         | 1         | true     | WARL   | 0x0         | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc1_selftest | 12         | 1         | true     | WARL   | 0x0         | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |

|                |    |   |      |      |     |                                                                                                                                           |
|----------------|----|---|------|------|-----|-------------------------------------------------------------------------------------------------------------------------------------------|
| proc1_rsvd     | 13 | 3 | true | WARL | 0x0 |                                                                                                                                           |
| proc2_pass     | 16 | 1 | true | WARL | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc2_leaky    | 17 | 1 | true | WARL | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc2_table    | 18 | 1 | true | WARL | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc2_lfsr     | 19 | 1 | true | WARL | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc2_selftest | 20 | 1 | true | WARL | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc2_rsvd     | 21 | 3 | true | WARL | 0x0 |                                                                                                                                           |
| proc3_pass     | 24 | 1 | true | WARL | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc3_leaky    | 25 | 1 | true | WARL | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc3_table    | 26 | 1 | true | WARL | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc3_lfsr     | 27 | 1 | true | WARL | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc3_selftest | 28 | 1 | true | WARL | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc3_rsvd     | 29 | 3 | true | WARL | 0x0 |                                                                                                                                           |
| proc4_pass     | 32 | 1 | true | WARL | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc4_leaky    | 33 | 1 | true | WARL | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc4_table    | 34 | 1 | true | WARL | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc4_lfsr     | 35 | 1 | true | WARL | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc4_selftest | 36 | 1 | true | WARL | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc4_rsvd     | 37 | 3 | true | WARL | 0x0 |                                                                                                                                           |
| proc5_pass     | 40 | 1 | true | WARL | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc5_leaky    | 41 | 1 | true | WARL | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |

|                |    |   |      |      |     |                                                                                                                                           |
|----------------|----|---|------|------|-----|-------------------------------------------------------------------------------------------------------------------------------------------|
| proc5_table    | 42 | 1 | true | WARL | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc5_lfsr     | 43 | 1 | true | WARL | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc5_selftest | 44 | 1 | true | WARL | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc5_rsvd     | 45 | 3 | true | WARL | 0x0 |                                                                                                                                           |
| proc6_pass     | 48 | 1 | true | WARL | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc6_leaky    | 49 | 1 | true | WARL | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc6_table    | 50 | 1 | true | WARL | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc6_lfsr     | 51 | 1 | true | WARL | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc6_selftest | 52 | 1 | true | WARL | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc6_rsvd     | 53 | 3 | true | WARL | 0x0 |                                                                                                                                           |
| proc7_pass     | 56 | 1 | true | WARL | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc7_leaky    | 57 | 1 | true | WARL | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc7_table    | 58 | 1 | true | WARL | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc7_lfsr     | 59 | 1 | true | WARL | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc7_selftest | 60 | 1 | true | WARL | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc7_rsvd     | 61 | 3 | true | WARL | 0x0 |                                                                                                                                           |

### sc\_ras\_tpm\_main\_proc\_capability\_1

**Address:** 0x421A0688

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Capabilities possible with each of the main processors: 8-15

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                               |
|------------|------------|-----------|----------|-----------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| proc8_pass | 0          | 1         | true     | Read-Only | 0x0         | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |

|                 |    |   |      |           |     |                                                                                                                                           |
|-----------------|----|---|------|-----------|-----|-------------------------------------------------------------------------------------------------------------------------------------------|
| proc8_leaky     | 1  | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc8_table     | 2  | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc8_lfsr      | 3  | 1 | true | Read-Only | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc8_selftest  | 4  | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc8_rsvd      | 5  | 3 | true | Read-Only | 0x0 |                                                                                                                                           |
| proc9_pass      | 8  | 1 | true | Read-Only | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc9_leaky     | 9  | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc9_table     | 10 | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc9_lfsr      | 11 | 1 | true | Read-Only | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc9_selftest  | 12 | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc9_rsvd      | 13 | 3 | true | Read-Only | 0x0 |                                                                                                                                           |
| proc10_pass     | 16 | 1 | true | Read-Only | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc10_leaky    | 17 | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc10_table    | 18 | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc10_lfsr     | 19 | 1 | true | Read-Only | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc10_selftest | 20 | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc10_rsvd     | 21 | 3 | true | Read-Only | 0x0 |                                                                                                                                           |
| proc11_pass     | 24 | 1 | true | Read-Only | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc11_leaky    | 25 | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc11_table    | 26 | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc11_lfsr     | 27 | 1 | true | Read-Only | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |

|                 |    |   |      |           |     |                                                                                                                                           |
|-----------------|----|---|------|-----------|-----|-------------------------------------------------------------------------------------------------------------------------------------------|
| proc11_selftest | 28 | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc11_rsvd     | 29 | 3 | true | Read-Only | 0x0 |                                                                                                                                           |
| proc12_pass     | 32 | 1 | true | Read-Only | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc12_leaky    | 33 | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc12_table    | 34 | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc12_lfsr     | 35 | 1 | true | Read-Only | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc12_selftest | 36 | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc12_rsvd     | 37 | 3 | true | Read-Only | 0x0 |                                                                                                                                           |
| proc13_pass     | 40 | 1 | true | Read-Only | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc13_leaky    | 41 | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc13_table    | 42 | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc13_lfsr     | 43 | 1 | true | Read-Only | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc13_selftest | 44 | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc13_rsvd     | 45 | 3 | true | Read-Only | 0x0 |                                                                                                                                           |
| proc14_pass     | 48 | 1 | true | Read-Only | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc14_leaky    | 49 | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc14_table    | 50 | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc14_lfsr     | 51 | 1 | true | Read-Only | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc14_selftest | 52 | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc14_rsvd     | 53 | 3 | true | Read-Only | 0x0 |                                                                                                                                           |
| proc15_pass     | 56 | 1 | true | Read-Only | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |

|                 |    |   |      |           |     |                                                                                                   |
|-----------------|----|---|------|-----------|-----|---------------------------------------------------------------------------------------------------|
| proc15_leaky    | 57 | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                    |
| proc15_table    | 58 | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.          |
| proc15_lfsr     | 59 | 1 | true | Read-Only | 0x0 | The lfsr function where one or more LFSR polynomials are applied to determine the trigger output. |
| proc15_selftest | 60 | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.   |
| proc15_rsvd     | 61 | 3 | true | Read-Only | 0x0 |                                                                                                   |

## sc\_ras\_tpm\_pre\_proc\_sts

**Address:** 0x421A0690

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** TPM Pre processor status: 0-15

| Field Name    | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                    |
|---------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------------------------------------------|
| proc0_state_i | 0          | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc0_state_o | 2          | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc1_state_i | 4          | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc1_state_o | 6          | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc2_state_i | 8          | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc2_state_o | 10         | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc3_state_i | 12         | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc3_state_o | 14         | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc4_state_i | 16         | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc4_state_o | 18         | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc5_state_i | 20         | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |

|                |    |   |      |      |     |                                                                                                |
|----------------|----|---|------|------|-----|------------------------------------------------------------------------------------------------|
| proc5_state_o  | 22 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc6_state_i  | 24 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc6_state_o  | 26 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc7_state_i  | 28 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc7_state_o  | 30 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc8_state_i  | 32 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc8_state_o  | 34 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc9_state_i  | 36 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc9_state_o  | 38 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc10_state_i | 40 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc10_state_o | 42 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc11_state_i | 44 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc11_state_o | 46 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc12_state_i | 48 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc12_state_o | 50 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc13_state_i | 52 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc13_state_o | 54 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc14_state_i | 56 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc14_state_o | 58 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |

|                |    |   |      |      |     |                                                                                                |
|----------------|----|---|------|------|-----|------------------------------------------------------------------------------------------------|
| proc15_state_i | 60 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc15_state_o | 62 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |

## sc\_ras\_tpm\_main\_proc\_sts\_0

**Address:** 0x421A0698

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** TPM Main processor status: 0-7

| Field Name       | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                                                                           |
|------------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc0_state_a    | 0          | 2         | true     | WARL   | 0x0         | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc0_state_b    | 2          | 2         | true     | WARL   | 0x0         | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc0_state_meta | 4          | 2         | true     | WARL   | 0x0         | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc0_state_o    | 6          | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc1_state_a    | 8          | 2         | true     | WARL   | 0x0         | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc1_state_b    | 10         | 2         | true     | WARL   | 0x0         | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc1_state_meta | 12         | 2         | true     | WARL   | 0x0         | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc1_state_o    | 14         | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc2_state_a    | 16         | 2         | true     | WARL   | 0x0         | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc2_state_b    | 18         | 2         | true     | WARL   | 0x0         | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc2_state_meta | 20         | 2         | true     | WARL   | 0x0         | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |

|                  |    |   |      |      |     |                                                                                                                                                                                                                                       |
|------------------|----|---|------|------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc2_state_o    | 22 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc3_state_a    | 24 | 2 | true | WARL | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc3_state_b    | 26 | 2 | true | WARL | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc3_state_meta | 28 | 2 | true | WARL | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc3_state_o    | 30 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc4_state_a    | 32 | 2 | true | WARL | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc4_state_b    | 34 | 2 | true | WARL | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc4_state_meta | 36 | 2 | true | WARL | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc4_state_o    | 38 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc5_state_a    | 40 | 2 | true | WARL | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc5_state_b    | 42 | 2 | true | WARL | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc5_state_meta | 44 | 2 | true | WARL | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc5_state_o    | 46 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc6_state_a    | 48 | 2 | true | WARL | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc6_state_b    | 50 | 2 | true | WARL | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc6_state_meta | 52 | 2 | true | WARL | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |

|                  |    |   |      |      |     |                                                                                                                                                                                                                                       |
|------------------|----|---|------|------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc6_state_o    | 54 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc7_state_a    | 56 | 2 | true | WARL | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc7_state_b    | 58 | 2 | true | WARL | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc7_state_meta | 60 | 2 | true | WARL | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc7_state_o    | 62 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |

### sc\_ras\_tpm\_main\_proc\_sts\_1

**Address:** 0x421A06A0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** TPM Main processor status: 8-15

| Field Name       | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                                           |
|------------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc8_state_a    | 0          | 2         | true     | Read-Only | 0x0         | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc8_state_b    | 2          | 2         | true     | Read-Only | 0x0         | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc8_state_meta | 4          | 2         | true     | Read-Only | 0x0         | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc8_state_o    | 6          | 2         | true     | Read-Only | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc9_state_a    | 8          | 2         | true     | Read-Only | 0x0         | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc9_state_b    | 10         | 2         | true     | Read-Only | 0x0         | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc9_state_meta | 12         | 2         | true     | Read-Only | 0x0         | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc9_state_o    | 14         | 2         | true     | Read-Only | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |

|                   |    |   |      |           |     |                                                                                                                                                                                                                                       |
|-------------------|----|---|------|-----------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc10_state_a    | 16 | 2 | true | Read-Only | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc10_state_b    | 18 | 2 | true | Read-Only | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc10_state_meta | 20 | 2 | true | Read-Only | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc10_state_o    | 22 | 2 | true | Read-Only | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc11_state_a    | 24 | 2 | true | Read-Only | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc11_state_b    | 26 | 2 | true | Read-Only | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc11_state_meta | 28 | 2 | true | Read-Only | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc11_state_o    | 30 | 2 | true | Read-Only | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc12_state_a    | 32 | 2 | true | Read-Only | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc12_state_b    | 34 | 2 | true | Read-Only | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc12_state_meta | 36 | 2 | true | Read-Only | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc12_state_o    | 38 | 2 | true | Read-Only | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc13_state_a    | 40 | 2 | true | Read-Only | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc13_state_b    | 42 | 2 | true | Read-Only | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc13_state_meta | 44 | 2 | true | Read-Only | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc13_state_o    | 46 | 2 | true | Read-Only | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |

|                   |    |   |      |           |     |                                                                                                                                                                                                                                       |
|-------------------|----|---|------|-----------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc14_state_a    | 48 | 2 | true | Read-Only | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc14_state_b    | 50 | 2 | true | Read-Only | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc14_state_meta | 52 | 2 | true | Read-Only | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc14_state_o    | 54 | 2 | true | Read-Only | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc15_state_a    | 56 | 2 | true | Read-Only | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc15_state_b    | 58 | 2 | true | Read-Only | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc15_state_meta | 60 | 2 | true | Read-Only | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc15_state_o    | 62 | 2 | true | Read-Only | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |

### sc\_ras\_tpm\_post\_proc\_sts

**Address:** 0x421A06A8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** TPM Post processor status: 0-15

| Field Name    | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                    |
|---------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------------------------------------------|
| proc0_state_i | 0          | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc0_state_o | 2          | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc1_state_i | 4          | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc1_state_o | 6          | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc2_state_i | 8          | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |

|                |    |   |      |      |     |                                                                                                |
|----------------|----|---|------|------|-----|------------------------------------------------------------------------------------------------|
| proc2_state_o  | 10 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc3_state_i  | 12 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc3_state_o  | 14 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc4_state_i  | 16 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc4_state_o  | 18 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc5_state_i  | 20 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc5_state_o  | 22 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc6_state_i  | 24 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc6_state_o  | 26 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc7_state_i  | 28 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc7_state_o  | 30 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc8_state_i  | 32 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc8_state_o  | 34 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc9_state_i  | 36 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc9_state_o  | 38 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc10_state_i | 40 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc10_state_o | 42 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc11_state_i | 44 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc11_state_o | 46 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |

|                |    |   |      |      |     |                                                                                                |
|----------------|----|---|------|------|-----|------------------------------------------------------------------------------------------------|
| proc12_state_i | 48 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc12_state_o | 50 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc13_state_i | 52 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc13_state_o | 54 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc14_state_i | 56 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc14_state_o | 58 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc15_state_i | 60 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc15_state_o | 62 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |

### sc\_ras\_tpm\_main\_proc\_values\_0

**Address:** 0x421A06B0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Provides information on the details of the processor not available in the status registers and unique to the processor's selected function: pass-thru: reserved, leaky bucket: current counter value, LFSR: current LFSR value, Table: last allocated value, Self-test: FSM state identifier

| Field Name  | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|-------------|------------|-----------|----------|--------|-------------|-------------|
| proc0_value | 0          | 16        | false    | WARL   | 0x0         |             |
| proc1_value | 16         | 16        | false    | WARL   | 0x0         |             |
| proc2_value | 32         | 16        | false    | WARL   | 0x0         |             |
| proc3_value | 48         | 16        | false    | WARL   | 0x0         |             |

### scb\_ras\_inj\_hdr

**Address:** 0x421A0DC0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Fault Injection Header

| Field Name      | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                           |
|-----------------|------------|-----------|----------|--------|-------------|-----------------------------------------------------------------------|
| revision_min or | 0          | 4         | false    | WARL   | 0x0         | Indicates that the meaning or behavior of some registers have changed |

|                  |    |    |       |      |     |                                                                                                                                                                                         |
|------------------|----|----|-------|------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| revision_major   | 4  | 4  | false | WARL | 0x0 | Indicates that the organization and/or meaning of later fields and/or registers have changed                                                                                            |
| data_w           | 8  | 4  | false | WARL | 0x0 | Indicates the number of bits, as a power of 2, of the corruption vectors associated with the injection module                                                                           |
| check_w          | 12 | 4  | false | WARL | 0x0 | Indicates the number of bits, as a power of 2, of the corruption vectors associated with the injection module                                                                           |
| presentations    | 16 | 16 | false | WARL | 0x0 | Bit vector of which data and check corruption vector presentations are available                                                                                                        |
| pre_processors   | 32 | 4  | false | WARL | 0x0 | Encodes the number of pre-processor instances (and hence total available trigger count)                                                                                                 |
| main_processors  | 36 | 4  | false | WARL | 0x0 | Encodes the number of main-processors                                                                                                                                                   |
| post_processors  | 40 | 4  | false | WARL | 0x0 | Encodes the number of post-processors (and hence the total available trigger outputs)                                                                                                   |
| rsvd_processors  | 44 | 4  | false | WARL | 0x0 | Reserved encoding for processors                                                                                                                                                        |
| injector_enabled | 48 | 1  | false | WARL | 0x0 | The injector is enabled and will inject faults when the trigger is asserted and processed according to the provided configurations and setup. This is an overall gate for the injector. |
| rsvd63_49        | 49 | 15 | false | WARL | 0x0 |                                                                                                                                                                                         |

### scb\_ras\_inj\_init

**Address:** 0x421A0DC8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Initialisation: write 1 to initialize behavior for the associated Trigger Processor Module and any injector internal state

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| pre        | 0          | 16        | false    | WARL   | 0x0         |             |
| main       | 16         | 16        | false    | WARL   | 0x0         |             |
| post       | 32         | 16        | false    | WARL   | 0x0         |             |
| injector   | 48         | 16        | false    | WARL   | 0x0         |             |

### scb\_ras\_inj\_data\_match\_0

**Address:** 0x421A0DD0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Trigger Data Match Value 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|       |   |    |       |      |     |  |
|-------|---|----|-------|------|-----|--|
| match | 0 | 64 | false | WARL | 0x0 |  |
|-------|---|----|-------|------|-----|--|

### scb\_ras\_inj\_data\_mask\_0

**Address:** 0x421A0DD8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Trigger Data Match Mask 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | false    | WARL   | 0x0         |             |

### scb\_ras\_inj\_data\_match\_1

**Address:** 0x421A0DE0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Trigger Data Match Value 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| match      | 0          | 64        | false    | WARL   | 0x0         |             |

### scb\_ras\_inj\_data\_mask\_1

**Address:** 0x421A0DE8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Trigger Data Match Mask 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | false    | WARL   | 0x0         |             |

### scb\_ras\_inj\_data\_match\_2

**Address:** 0x421A0DF0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Trigger Data Match Value 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| match      | 0          | 64        | false    | WARL   | 0x0         |             |

### scb\_ras\_inj\_data\_mask\_2

**Address:** 0x421A0DF8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Trigger Data Match Mask 2

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| mask       | 0          | 64        | false    | WARL   | 0x0         |             |

## scb\_ras\_inj\_0\_1\_cfg

**Address:** 0x421A0E00

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Injection Configuration register 0-1. Injector 0 if for array writes (encode), and Injector 1 is for array reads (decode)

| Field Name               | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                           |
|--------------------------|------------|-----------|----------|--------|-------------|-----------------------------------------------------------------------------------------------------------------------|
| inj0_acb_rbuff_rd_select | 0          | 1         | false    | WARL   | 0x0         | Enables ACB rdat buffer read (decode) to receive the injection when the trigger occurs.                               |
| inj0_acb_rbuff_wr_select | 1          | 1         | false    | WARL   | 0x0         | Enables ACB rdat buffer write (encode) to receive the injection when the trigger occurs.                              |
| inj0_structur<br>e_rsvd  | 2          | 6         | false    | WARL   | 0x0         | Identifies which of the structures associated with the injector are to receive the injection when the trigger occurs. |
| inj0_rsvd                | 8          | 8         | false    | WARL   | 0x0         |                                                                                                                       |
| inj0_c_data_<br>pld      | 16         | 4         | false    | WARL   | 0x0         | Identifies what is to be presented to the encode or decode block selected by structure                                |
| inj0_c_data_<br>pt       | 20         | 4         | false    | WARL   | 0x0         | Determines how the corrupt data pld field is presented to the encode or decode block selected by structure.           |
| inj0_c_check_<br>pld     | 24         | 4         | false    | WARL   | 0x0         | Identifies what is to be presented to the encode or decode block selected by structure                                |
| inj0_c_check_<br>pt      | 28         | 4         | false    | WARL   | 0x0         | Determines how the corrupt check pld field is presented to the encode or decode block selected by structure.          |
| inj1_acb_wbuff_rd_select | 32         | 1         | false    | WARL   | 0x0         | Enables ACB wdat buffer read (decode) to receive the injection when the trigger occurs.                               |
| inj1_acb_wbuff_wr_select | 33         | 1         | false    | WARL   | 0x0         | Enables ACB wdat buffer write (encode) to receive the injection when the trigger occurs.                              |
| inj1_structur<br>e_rsvd  | 34         | 6         | false    | WARL   | 0x0         | Identifies which of the structures associated with the injector are to receive the injection when the trigger occurs. |
| inj1_rsvd                | 40         | 8         | false    | WARL   | 0x0         |                                                                                                                       |
| inj1_c_data_<br>pld      | 48         | 4         | false    | WARL   | 0x0         | Identifies what is to be presented to the encode or decode block selected by structure                                |

|                  |    |   |       |      |     |                                                                                                              |
|------------------|----|---|-------|------|-----|--------------------------------------------------------------------------------------------------------------|
| inj1_c_data_pt   | 52 | 4 | false | WARL | 0x0 | Determines how the corrupt data pld field is presented to the encode or decode block selected by structure.  |
| inj1_c_check_pld | 56 | 4 | false | WARL | 0x0 | Identifies what is to be presented to the encode or decode block selected by structure                       |
| inj1_c_check_pt  | 60 | 4 | false | WARL | 0x0 | Determines how the corrupt check pld field is presented to the encode or decode block selected by structure. |

### scb\_ras\_tpm\_pre\_proc\_cfg\_0

**Address:** 0x421A0E40

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Pre processor 0-7

| Field Name     | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                             |
|----------------|------------|-----------|----------|--------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc0_function | 0          | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc0_meta     | 2          | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc0_rsvd     | 4          | 4         | false    | WARL   | 0x0         |                                                                                                                                                                         |
| proc1_function | 8          | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc1_meta     | 10         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc1_rsvd     | 12         | 4         | false    | WARL   | 0x0         |                                                                                                                                                                         |
| proc2_function | 16         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc2_meta     | 18         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc2_rsvd     | 20         | 4         | false    | WARL   | 0x0         |                                                                                                                                                                         |
| proc3_function | 24         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc3_meta     | 26         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc3_rsvd     | 28         | 4         | false    | WARL   | 0x0         |                                                                                                                                                                         |
| proc4_function | 32         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |

|                |    |   |       |      |     |                                                                                                                                                                         |
|----------------|----|---|-------|------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc4_meta     | 34 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc4_rsvd     | 36 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc5_function | 40 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor: 0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                    |
| proc5_meta     | 42 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc5_rsvd     | 44 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc6_function | 48 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor: 0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                    |
| proc6_meta     | 50 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc6_rsvd     | 52 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc7_function | 56 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor: 0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                    |
| proc7_meta     | 58 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc7_rsvd     | 60 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |

### sch\_ras\_tpm\_pre\_proc\_cfg\_1

**Address:** 0x421A0E48

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Pre processor 8-15

| Field Name     | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                             |
|----------------|------------|-----------|----------|--------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc8_function | 0          | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor: 0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                    |
| proc8_meta     | 2          | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc8_rsvd     | 4          | 4         | false    | WARL   | 0x0         |                                                                                                                                                                         |
| proc9_function | 8          | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor: 0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                    |
| proc9_meta     | 10         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc9_rsvd     | 12         | 4         | false    | WARL   | 0x0         |                                                                                                                                                                         |

|                 |    |   |       |      |     |                                                                                                                                                                         |
|-----------------|----|---|-------|------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc10_function | 16 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc10_meta     | 18 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc10_rsvd     | 20 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc11_function | 24 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc11_meta     | 26 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc11_rsvd     | 28 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc12_function | 32 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc12_meta     | 34 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc12_rsvd     | 36 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc13_function | 40 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc13_meta     | 42 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc13_rsvd     | 44 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc14_function | 48 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc14_meta     | 50 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc14_rsvd     | 52 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |
| proc15_function | 56 | 2 | false | WARL | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: rising edge detector, 3: falling edge detector                                                 |
| proc15_meta     | 58 | 2 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, count: threshold as a power of 2 (i.e. 1,2,4, or 8), rising edge detector: ignored, falling edge detector: ignored |
| proc15_rsvd     | 60 | 4 | false | WARL | 0x0 |                                                                                                                                                                         |

### sch\_ras\_tpm\_main\_proc\_cfg\_0

**Address:** 0x421A0E50

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Main processor 0-3

| Field Name      | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|------------|-----------|----------|--------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc0_trigger_a | 0          | 4         | false    | WARL   | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc0_trigger_b | 4          | 4         | false    | WARL   | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc0_function  | 8          | 4         | false    | WARL   | 0x0         | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc0_meta      | 12         | 4         | false    | WARL   | 0x0         | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc1_trigger_a | 16         | 4         | false    | WARL   | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc1_trigger_b | 20         | 4         | false    | WARL   | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc1_function  | 24         | 4         | false    | WARL   | 0x0         | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc1_meta      | 28         | 4         | false    | WARL   | 0x0         | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc2_trigger_a | 32         | 4         | false    | WARL   | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc2_trigger_b | 36         | 4         | false    | WARL   | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |

|                 |    |   |       |      |     |                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|----|---|-------|------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc2_function  | 40 | 4 | false | WARL | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc2_meta      | 44 | 4 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc3_trigger_a | 48 | 4 | false | WARL | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc3_trigger_b | 52 | 4 | false | WARL | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc3_function  | 56 | 4 | false | WARL | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc3_meta      | 60 | 4 | false | WARL | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |

### scb\_ras\_tpm\_main\_proc\_cfg\_1

**Address:** 0x421A0E58

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Main processor 4-7

| Field Name      | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                            |
|-----------------|------------|-----------|----------|-----------|-------------|----------------------------------------------------------------------------------------------------------------------------------------|
| proc4_trigger_a | 0          | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires. |
| proc4_trigger_b | 4          | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires. |

|                 |    |   |       |           |     |                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|----|---|-------|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc4_function  | 8  | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc4_meta      | 12 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc5_trigger_a | 16 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc5_trigger_b | 20 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc5_function  | 24 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc5_meta      | 28 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc6_trigger_a | 32 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc6_trigger_b | 36 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc6_function  | 40 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc6_meta      | 44 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |

|                 |    |   |       |           |     |                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|----|---|-------|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc7_trigger_a | 48 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc7_trigger_b | 52 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc7_function  | 56 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc7_meta      | 60 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |

## sch\_ras\_tpm\_main\_proc\_cfg\_2

**Address:** 0x421A0E60

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Main processor 8-11

| Field Name      | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|------------|-----------|----------|-----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc8_trigger_a | 0          | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc8_trigger_b | 4          | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc8_function  | 8          | 4         | false    | Read-Only | 0x0         | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc8_meta      | 12         | 4         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc9_trigger_a | 16         | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |

|                  |    |   |       |           |     |                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|----|---|-------|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc9_trigger_b  | 20 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc9_function   | 24 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc9_meta       | 28 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc10_trigger_a | 32 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc10_trigger_b | 36 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc10_function  | 40 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc10_meta      | 44 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc11_trigger_a | 48 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc11_trigger_b | 52 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc11_function  | 56 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |

|             |    |   |       |           |     |                                                                                                                                                                                                                                           |
|-------------|----|---|-------|-----------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc11_meta | 60 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD |
|-------------|----|---|-------|-----------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### scb\_ras\_tpm\_main\_proc\_cfg\_3

**Address:** 0x421A0E68

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Main processor 12-15

| Field Name       | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|------------|-----------|----------|-----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc12_trigger_a | 0          | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc12_trigger_b | 4          | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc12_function  | 8          | 4         | false    | Read-Only | 0x0         | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc12_meta      | 12         | 4         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc13_trigger_a | 16         | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc13_trigger_b | 20         | 4         | false    | Read-Only | 0x0         | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc13_function  | 24         | 4         | false    | Read-Only | 0x0         | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |

|                  |    |   |       |           |     |                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|----|---|-------|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc13_meta      | 28 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc14_trigger_a | 32 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc14_trigger_b | 36 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc14_function  | 40 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc14_meta      | 44 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |
| proc15_trigger_a | 48 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.a for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc15_trigger_b | 52 | 4 | false | Read-Only | 0x0 | Encodes the selected pre-processor trigger.o to associated with trigger.b for this main-processor module. Pre-processor 0 never fires.                                                                                                                                                                                                                                                           |
| proc15_function  | 56 | 4 | false | Read-Only | 0x0 | Selects the function of the main-processor: 0: pass-thru 1: reserved, 2: leaky bucket (width set by parameter), 3: tracker (width set by parameter) using the data provided, 4: reserved - default to pass-thru, 5: LFSR (polynomials set by parameters) initialized by the data provided, 6: reserved - default to pass-thru, 7: self test (FSM steps through the test sequence when triggered) |
| proc15_meta      | 60 | 4 | false | Read-Only | 0x0 | Function specific configuration: pass-thru: ignored, leaky bucket: threshold as a power of 2, Table: mask selection from those defined at instantiation, LFSR: LFSR equation selector from those defined at instantiation, Self-test: TBD                                                                                                                                                        |

### sch\_ras\_tpm\_post\_proc\_cfg\_0

**Address:** 0x421A0E70

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Post processor 0-7

| Field Name     | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                        |
|----------------|------------|-----------|----------|--------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| proc0_function | 0          | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc0_meta     | 2          | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc0_rsvd     | 4          | 4         | false    | WARL   | 0x0         |                                                                                                                                                    |
| proc1_function | 8          | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc1_meta     | 10         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc1_rsvd     | 12         | 4         | false    | WARL   | 0x0         |                                                                                                                                                    |
| proc2_function | 16         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc2_meta     | 18         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc2_rsvd     | 20         | 4         | false    | WARL   | 0x0         |                                                                                                                                                    |
| proc3_function | 24         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc3_meta     | 26         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc3_rsvd     | 28         | 4         | false    | WARL   | 0x0         |                                                                                                                                                    |
| proc4_function | 32         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc4_meta     | 34         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc4_rsvd     | 36         | 4         | false    | WARL   | 0x0         |                                                                                                                                                    |
| proc5_function | 40         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc5_meta     | 42         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc5_rsvd     | 44         | 4         | false    | WARL   | 0x0         |                                                                                                                                                    |
| proc6_function | 48         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc6_meta     | 50         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc6_rsvd     | 52         | 4         | false    | WARL   | 0x0         |                                                                                                                                                    |
| proc7_function | 56         | 2         | false    | WARL   | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc7_meta     | 58         | 2         | false    | WARL   | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |

|            |    |   |       |      |     |  |
|------------|----|---|-------|------|-----|--|
| proc7_rsvd | 60 | 4 | false | WARL | 0x0 |  |
|------------|----|---|-------|------|-----|--|

## scb\_ras\_tpm\_post\_proc\_cfg\_1

**Address:** 0x421A0E78

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Configuration register of FIM-TPM Post processor 8-15

| Field Name      | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                        |
|-----------------|------------|-----------|----------|-----------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| proc8_function  | 0          | 2         | false    | Read-Only | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc8_meta      | 2          | 2         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc8_rsvd      | 4          | 4         | false    | Read-Only | 0x0         |                                                                                                                                                    |
| proc9_function  | 8          | 2         | false    | Read-Only | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc9_meta      | 10         | 2         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc9_rsvd      | 12         | 4         | false    | Read-Only | 0x0         |                                                                                                                                                    |
| proc10_function | 16         | 2         | false    | Read-Only | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc10_meta     | 18         | 2         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc10_rsvd     | 20         | 4         | false    | Read-Only | 0x0         |                                                                                                                                                    |
| proc11_function | 24         | 2         | false    | Read-Only | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc11_meta     | 26         | 2         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc11_rsvd     | 28         | 4         | false    | Read-Only | 0x0         |                                                                                                                                                    |
| proc12_function | 32         | 2         | false    | Read-Only | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc12_meta     | 34         | 2         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc12_rsvd     | 36         | 4         | false    | Read-Only | 0x0         |                                                                                                                                                    |
| proc13_function | 40         | 2         | false    | Read-Only | 0x0         | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc13_meta     | 42         | 2         | false    | Read-Only | 0x0         | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc13_rsvd     | 44         | 4         | false    | Read-Only | 0x0         |                                                                                                                                                    |

|                 |    |   |       |           |     |                                                                                                                                                    |
|-----------------|----|---|-------|-----------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------|
| proc14_function | 48 | 2 | false | Read-Only | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc14_meta     | 50 | 2 | false | Read-Only | 0x0 | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc14_rsvd     | 52 | 4 | false | Read-Only | 0x0 |                                                                                                                                                    |
| proc15_function | 56 | 2 | false | Read-Only | 0x0 | Selects the function of the pre-processor:<br>0: pass-thru, 1: count, 2: invert, 3: 1 shot                                                         |
| proc15_meta     | 58 | 2 | false | Read-Only | 0x0 | Function specific configuration: pass-thru:<br>ignored, count: threshold as a power of 2<br>(i.e. 1,2,4, or 8), pulse: ignored, 1 shot:<br>ignored |
| proc15_rsvd     | 60 | 4 | false | Read-Only | 0x0 |                                                                                                                                                    |

### scb\_ras\_tpm\_main\_proc\_capability\_0

**Address:** 0x421A0E80

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Capabilities possible with each of the main processors: 0-7

| Field Name     | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                               |
|----------------|------------|-----------|----------|--------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| proc0_pass     | 0          | 1         | true     | WARL   | 0x0         | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc0_leaky    | 1          | 1         | true     | WARL   | 0x0         | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc0_table    | 2          | 1         | true     | WARL   | 0x0         | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc0_lfsr     | 3          | 1         | true     | WARL   | 0x0         | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc0_selftest | 4          | 1         | true     | WARL   | 0x0         | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc0_rsvd     | 5          | 3         | true     | WARL   | 0x0         |                                                                                                                                           |
| proc1_pass     | 8          | 1         | true     | WARL   | 0x0         | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc1_leaky    | 9          | 1         | true     | WARL   | 0x0         | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc1_table    | 10         | 1         | true     | WARL   | 0x0         | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc1_lfsr     | 11         | 1         | true     | WARL   | 0x0         | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc1_selftest | 12         | 1         | true     | WARL   | 0x0         | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |

|                |    |   |      |      |     |                                                                                                                                           |
|----------------|----|---|------|------|-----|-------------------------------------------------------------------------------------------------------------------------------------------|
| proc1_rsvd     | 13 | 3 | true | WARL | 0x0 |                                                                                                                                           |
| proc2_pass     | 16 | 1 | true | WARL | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc2_leaky    | 17 | 1 | true | WARL | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc2_table    | 18 | 1 | true | WARL | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc2_lfsr     | 19 | 1 | true | WARL | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc2_selftest | 20 | 1 | true | WARL | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc2_rsvd     | 21 | 3 | true | WARL | 0x0 |                                                                                                                                           |
| proc3_pass     | 24 | 1 | true | WARL | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc3_leaky    | 25 | 1 | true | WARL | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc3_table    | 26 | 1 | true | WARL | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc3_lfsr     | 27 | 1 | true | WARL | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc3_selftest | 28 | 1 | true | WARL | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc3_rsvd     | 29 | 3 | true | WARL | 0x0 |                                                                                                                                           |
| proc4_pass     | 32 | 1 | true | WARL | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc4_leaky    | 33 | 1 | true | WARL | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc4_table    | 34 | 1 | true | WARL | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc4_lfsr     | 35 | 1 | true | WARL | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc4_selftest | 36 | 1 | true | WARL | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc4_rsvd     | 37 | 3 | true | WARL | 0x0 |                                                                                                                                           |
| proc5_pass     | 40 | 1 | true | WARL | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc5_leaky    | 41 | 1 | true | WARL | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |

|                |    |   |      |      |     |                                                                                                                                           |
|----------------|----|---|------|------|-----|-------------------------------------------------------------------------------------------------------------------------------------------|
| proc5_table    | 42 | 1 | true | WARL | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc5_lfsr     | 43 | 1 | true | WARL | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc5_selftest | 44 | 1 | true | WARL | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc5_rsvd     | 45 | 3 | true | WARL | 0x0 |                                                                                                                                           |
| proc6_pass     | 48 | 1 | true | WARL | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc6_leaky    | 49 | 1 | true | WARL | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc6_table    | 50 | 1 | true | WARL | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc6_lfsr     | 51 | 1 | true | WARL | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc6_selftest | 52 | 1 | true | WARL | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc6_rsvd     | 53 | 3 | true | WARL | 0x0 |                                                                                                                                           |
| proc7_pass     | 56 | 1 | true | WARL | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc7_leaky    | 57 | 1 | true | WARL | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc7_table    | 58 | 1 | true | WARL | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc7_lfsr     | 59 | 1 | true | WARL | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc7_selftest | 60 | 1 | true | WARL | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc7_rsvd     | 61 | 3 | true | WARL | 0x0 |                                                                                                                                           |

### scb\_ras\_tpm\_main\_proc\_capability\_1

**Address:** 0x421A0E88

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Capabilities possible with each of the main processors: 8-15

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                               |
|------------|------------|-----------|----------|-----------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| proc8_pass | 0          | 1         | true     | Read-Only | 0x0         | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |

|                 |    |   |      |           |     |                                                                                                                                           |
|-----------------|----|---|------|-----------|-----|-------------------------------------------------------------------------------------------------------------------------------------------|
| proc8_leaky     | 1  | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc8_table     | 2  | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc8_lfsr      | 3  | 1 | true | Read-Only | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc8_selftest  | 4  | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc8_rsvd      | 5  | 3 | true | Read-Only | 0x0 |                                                                                                                                           |
| proc9_pass      | 8  | 1 | true | Read-Only | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc9_leaky     | 9  | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc9_table     | 10 | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc9_lfsr      | 11 | 1 | true | Read-Only | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc9_selftest  | 12 | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc9_rsvd      | 13 | 3 | true | Read-Only | 0x0 |                                                                                                                                           |
| proc10_pass     | 16 | 1 | true | Read-Only | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc10_leaky    | 17 | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc10_table    | 18 | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc10_lfsr     | 19 | 1 | true | Read-Only | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc10_selftest | 20 | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc10_rsvd     | 21 | 3 | true | Read-Only | 0x0 |                                                                                                                                           |
| proc11_pass     | 24 | 1 | true | Read-Only | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc11_leaky    | 25 | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc11_table    | 26 | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc11_lfsr     | 27 | 1 | true | Read-Only | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |

|                 |    |   |      |           |     |                                                                                                                                           |
|-----------------|----|---|------|-----------|-----|-------------------------------------------------------------------------------------------------------------------------------------------|
| proc11_selftest | 28 | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc11_rsvd     | 29 | 3 | true | Read-Only | 0x0 |                                                                                                                                           |
| proc12_pass     | 32 | 1 | true | Read-Only | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc12_leaky    | 33 | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc12_table    | 34 | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc12_lfsr     | 35 | 1 | true | Read-Only | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc12_selftest | 36 | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc12_rsvd     | 37 | 3 | true | Read-Only | 0x0 |                                                                                                                                           |
| proc13_pass     | 40 | 1 | true | Read-Only | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc13_leaky    | 41 | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc13_table    | 42 | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc13_lfsr     | 43 | 1 | true | Read-Only | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc13_selftest | 44 | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc13_rsvd     | 45 | 3 | true | Read-Only | 0x0 |                                                                                                                                           |
| proc14_pass     | 48 | 1 | true | Read-Only | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |
| proc14_leaky    | 49 | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                                                            |
| proc14_table    | 50 | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.                                                  |
| proc14_lfsr     | 51 | 1 | true | Read-Only | 0x0 | The lfsr fuction where one or more LFSR polynomials are applied to determine the trigger output.                                          |
| proc14_selftest | 52 | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.                                           |
| proc14_rsvd     | 53 | 3 | true | Read-Only | 0x0 |                                                                                                                                           |
| proc15_pass     | 56 | 1 | true | Read-Only | 0x0 | The pass-thru function is provided. This is true for every instance fo the trigger main-processor since this is required to be supported. |

|                 |    |   |      |           |     |                                                                                                   |
|-----------------|----|---|------|-----------|-----|---------------------------------------------------------------------------------------------------|
| proc15_leaky    | 57 | 1 | true | Read-Only | 0x0 | The leaky bucket function where trigger.a increments and trigger.b decrements.                    |
| proc15_table    | 58 | 1 | true | Read-Only | 0x0 | The table function where data provided with the trigger.a input is allocated to a table.          |
| proc15_lfsr     | 59 | 1 | true | Read-Only | 0x0 | The lfsr function where one or more LFSR polynomials are applied to determine the trigger output. |
| proc15_selftest | 60 | 1 | true | Read-Only | 0x0 | The self-test function that sequences through a self-test sequence at each trigger.a assertion.   |
| proc15_rsvd     | 61 | 3 | true | Read-Only | 0x0 |                                                                                                   |

## sch\_ras\_tpm\_pre\_proc\_sts

**Address:** 0x421A0E90

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** TPM Pre processor status: 0-15

| Field Name    | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                    |
|---------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------------------------------------------|
| proc0_state_i | 0          | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc0_state_o | 2          | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc1_state_i | 4          | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc1_state_o | 6          | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc2_state_i | 8          | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc2_state_o | 10         | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc3_state_i | 12         | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc3_state_o | 14         | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc4_state_i | 16         | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc4_state_o | 18         | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc5_state_i | 20         | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |

|                |    |   |      |      |     |                                                                                                |
|----------------|----|---|------|------|-----|------------------------------------------------------------------------------------------------|
| proc5_state_o  | 22 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc6_state_i  | 24 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc6_state_o  | 26 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc7_state_i  | 28 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc7_state_o  | 30 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc8_state_i  | 32 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc8_state_o  | 34 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc9_state_i  | 36 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc9_state_o  | 38 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc10_state_i | 40 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc10_state_o | 42 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc11_state_i | 44 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc11_state_o | 46 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc12_state_i | 48 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc12_state_o | 50 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc13_state_i | 52 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc13_state_o | 54 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc14_state_i | 56 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc14_state_o | 58 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |

|                |    |   |      |      |     |                                                                                                |
|----------------|----|---|------|------|-----|------------------------------------------------------------------------------------------------|
| proc15_state_i | 60 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc15_state_o | 62 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |

### scb\_ras\_tpm\_main\_proc\_sts\_0

**Address:** 0x421A0E98

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** TPM Main processor status: 0-7

| Field Name       | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                                                                           |
|------------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc0_state_a    | 0          | 2         | true     | WARL   | 0x0         | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc0_state_b    | 2          | 2         | true     | WARL   | 0x0         | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc0_state_meta | 4          | 2         | true     | WARL   | 0x0         | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc0_state_o    | 6          | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc1_state_a    | 8          | 2         | true     | WARL   | 0x0         | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc1_state_b    | 10         | 2         | true     | WARL   | 0x0         | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc1_state_meta | 12         | 2         | true     | WARL   | 0x0         | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc1_state_o    | 14         | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc2_state_a    | 16         | 2         | true     | WARL   | 0x0         | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc2_state_b    | 18         | 2         | true     | WARL   | 0x0         | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc2_state_meta | 20         | 2         | true     | WARL   | 0x0         | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |

|                  |    |   |      |      |     |                                                                                                                                                                                                                                       |
|------------------|----|---|------|------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc2_state_o    | 22 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc3_state_a    | 24 | 2 | true | WARL | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc3_state_b    | 26 | 2 | true | WARL | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc3_state_meta | 28 | 2 | true | WARL | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc3_state_o    | 30 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc4_state_a    | 32 | 2 | true | WARL | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc4_state_b    | 34 | 2 | true | WARL | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc4_state_meta | 36 | 2 | true | WARL | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc4_state_o    | 38 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc5_state_a    | 40 | 2 | true | WARL | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc5_state_b    | 42 | 2 | true | WARL | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc5_state_meta | 44 | 2 | true | WARL | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc5_state_o    | 46 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc6_state_a    | 48 | 2 | true | WARL | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc6_state_b    | 50 | 2 | true | WARL | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc6_state_meta | 52 | 2 | true | WARL | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |

|                  |    |   |      |      |     |                                                                                                                                                                                                                                       |
|------------------|----|---|------|------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc6_state_o    | 54 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc7_state_a    | 56 | 2 | true | WARL | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc7_state_b    | 58 | 2 | true | WARL | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc7_state_meta | 60 | 2 | true | WARL | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc7_state_o    | 62 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |

### scb\_ras\_tpm\_main\_proc\_sts\_1

**Address:** 0x421A0EA0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** TPM Main processor status: 8-15

| Field Name       | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                                           |
|------------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc8_state_a    | 0          | 2         | true     | Read-Only | 0x0         | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc8_state_b    | 2          | 2         | true     | Read-Only | 0x0         | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc8_state_meta | 4          | 2         | true     | Read-Only | 0x0         | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc8_state_o    | 6          | 2         | true     | Read-Only | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc9_state_a    | 8          | 2         | true     | Read-Only | 0x0         | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc9_state_b    | 10         | 2         | true     | Read-Only | 0x0         | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc9_state_meta | 12         | 2         | true     | Read-Only | 0x0         | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc9_state_o    | 14         | 2         | true     | Read-Only | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |

|                   |    |   |      |           |     |                                                                                                                                                                                                                                       |
|-------------------|----|---|------|-----------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc10_state_a    | 16 | 2 | true | Read-Only | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc10_state_b    | 18 | 2 | true | Read-Only | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc10_state_meta | 20 | 2 | true | Read-Only | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc10_state_o    | 22 | 2 | true | Read-Only | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc11_state_a    | 24 | 2 | true | Read-Only | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc11_state_b    | 26 | 2 | true | Read-Only | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc11_state_meta | 28 | 2 | true | Read-Only | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc11_state_o    | 30 | 2 | true | Read-Only | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc12_state_a    | 32 | 2 | true | Read-Only | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc12_state_b    | 34 | 2 | true | Read-Only | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc12_state_meta | 36 | 2 | true | Read-Only | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc12_state_o    | 38 | 2 | true | Read-Only | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc13_state_a    | 40 | 2 | true | Read-Only | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc13_state_b    | 42 | 2 | true | Read-Only | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc13_state_meta | 44 | 2 | true | Read-Only | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc13_state_o    | 46 | 2 | true | Read-Only | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |

|                   |    |   |      |           |     |                                                                                                                                                                                                                                       |
|-------------------|----|---|------|-----------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proc14_state_a    | 48 | 2 | true | Read-Only | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc14_state_b    | 50 | 2 | true | Read-Only | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc14_state_meta | 52 | 2 | true | Read-Only | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc14_state_o    | 54 | 2 | true | Read-Only | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |
| proc15_state_a    | 56 | 2 | true | Read-Only | 0x0 | The current state of the trigger.a input [0] is set to 1 once trigger.a asserts. [1] reserved                                                                                                                                         |
| proc15_state_b    | 58 | 2 | true | Read-Only | 0x0 | The current state of the trigger.b input [0] is set to 1 once trigger.b asserts. [1] reserved                                                                                                                                         |
| proc15_state_meta | 60 | 2 | true | Read-Only | 0x0 | The meta field takes on meaning according to the function configured for the main processor, leaky-bucket: 0: empty 1: mid 2: full 3: reserved, table: 0: empty 1: mid 2: full 3: reserved, lfsr: 0: reserved, Self-Test: 0: reserved |
| proc15_state_o    | 62 | 2 | true | Read-Only | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved                                                                                                                                        |

### scb\_ras\_tpm\_post\_proc\_sts

**Address:** 0x421A0EA8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** TPM Post processor status: 0-15

| Field Name    | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                    |
|---------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------------------------------------------|
| proc0_state_i | 0          | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc0_state_o | 2          | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc1_state_i | 4          | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc1_state_o | 6          | 2         | true     | WARL   | 0x0         | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc2_state_i | 8          | 2         | true     | WARL   | 0x0         | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |

|                |    |   |      |      |     |                                                                                                |
|----------------|----|---|------|------|-----|------------------------------------------------------------------------------------------------|
| proc2_state_o  | 10 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc3_state_i  | 12 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc3_state_o  | 14 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc4_state_i  | 16 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc4_state_o  | 18 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc5_state_i  | 20 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc5_state_o  | 22 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc6_state_i  | 24 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc6_state_o  | 26 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc7_state_i  | 28 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc7_state_o  | 30 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc8_state_i  | 32 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc8_state_o  | 34 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc9_state_i  | 36 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc9_state_o  | 38 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc10_state_i | 40 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc10_state_o | 42 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc11_state_i | 44 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc11_state_o | 46 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |

|                |    |   |      |      |     |                                                                                                |
|----------------|----|---|------|------|-----|------------------------------------------------------------------------------------------------|
| proc12_state_i | 48 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc12_state_o | 50 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc13_state_i | 52 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc13_state_o | 54 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc14_state_i | 56 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc14_state_o | 58 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |
| proc15_state_i | 60 | 2 | true | WARL | 0x0 | The current state of the trigger.i input [0] is set to 1 once trigger.i asserts. [1] reserved  |
| proc15_state_o | 62 | 2 | true | WARL | 0x0 | The current state of the trigger.o output [0] is set to 1 once trigger.o asserts. [1] reserved |

### scb\_ras\_tpm\_main\_proc\_values\_0

**Address:** 0x421A0EB0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Provides information on the details of the processor not available in the status registers and unique to the processor's selected function: pass-thru: reserved, leaky bucket: current counter value, LFSR: current LFSR value, Table: last allocated value, Self-test: FSM state identifier

| Field Name  | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|-------------|------------|-----------|----------|--------|-------------|-------------|
| proc0_value | 0          | 16        | false    | WARL   | 0x0         |             |
| proc1_value | 16         | 16        | false    | WARL   | 0x0         |             |
| proc2_value | 32         | 16        | false    | WARL   | 0x0         |             |
| proc3_value | 48         | 16        | false    | WARL   | 0x0         |             |

### scb\_hardware\_assertion

**Address:** 0x421A0EC0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Used to log SCB hardware assertion fails

| Field Name  | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                   |
|-------------|------------|-----------|----------|-----------|-------------|-----------------------------------------------|
| hwatype     | 0          | 10        | true     | Read-Only | 0x0         | HWA identifier. 0 indicates no failure        |
| timestamp_v | 10         | 1         | true     | Read-Only | 0x0         | If 1, indicates that timestamp field is valid |

|           |    |    |      |           |     |                                                                                                                            |
|-----------|----|----|------|-----------|-----|----------------------------------------------------------------------------------------------------------------------------|
| timestamp | 11 | 27 | true | Read-Only | 0x0 | RAS timestamp stub of when assertion failure occurred. To directly compare to logged RERI timestamps, left shift by 5 bits |
| rsvd      | 38 | 26 | true | Read-Only | 0x0 |                                                                                                                            |

### scb\_error\_header\_vendor

**Address:** 0x421A0F00

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000821

**Description:** Vendor and implementation ID

| Field Name          | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                 |
|---------------------|------------|-----------|----------|-----------|-------------|-------------------------------------------------------------------------------------------------------------|
| vendor_id           | 0          | 32        | false    | Read-Only | 0x821       | Provides the JEDEC manufacturer ID of the provider of the component hosting the error bank                  |
| imp_id              | 32         | 16        | false    | Read-Only | 0x0         | Provides a unique identity, defined by the vendor, to identify the component and revisions of the component |
| wpri_reserved_63_48 | 48         | 16        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                |

### scb\_error\_header\_bank\_info

**Address:** 0x421A0F08

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0100000000420000

**Description:** Error bank information

| Field Name          | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                 |
|---------------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| inst_id             | 0          | 16        | false    | Read-Only | 0x0         | Identifies a unique instance of an error bank within a package or die, ideally in the whole system. Not being used, tied to 0                                                               |
| n_err_recs          | 16         | 6         | false    | Read-Only | 0x2         | Indicates the number of error records implemented by the error bank                                                                                                                         |
| layout              | 22         | 2         | false    | Read-Only | 0x1         | Indicates the layout of the registers in the error bank and the error records. The layout encoding 0 indicates the registers are arranged and have meaning as defined by the specification. |
| wpri_reserved_55_24 | 24         | 32        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                |
| version             | 56         | 8         | false    | Read-Only | 0x1         | Version of the arch register layout specification implemented by the error bank                                                                                                             |

### scb\_error\_header\_valid\_summary

**Address:** 0x421A0F10

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000001

**Description:** Summary of valid error records

| Field Name   | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                    |
|--------------|------------|-----------|----------|-----------|-------------|------------------------------------------------------------------------------------------------|
| sv           | 0          | 1         | false    | Read-Only | 0x1         | Indicates that the valid_bitmap provides a summary of the valid bits from the status registers |
| valid_bitmap | 1          | 63        | true     | Read-Only | 0x0         | Summary of the valid bits from status registers in the error bank                              |

### scb\_error\_header\_reserved\_0

**Address:** 0x421A0F18

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Reserved for future RISC-V RERI standard use

| Field Name    | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                  |
|---------------|------------|-----------|----------|-----------|-------------|----------------------------------------------|
| reserved_63_0 | 0          | 64        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use |

### scb\_error\_header\_reserved\_1

**Address:** 0x421A0F20

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Reserved for future RISC-V RERI standard use

| Field Name    | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                  |
|---------------|------------|-----------|----------|-----------|-------------|----------------------------------------------|
| reserved_63_0 | 0          | 64        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use |

### scb\_error\_header\_reserved\_2

**Address:** 0x421A0F28

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Reserved for future RISC-V RERI standard use

| Field Name    | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                  |
|---------------|------------|-----------|----------|-----------|-------------|----------------------------------------------|
| reserved_63_0 | 0          | 64        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use |

### scb\_error\_header\_reserved\_3

**Address:** 0x421A0F30

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Reserved for future RISC-V RERI standard use

| Field Name    | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                  |
|---------------|------------|-----------|----------|-----------|-------------|----------------------------------------------|
| reserved_63_0 | 0          | 64        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use |

### scb\_error\_header\_custom\_0

**Address:** 0x421A0F38

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Reserved for custom use

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description             |
|------------|------------|-----------|----------|-----------|-------------|-------------------------|
| unused     | 0          | 64        | false    | Read-Only | 0x0         | Reserved for custom use |

### scb\_error\_record\_0\_control

**Address:** 0x421A0F40

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000001

**Description:** Control register of error record 0

| Field Name         | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                        |
|--------------------|------------|-----------|----------|-----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| else               | 0          | 1         | false    | WARL      | 0x1         | If 1, the hardware unit logs and signals errors in the error record. If 0, the hardware unit will not log RAS errors                                                                                               |
| cece               | 1          | 1         | false    | WARL      | 0x0         | If 1, hardware will count corrected errors in the cec field of the status register. If 0, cec does not count and retains its value.                                                                                |
| ces                | 2          | 2         | false    | WARL      | 0x0         | Used to enable signaling of corrected errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal.            |
| ueds               | 4          | 2         | false    | WARL      | 0x0         | Used to enable signaling of uncorrected deferred errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| uecs               | 6          | 2         | false    | WARL      | 0x0         | Used to enable signaling of uncorrected critical errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| wpri_reserved_31_8 | 8          | 24        | false    | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| eid                | 32         | 16        | true     | WARL      | 0x0         | Error injection delay                                                                                                                                                                                              |
| sinv               | 48         | 1         | true     | WARL      | 0x0         | Status register invalidate. When written to 1, causes the valid (v) field of the status register to be cleared.                                                                                                    |

|                     |    |    |       |           |     |                                                                        |
|---------------------|----|----|-------|-----------|-----|------------------------------------------------------------------------|
| srdp                | 49 | 1  | true  | WARL      | 0x0 | Set read in progress. When 1, causes read in progress (rdip) to be set |
| wpri_reserved_55_48 | 50 | 10 | false | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                           |
| custom_unused       | 60 | 4  | false | Read-Only | 0x0 | Reserved                                                               |

### sch\_error\_record\_0\_status

**Address:** 0x421A0F48

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Status register of error record 0

| Field Name          | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                            |
|---------------------|------------|-----------|----------|-----------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v                   | 0          | 1         | true     | WARL      | 0x0         | If 1, error record holds a valid error                                                                                                                                                 |
| ce                  | 1          | 1         | true     | WARL      | 0x0         | Set to 1 if the detected error was corrected                                                                                                                                           |
| ued                 | 2          | 1         | true     | WARL      | 0x0         | Set to 1 if the error could not be corrected but was deferred                                                                                                                          |
| uec                 | 3          | 1         | true     | WARL      | 0x0         | Set to 1 if the error could not be corrected or deferred and needs critical handling by RAS handler                                                                                    |
| pri                 | 4          | 2         | true     | WARL      | 0x0         | Priority of the error. Indicates relative priority among errors of the same class.                                                                                                     |
| mo                  | 6          | 1         | true     | WARL      | 0x0         | Set to 1 if more errors of the same class and severity occur                                                                                                                           |
| c                   | 7          | 1         | true     | WARL      | 0x0         | Containable bit. If 1, indicates that the error has not propagated beyond the boundaries of the hardware unit                                                                          |
| tt                  | 8          | 3         | true     | WARL      | 0x0         | Transaction type. 0=Unspecified or NA, 1=UNSPECIFIED, 2-3=Reserved for future standard use, 4=Explicit read, 5=Explicit write, 6=Implicit read, 7=Implicit write                       |
| iv                  | 11         | 1         | true     | WARL      | 0x0         | Information valid, set to 1 if error reports additional information in the info register                                                                                               |
| at                  | 12         | 4         | true     | WARL      | 0x0         | Address type, indicates the type of address reported in the addr register. 0=None, 1=Supervisor Physical Address, 2=Guest Physical Address, 3=Virtual Address, 4-15=Component Specific |
| siv                 | 16         | 1         | true     | WARL      | 0x0         | Supplemental information valid, set to 1 if error reports supplemental information in the suppl_info register                                                                          |
| tsv                 | 17         | 1         | true     | WARL      | 0x0         | Timestamp valid, set to 1 if the error record holds a valid timestamp in the timestamp register                                                                                        |
| lv                  | 18         | 1         | true     | WARL      | 0x0         | Locator valid, set to 1 if the error record holds a valid timestamp in the locator register                                                                                            |
| wpri_reserved_19_19 | 19         | 1         | true     | Read-Only | 0x0         | Reserved for future RISC-V RERI standard use                                                                                                                                           |

|                     |    |    |      |           |     |                                                                                                                                                                                                               |
|---------------------|----|----|------|-----------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| scrub               | 20 | 1  | true | WARL      | 0x0 | Set to 1 to indicate that the storage location has been updated with the corrected value. Only valid for corrected errors.                                                                                    |
| ceco                | 21 | 1  | true | WARL      | 0x0 | Corrected error counter overflow                                                                                                                                                                              |
| wpri_reserved_22_22 | 22 | 1  | true | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                  |
| rdip                | 23 | 1  | true | WARL      | 0x0 | Read in progress. Set to 1 when a new error is recorded in an invalid status register. Cleared to 0 when a valid status is overwritten. When control.sinv is set, status.v is cleared to 0 only if rdip is 1. |
| ec                  | 24 | 8  | true | WARL      | 0x0 | Error code, provides a description of the detected error. Encodings defined in section 2.7 of RISC-V RERI spec                                                                                                |
| le                  | 32 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the locator register value                                                                                                                               |
| ie                  | 36 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the information register value                                                                                                                           |
| se                  | 40 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the supplemental info register value                                                                                                                     |
| te                  | 44 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the timestamp register value                                                                                                                             |
| cec                 | 48 | 16 | true | WARL      | 0x0 | Corrected error counter. Incremented on each corrected error if cece is set to 1.                                                                                                                             |

### sch\_error\_record\_0\_addr

**Address:** 0x421A0F50

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Address register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                |
|------------|------------|-----------|----------|--------|-------------|--------------------------------------------|
| address    | 0          | 64        | true     | WARL   | 0x0         | Address associated with the detected error |

### sch\_error\_record\_0\_info

**Address:** 0x421A0F58

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Information register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                           |
|------------|------------|-----------|----------|--------|-------------|---------------------------------------|
| info       | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom error information |

### sch\_error\_record\_0\_suppl\_info

**Address:** 0x421A0F60

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Supplemental information register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                        |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------------|
| suppl_info | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom supplemental error information |

### sch\_error\_record\_0\_timestamp

**Address:** 0x421A0F68

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Timestamp register of error record 0

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                   |
|------------|------------|-----------|----------|--------|-------------|-------------------------------|
| timestamp  | 0          | 64        | true     | WARL   | 0x0         | Timestamp of the error record |

### sch\_error\_record\_0\_locator

**Address:** 0x421A0F70

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Locator register of error record 0

| Field Name  | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                          |
|-------------|------------|-----------|----------|--------|-------------|--------------------------------------------------------------------------------------|
| description | 0          | 32        | true     | WARL   | 0x0         | Contains description of the error event telling what happened [descriptor, instance] |
| location    | 32         | 32        | true     | WARL   | 0x0         | Contains location of the error event telling where it happened [chip, ip, instance]  |

### sch\_error\_record\_1\_control

**Address:** 0x421A0F80

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000001

**Description:** Control register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                         |
|------------|------------|-----------|----------|--------|-------------|-------------------------------------------------------------------------------------------------------------------------------------|
| else       | 0          | 1         | false    | WARL   | 0x1         | If 1, the hardware unit logs and signals errors in the error record. If 0, the hardware unit will not log RAS errors                |
| cece       | 1          | 1         | false    | WARL   | 0x0         | If 1, hardware will count corrected errors in the cec field of the status register. If 0, cec does not count and retains its value. |

|                     |    |    |       |           |     |                                                                                                                                                                                                                    |
|---------------------|----|----|-------|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ces                 | 2  | 2  | false | WARL      | 0x0 | Used to enable signaling of corrected errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal.            |
| ueds                | 4  | 2  | false | WARL      | 0x0 | Used to enable signaling of uncorrected deferred errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| uecs                | 6  | 2  | false | WARL      | 0x0 | Used to enable signaling of uncorrected critical errors. If 0, disabled. If 1, signal using low-priority RAS signal. If 2, signal using high-priority RAS signal. If 3, signal using platform specific RAS signal. |
| wpri_reserved_31_8  | 8  | 24 | false | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| eid                 | 32 | 16 | true  | WARL      | 0x0 | Error injection delay                                                                                                                                                                                              |
| sinv                | 48 | 1  | true  | WARL      | 0x0 | Status register invalidate. When written to 1, causes the valid (v) field of the status register to be cleared.                                                                                                    |
| srdp                | 49 | 1  | true  | WARL      | 0x0 | Set read in progress. When 1, causes read in progress (rdip) to be set                                                                                                                                             |
| wpri_reserved_55_48 | 50 | 10 | false | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                       |
| custom_unused       | 60 | 4  | false | Read-Only | 0x0 | Reserved                                                                                                                                                                                                           |

## sch\_error\_record\_1\_status

**Address:** 0x421A0F88

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Status register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                      |
|------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v          | 0          | 1         | true     | WARL   | 0x0         | If 1, error record holds a valid error                                                                                                                           |
| ce         | 1          | 1         | true     | WARL   | 0x0         | Set to 1 if the detected error was corrected                                                                                                                     |
| ued        | 2          | 1         | true     | WARL   | 0x0         | Set to 1 if the error could not be corrected but was deferred                                                                                                    |
| uec        | 3          | 1         | true     | WARL   | 0x0         | Set to 1 if the error could not be corrected or deferred and needs critical handling by RAS handler                                                              |
| pri        | 4          | 2         | true     | WARL   | 0x0         | Priority of the error. Indicates relative priority among errors of the same class.                                                                               |
| mo         | 6          | 1         | true     | WARL   | 0x0         | Set to 1 if more errors of the same class and severity occur                                                                                                     |
| c          | 7          | 1         | true     | WARL   | 0x0         | Containable bit. If 1, indicates that the error has not propagated beyond the boundaries of the hardware unit                                                    |
| tt         | 8          | 3         | true     | WARL   | 0x0         | Transaction type. 0=Unspecified or NA, 1=UNSPECIFIED, 2-3=Reserved for future standard use, 4=Explicit read, 5=Explicit write, 6=Implicit read, 7=Implicit write |

|                     |    |    |      |           |     |                                                                                                                                                                                                               |
|---------------------|----|----|------|-----------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| iv                  | 11 | 1  | true | WARL      | 0x0 | Information valid, set to 1 if error reports additional information in the info register                                                                                                                      |
| at                  | 12 | 4  | true | WARL      | 0x0 | Address type, indicates the type of address reported in the addr register. 0=None, 1=Supervisor Physical Address, 2=Guest Physical Address, 3=Virtual Address, 4-15=Component Specific                        |
| siv                 | 16 | 1  | true | WARL      | 0x0 | Supplemental information valid, set to 1 if error reports supplemental information in the suppl_info register                                                                                                 |
| tsv                 | 17 | 1  | true | WARL      | 0x0 | Timestamp valid, set to 1 if the error record holds a valid timestamp in the timestamp register                                                                                                               |
| lv                  | 18 | 1  | true | WARL      | 0x0 | Locator valid, set to 1 if the error record holds a valid timestamp in the locator register                                                                                                                   |
| wpri_reserved_19_19 | 19 | 1  | true | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                  |
| scrub               | 20 | 1  | true | WARL      | 0x0 | Set to 1 to indicate that the storage location has been updated with the corrected value. Only valid for corrected errors.                                                                                    |
| ceco                | 21 | 1  | true | WARL      | 0x0 | Corrected error counter overflow                                                                                                                                                                              |
| wpri_reserved_22_22 | 22 | 1  | true | Read-Only | 0x0 | Reserved for future RISC-V RERI standard use                                                                                                                                                                  |
| rdip                | 23 | 1  | true | WARL      | 0x0 | Read in progress. Set to 1 when a new error is recorded in an invalid status register. Cleared to 0 when a valid status is overwritten. When control.sinv is set, status.v is cleared to 0 only if rdip is 1. |
| ec                  | 24 | 8  | true | WARL      | 0x0 | Error code, provides a description of the detected error. Encodings defined in section 2.7 of RISC-V RERI spec. For memc, what values to use?                                                                 |
| le                  | 32 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the locator register value                                                                                                                               |
| ie                  | 36 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the information register value                                                                                                                           |
| se                  | 40 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the supplemental info register value                                                                                                                     |
| te                  | 44 | 4  | true | WARL      | 0x0 | The encoding that describes how to interpret/process the timestamp register value                                                                                                                             |
| cec                 | 48 | 16 | true | WARL      | 0x0 | Corrected error counter. Incremented on each corrected error if cece is set to 1.                                                                                                                             |

### scb\_error\_record\_1\_addr

**Address:** 0x421A0F90

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Address register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|         |   |    |      |      |     |                                            |
|---------|---|----|------|------|-----|--------------------------------------------|
| address | 0 | 64 | true | WARL | 0x0 | Address associated with the detected error |
|---------|---|----|------|------|-----|--------------------------------------------|

### sch\_error\_record\_1\_info

**Address:** 0x421A0F98

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Information register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                           |
|------------|------------|-----------|----------|--------|-------------|---------------------------------------|
| info       | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom error information |

### sch\_error\_record\_1\_suppl\_info

**Address:** 0x421A0FA0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Supplemental information register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                        |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------------|
| suppl_info | 0          | 64        | true     | WARL   | 0x0         | Reserved for custom supplemental error information |

### sch\_error\_record\_1\_timestamp

**Address:** 0x421A0FA8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Timestamp register of error record 1

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                   |
|------------|------------|-----------|----------|--------|-------------|-------------------------------|
| timestamp  | 0          | 64        | true     | WARL   | 0x0         | Timestamp of the error record |

### sch\_error\_record\_1\_locator

**Address:** 0x421A0FB0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** Locator register of error record 1

| Field Name  | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                          |
|-------------|------------|-----------|----------|--------|-------------|--------------------------------------------------------------------------------------|
| description | 0          | 32        | true     | WARL   | 0x0         | Contains description of the error event telling what happened [descriptor, instance] |
| location    | 32         | 32        | true     | WARL   | 0x0         | Contains location of the error event telling where it happened [chip, ip, instance]  |

## sch\_acb\_cfg

**Address:** 0x421A0FC0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000040

**Description:** None

| Field Name            | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                              |
|-----------------------|------------|-----------|----------|-----------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| slow_ordered          | 0          | 1         | false    | WARL      | 0x0         | If 1, disables streaming ordered requests. ACB will not accept a new req with the same AXID as an outstanding request.                                   |
| cacheable_enable      | 1          | 1         | false    | WARL      | 0x0         | If 0, Generate non-snoopable request opcode, ReadNoSnp/WriteNoSnp for scratchpad requests. If 1, Generate snoopable request opcode, ReadOnce/WriteUnique |
| common_mr_rd_nw_id    | 2          | 1         | false    | WARL      | 0x0         | Select which SC network read requests to common SC MMRs are sent to                                                                                      |
| common_mr_rd_slice_id | 3          | 3         | false    | WARL      | 0x0         | Select which SC slice read requests to common SC MMRs are sent to                                                                                        |
| hwa_write_policy      | 6          | 2         | false    | WARL      | 0x1         | Determines how new hardware assertions are logged in sch_hardware_assertion. 0=disabled, 1=log first, 2=log latest, 3=log worst (highest hwatype val)    |
| unused                | 8          | 56        | false    | Read-Only | 0x0         |                                                                                                                                                          |

## sch\_cab\_chicken

**Address:** 0x421A0FC8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** None

| Field Name                            | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                       |
|---------------------------------------|------------|-----------|----------|-----------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| urt_disable                           | 0          | 16        | false    | WARL      | 0x0         | Disable URTs used for protocol conversion. It decides number of URTs in parallel operation                                                                        |
| coarse_clock_gating_disable           | 16         | 1         | false    | WARL      | 0x0         | If 1, CAB input clocks will be free running                                                                                                                       |
| coarse_clock_gating_hysteresis_enable | 17         | 1         | false    | WARL      | 0x0         | If 1, enables hysteresis for gated CAB input clocks                                                                                                               |
| fine_clock_gating_disable             | 18         | 1         | false    | WARL      | 0x0         | If 1, all power gated clocks inside CAB will be free running                                                                                                      |
| fine_clock_gating_hysteresis_enable   | 19         | 1         | false    | WARL      | 0x0         | Reserved for future clock gating feature. If 1, enables hysteresis for power gated clocks inside CAB. Hysteresis is not currently used for CAB fine clock gating. |
| rsvd                                  | 20         | 8         | false    | WARL      | 0x0         |                                                                                                                                                                   |
| unused                                | 28         | 36        | false    | Read-Only | 0x0         |                                                                                                                                                                   |

## sch\_acb\_chicken

**Address:** 0x421A0FD0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** None

| Field Name                            | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                          |
|---------------------------------------|------------|-----------|----------|-----------|-------------|------------------------------------------------------------------------------------------------------|
| rd_urt_disable                        | 0          | 16        | false    | WARL      | 0x0         | Disable rd URTs used for protocol conversion. It decides number of URTs in parallel operation        |
| wr_urt_disable                        | 16         | 16        | false    | WARL      | 0x0         | Disable wr URTs used for protocol conversion. It decides number of URTs in parallel operation        |
| io_coherency_disable                  | 32         | 1         | false    | WARL      | 0x0         | If 1, ACB will respond with DECERR when it receives a request outside the SC/SCB domains (MMR or SP) |
| coarse_clock_gating_disable           | 33         | 1         | false    | WARL      | 0x0         | If 1, ACB input clocks will be free running                                                          |
| coarse_clock_gating_hysteresis_enable | 34         | 1         | false    | WARL      | 0x0         | If 1, enables hysteresis for gated ACB input clocks                                                  |
| fine_clock_gating_disable             | 35         | 1         | false    | WARL      | 0x0         | If 1, all power gated clocks inside ACB will be free running                                         |
| fine_clock_gating_hysteresis_enable   | 36         | 1         | false    | WARL      | 0x0         | If 1, enables hysteresis for power gated clocks inside ACB                                           |
| ras_low_priority_signal_disable       | 37         | 1         | false    | WARL      | 0x0         | If 1, disables ras low priority signaling                                                            |
| ras_high_priority_signal_disable      | 38         | 1         | false    | WARL      | 0x0         | If 1, disables ras high priority signaling                                                           |
| ras_platform_signal_disable           | 39         | 1         | false    | WARL      | 0x0         | If 1, disables ras platform signaling                                                                |
| rsvd                                  | 40         | 8         | false    | WARL      | 0x0         |                                                                                                      |
| unused                                | 48         | 16        | false    | Read-Only | 0x0         |                                                                                                      |

## sch\_fuse

**Address:** 0x421A0FD8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** None

| Field Name    | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                              |
|---------------|------------|-----------|----------|-----------|-------------|--------------------------------------------------------------------------|
| harvest_starp | 0          | 8         | false    | WARL      | 0x0         | 1 bit / 4 ways (total 8 bit / 32 ways possible)<br>1: enable 0: disabled |
| reserved_9_8  | 8          | 2         | false    | Read-Only | 0x0         |                                                                          |

|                           |    |    |       |           |     |                                                                                                                                                                   |
|---------------------------|----|----|-------|-----------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| export_restriction_enable | 10 | 1  | false | WARL      | 0x0 | Export restriction enable.                                                                                                                                        |
| io_coherency_disable      | 11 | 1  | false | WARL      | 0x0 | If 1, ACB will respond with DECERR when it receives a request outside the SC/SCB domains (MMR or SP)                                                              |
| reserved_15_12            | 12 | 4  | false | Read-Only | 0x0 |                                                                                                                                                                   |
| core_0_enable             | 16 | 1  | false | WARL      | 0x0 |                                                                                                                                                                   |
| core_0_vid                | 17 | 3  | false | WARL      | 0x0 |                                                                                                                                                                   |
| core_1_enable             | 20 | 1  | false | WARL      | 0x0 |                                                                                                                                                                   |
| core_1_vid                | 21 | 3  | false | WARL      | 0x0 |                                                                                                                                                                   |
| core_2_enable             | 24 | 1  | false | WARL      | 0x0 |                                                                                                                                                                   |
| core_2_vid                | 25 | 3  | false | WARL      | 0x0 |                                                                                                                                                                   |
| core_3_enable             | 28 | 1  | false | WARL      | 0x0 |                                                                                                                                                                   |
| core_3_vid                | 29 | 3  | false | WARL      | 0x0 |                                                                                                                                                                   |
| core_4_enable             | 32 | 1  | false | WARL      | 0x0 |                                                                                                                                                                   |
| core_4_vid                | 33 | 3  | false | WARL      | 0x0 |                                                                                                                                                                   |
| core_5_enable             | 36 | 1  | false | WARL      | 0x0 |                                                                                                                                                                   |
| core_5_vid                | 37 | 3  | false | WARL      | 0x0 |                                                                                                                                                                   |
| core_6_enable             | 40 | 1  | false | WARL      | 0x0 |                                                                                                                                                                   |
| core_6_vid                | 41 | 3  | false | WARL      | 0x0 |                                                                                                                                                                   |
| core_7_enable             | 44 | 1  | false | WARL      | 0x0 |                                                                                                                                                                   |
| core_7_vid                | 45 | 3  | false | WARL      | 0x0 |                                                                                                                                                                   |
| debug_enable              | 48 | 2  | false | WARL      | 0x0 | Debug enable. 00 - No Debug 01 - Default (JTAG based debug) 10 - AXI based debug 11 - Support both enabled at same time. Same cycle collision causes error packet |
| cla_enable                | 50 | 1  | false | WARL      | 0x0 | CLA enable.                                                                                                                                                       |
| trace_enable              | 51 | 1  | false | WARL      | 0x0 | Trace enable.                                                                                                                                                     |
| dst_enable                | 52 | 1  | false | WARL      | 0x0 | DST enable.                                                                                                                                                       |
| reserved_62_53            | 53 | 10 | false | Read-Only | 0x0 |                                                                                                                                                                   |
| lock                      | 63 | 1  | false | WARL      | 0x0 | Lock bit                                                                                                                                                          |

## sch\_unused\_6

**Address:** 0x421A0FF0

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** None

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|        |   |    |       |           |     |  |
|--------|---|----|-------|-----------|-----|--|
| unused | 0 | 64 | false | Read-Only | 0x0 |  |
|--------|---|----|-------|-----------|-----|--|

### sch\_unused\_7

**Address:** 0x421A0FF8

**cluster\_mask:** 0x01E00000

**slice\_mask:** 0x0000F000

**Size:** 64 bits

**Reset Value:** 0x0000000000000000

**Description:** None

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description |
|------------|------------|-----------|----------|-----------|-------------|-------------|
| unused     | 0          | 64        | false    | Read-Only | 0x0         |             |

### thub\_controller\_ip\_ver\_reg

**Address:** 0x0000

**Size:** 32 bits

**Reset Value:** 0x00200000

**Description:** THub Controller IP Version Reg

| Field Name    | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                 |
|---------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| step_version  | 0          | 10        | false    | Read Only | 0x0         | Represents step version of the IP. Incremented for bug fixes of the IP_x000B_Reset to '0' if either MAJOR_VERSION or MINOR_VERSION updates. |
| minor_version | 10         | 11        | false    | Read Only | 0x0         | Represents minor version of the IP. Incremented for feature changes of the IP_x000B_Reset to '0' if MAJOR_VERSION update.                   |
| major_version | 21         | 11        | false    | Read Only | 0x1         | Represents major version of the IP. Incremented for major architecture changes of the IP                                                    |

### thub\_control\_reg

**Address:** 0x0004

**Size:** 32 bits

**Reset Value:** 0x00000001

**Description:** THub control Reg

| Field Name      | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                         |
|-----------------|------------|-----------|----------|--------|-------------|---------------------------------------------------------------------------------------------------------------------|
| en_ts           | 0          | 1         | false    | WARL   | 0x1         | Temp Hub enable/disable control0x0: TempHub is disabled0x1: TempHub is enabled(default)                             |
| aggr_mode       | 1          | 1         | false    | WARL   | 0x0         | Sensor data aggregation mode0x0: Weighted average0x1: Max value                                                     |
| flush_req       | 2          | 1         | false    | WARL   | 0x0         | flush stale sensor data when returning from C* state0x0: Flush request de-asserted0x1: Flush request set            |
| tj_max_override | 3          | 1         | false    | WARL   | 0x0         | TJ Max override for debug. Used when TJ_MAX_TJ_SHUT_OVERRIDE_CTRL is set0x0: TJ Max de-asserted0x1: TJ Max asserted |

|                                  |    |    |       |      |     |                                                                                                                                                                                                                             |
|----------------------------------|----|----|-------|------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tj_shutdown_override             | 4  | 1  | false | WARL | 0x0 | TJ Shutdown override for debug. Used when TJ_MAX_TJ_SHUT_OVERRIDE_CTRL is set0x0: TJ Shutdown de-asserted0x1: TJ Shutdown asserted                                                                                          |
| sens_data_override               | 5  | 16 | false | WARL | 0x0 | Sensor data override for debug. Used when SENS_DATA_TO_CORE_OVERRIDE_CTRL or SENS_DATA_TO_CORE_OVERRIDE_CTRL are set                                                                                                        |
| sens_data_to_core_override_ctrl  | 21 | 1  | false | WARL | 0x0 | Sensors data to core override control0x0: Sensor data to core from sensors(default)0x1: Sensor data to core is from SENS_DATA_OVERRIDE                                                                                      |
| snapshot_override_ctrl           | 22 | 1  | false | WARL | 0x0 | Sensors snapshot data override control0x0: Sensor snapshot data controlled by sensors(default)0x1: Sensor snapshot data controlled by SENS_DATA_OVERRIDE                                                                    |
| tj_max_tj_shutdown_override_ctrl | 23 | 1  | false | WARL | 0x0 | TJ Max, TJ Shutdown override control0x0: TJ Max, TJ Shutdown controlled by sensors(default)0x1: TJ Max, TJ shutdown controlled by TJ_MAX_OVERRIDE, TJ_SHUTDOWN_OVERRIDE respectively                                        |
| calibration_mode                 | 24 | 1  | false | WARL | 0x0 | Calibration mode to override EN_TS, BJT_SEL_TS, SOC_TS control using thub_spare_control*_reg0x0: EN_TS, BJT_SEL_TS, SOC_TS controled by HW FSM(default)0x1: EN_TS, BJT_SEL_TS, SOC_TS controlled by thub_spare_control*_reg |
| spare                            | 25 | 7  | false | WARL | 0x0 | Reserved                                                                                                                                                                                                                    |

## thub\_sens\_weight0\_reg

**Address:** 0x0008

**Size:** 32 bits

**Reset Value:** 0x11111111

**Description:** THub Sensor weight 0 Reg

| Field Name   | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                 |
|--------------|------------|-----------|----------|--------|-------------|---------------------------------------------|
| sens0_weight | 0          | 4         | false    | WARL   | 0x1         | Sensor0 weight for weighted avg calculation |
| sens1_weight | 4          | 4         | false    | WARL   | 0x1         | Sensor1 weight for weighted avg calculation |
| sens2_weight | 8          | 4         | false    | WARL   | 0x1         | Sensor2 weight for weighted avg calculation |
| sens3_weight | 12         | 4         | false    | WARL   | 0x1         | Sensor3 weight for weighted avg calculation |
| sens4_weight | 16         | 4         | false    | WARL   | 0x1         | Sensor4 weight for weighted avg calculation |
| sens5_weight | 20         | 4         | false    | WARL   | 0x1         | Sensor5 weight for weighted avg calculation |
| sens6_weight | 24         | 4         | false    | WARL   | 0x1         | Sensor6 weight for weighted avg calculation |
| sens7_weight | 28         | 4         | false    | WARL   | 0x1         | Sensor7 weight for weighted avg calculation |

### thub\_sens\_weight1\_reg

**Address:** 0x000C

**Size:** 32 bits

**Reset Value:** 0x11111111

**Description:** THub Sensor weight 1 Reg

| Field Name    | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                  |
|---------------|------------|-----------|----------|--------|-------------|----------------------------------------------|
| sens8_weight  | 0          | 4         | false    | WARL   | 0x1         | Sensor8 weight for weighted avg calculation  |
| sens9_weight  | 4          | 4         | false    | WARL   | 0x1         | Sensor9 weight for weighted avg calculation  |
| sens10_weight | 8          | 4         | false    | WARL   | 0x1         | Sensor10 weight for weighted avg calculation |
| sens11_weight | 12         | 4         | false    | WARL   | 0x1         | Sensor11 weight for weighted avg calculation |
| sens12_weight | 16         | 4         | false    | WARL   | 0x1         | Sensor12 weight for weighted avg calculation |
| sens13_weight | 20         | 4         | false    | WARL   | 0x1         | Sensor13 weight for weighted avg calculation |
| sens14_weight | 24         | 4         | false    | WARL   | 0x1         | Sensor14 weight for weighted avg calculation |
| sens15_weight | 28         | 4         | false    | WARL   | 0x1         | Sensor15 weight for weighted avg calculation |

### thub\_sens\_disable\_reg

**Address:** 0x0010

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sensor disable Reg

| Field Name   | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                 |
|--------------|------------|-----------|----------|--------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sens_disable | 0          | 16        | false    | WARL   | 0x0         | Disables Sensor from sampling. One bit per sensor. Used to speed up sampling rate(during C* exit, powerup)0x0: Sensor enabled for sampling0x1: Sensor disabled for sampling |
| spare        | 16         | 16        | false    | WARL   | 0x0         | Reserved                                                                                                                                                                    |

### thub\_wait\_event\_0\_reg

**Address:** 0x0014

**Size:** 32 bits

**Reset Value:** 0x0001FFFF

**Description:** THub sensor wait event0 Reg

| Field Name           | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                        |
|----------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------------------------------|
| thub_counter_preset0 | 0          | 32        | false    | WARL   | 0x1FFFF     | THub preset value. Used for controlling EN_TS assertion to SOC_TS assertion timing |

### thub\_wait\_event\_1\_reg

**Address:** 0x0018

**Size:** 32 bits

**Reset Value:** 0x00001FFF

**Description:** THub sensor wait event1 Reg

| Field Name           | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                            |
|----------------------|------------|-----------|----------|--------|-------------|----------------------------------------------------------------------------------------|
| thub_counter_preset1 | 0          | 32        | false    | WARL   | 0x1FFF      | THub preset value. Used for controlling EOC_TS assertion to SOC_TS de-assertion timing |

### thub\_wait\_event\_2\_reg

**Address:** 0x001C

**Size:** 32 bits

**Reset Value:** 0x0000000F

**Description:** THub sensor wait event2 Reg

| Field Name           | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                         |
|----------------------|------------|-----------|----------|--------|-------------|-------------------------------------------------------------------------------------|
| thub_counter_preset2 | 0          | 32        | false    | WARL   | 0xF         | THub preset value. Used for controlling time interval between successive iterations |

### thub\_status\_reg

**Address:** 0x0020

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub Status Reg

| Field Name      | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                   |
|-----------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| conv_stat       | 0          | 1         | true     | Read Only | 0x0         | Indicates Temp sensor data conversion status 0x0: Conversion is not in progress 0x1: Conversion in progress                                                   |
| sens_sel        | 1          | 4         | true     | Read Only | 0x0         | Indicates Temp sensor # selected for data conversion. Need to qualify with CONV_STAT.                                                                         |
| eoc_stat        | 5          | 1         | true     | Read Only | 0x0         | Indicates Temp sensor data conversion complete status 0x0: Conversion is not completed 0x1: Conversion completed                                              |
| flush_ack       | 6          | 1         | true     | Read Only | 0x0         | ACK for Flush req. Asserted after FLUSH_REQ and after flush complete. De-asserted after FLUSH_REQ de-asserted 0x0: Flush ongoing/no flush 0x1: Flush complete |
| first_conv_done | 7          | 1         | true     | Read Only | 0x0         | Indicates first temp sensors data conversion complete after power-up, flush and C*exit                                                                        |
| spare           | 8          | 24        | true     | Read Only | 0x0         | Reserved                                                                                                                                                      |

### thub\_spare\_control\_0\_reg

**Address:** 0x0030

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub spare control 0 Reg

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| spare      | 0          | 32        | false    | WARL   | 0x0         | Spare bits  |

### thub\_spare\_control\_1\_reg

**Address:** 0x0034

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub spare control 1 Reg

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| spare      | 0          | 32        | false    | WARL   | 0x0         | Spare bits  |

### thub\_spare\_control\_2\_reg

**Address:** 0x0038

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub spare control 2 Reg

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
| spare      | 0          | 32        | false    | WARL   | 0x0         | Spare bits  |

### thub\_spare\_control\_3\_reg

**Address:** 0x003C

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub spare control 3 Reg

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                                                                                                                                                                                                  |
|------------|------------|-----------|----------|--------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spare      | 0          | 31        | false    | WARL   | 0x0         | Spare bits                                                                                                                                                                                                                                   |
| valid      | 31         | 1         | false    | WARL   | 0x0         | Valid bit for bus synchornization. Two way handshake protocol. To be set after writing all thub_spare_control_*_reg data and when thub_spare_status_1_reg[Ready] is low. To be de-asserted only when thub_spare_status_1_reg[Ready] is high. |

### thub\_spare\_status\_0\_reg

**Address:** 0x0040

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub spare status 0 Reg

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description |
|------------|------------|-----------|----------|-----------|-------------|-------------|
| spare      | 0          | 32        | true     | Read Only | 0x0         | Spare bits  |

### thub\_spare\_status\_1\_reg

**Address:** 0x0044

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub spare status 1 Reg

| Field Name | Bit Offset | Bit Width | Volatile | Access    | Reset Value | Description                                                                                                                                                                                                                           |
|------------|------------|-----------|----------|-----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spare      | 0          | 31        | true     | Read Only | 0x0         | Spare bits                                                                                                                                                                                                                            |
| ready      | 31         | 1         | true     | Read Only | 0x0         | Ready indication for bus synchornization. Indicates thub_spare_control_*_reg data sampled at destination. Two way handshake protocol. Set when thub_spare_control_3_reg[Valid] set, unset after thub_spare_control_3_reg[Valid] unset |

### thub\_threshold\_param\_reg

**Address:** 0x0050

**Size:** 32 bits

**Reset Value:** 0x0FFF0FFF

**Description:** THub Threshold Parameter Reg

| Field Name            | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|-----------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| tj_max_thres hold     | 0          | 15        | false    | WARL   | 0xFFF       | Threshold at which TJ Max is asserted                      |
| spare                 | 15         | 1         | false    | WARL   | 0x0         | Spare bits                                                 |
| tj_shutdown_threshold | 16         | 15        | false    | WARL   | 0xFFF       | Threshold at which TJ Shutdown is asserted                 |
| lock                  | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_cal\_control\_0\_reg

**Address:** 0x0054

**Size:** 32 bits

**Reset Value:** 0x07000000

**Description:** THub calibration conrol 0 Reg

| Field Name     | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                              |
|----------------|------------|-----------|----------|--------|-------------|----------------------------------------------------------|
| slope_con_vs   | 0          | 7         | false    | WARL   | 0x0         | Slope control for voltage output code                    |
| avg_mode_ts    | 7          | 2         | false    | WARL   | 0x0         | Average mode control ports for digital filter            |
| slope_con_ts   | 9          | 7         | false    | WARL   | 0x0         | Slope conol ports for temperature output code            |
| offset_con_ts  | 16         | 4         | false    | WARL   | 0x0         | offset control ports for temperature output code         |
| bgr_tc_con_t s | 20         | 2         | false    | WARL   | 0x0         | Reference voltage trimming ports for reference generator |
| bgr_i_con_ts   | 22         | 2         | false    | WARL   | 0x0         | BJT emitter current control ports in reference generator |
| slope_comp     | 24         | 4         | false    | WARL   | 0x7         | Slope compensation for low temperature                   |
| spare          | 28         | 2         | false    | WARL   | 0x0         | Spare bits                                               |

|          |    |   |       |      |     |                                                                                                |
|----------|----|---|-------|------|-----|------------------------------------------------------------------------------------------------|
| cal_mode | 30 | 1 | false | WARL | 0x0 | Selects 1-Point/2-point calibration0x0 : 1-Point calibration(Default)0x1 : 2-Point calibration |
| lock     | 31 | 1 | false | WARL | 0x0 | Lock Bit, once set write to the MMR is prevented until PoR                                     |

### thub\_cal\_control\_1\_reg

**Address:** 0x0058

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub calibration control 0 Reg

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| spare      | 0          | 31        | false    | WARL   | 0x0         | Spare bits                                                 |
| lock       | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_sens\_data\_cal\_0\_reg

**Address:** 0x005C

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name        | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|-------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| sens0_data_at_25c | 0          | 15        | false    | WARL   | 0x0         | Sensor 0 measured data at 25C                              |
| spare             | 15         | 1         | false    | WARL   | 0x0         | Spare bits                                                 |
| sens0_data_at_85c | 16         | 15        | false    | WARL   | 0x0         | Sensor 0 measured data at 85C                              |
| lock              | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_sens\_data\_cal\_1\_reg

**Address:** 0x0060

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name        | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|-------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| sens1_data_at_25c | 0          | 15        | false    | WARL   | 0x0         | Sensor 1 measured data at 25C                              |
| spare             | 15         | 1         | false    | WARL   | 0x0         | Spare bits                                                 |
| sens1_data_at_85c | 16         | 15        | false    | WARL   | 0x0         | Sensor 1 measured data at 85C                              |
| lock              | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_sens\_data\_cal\_2\_reg

**Address:** 0x0064

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name        | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|-------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| sens2_data_at_25c | 0          | 15        | false    | WARL   | 0x0         | Sensor 2 measured data at 25C                              |
| spare             | 15         | 1         | false    | WARL   | 0x0         | Spare bits                                                 |
| sens2_data_at_85c | 16         | 15        | false    | WARL   | 0x0         | Sensor 2 measured data at 85C                              |
| lock              | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_sens\_data\_cal\_3\_reg

**Address:** 0x0068

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name        | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|-------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| sens3_data_at_25c | 0          | 15        | false    | WARL   | 0x0         | Sensor 3 measured data at 25C                              |
| spare             | 15         | 1         | false    | WARL   | 0x0         | Spare bits                                                 |
| sens3_data_at_85c | 16         | 15        | false    | WARL   | 0x0         | Sensor 3 measured data at 85C                              |
| lock              | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_sens\_data\_cal\_4\_reg

**Address:** 0x006C

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name        | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|-------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| sens4_data_at_25c | 0          | 15        | false    | WARL   | 0x0         | Sensor 4 measured data at 25C                              |
| spare             | 15         | 1         | false    | WARL   | 0x0         | Spare bits                                                 |
| sens4_data_at_85c | 16         | 15        | false    | WARL   | 0x0         | Sensor 4 measured data at 85C                              |
| lock              | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_sens\_data\_cal\_5\_reg

**Address:** 0x0070

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description |
|------------|------------|-----------|----------|--------|-------------|-------------|
|------------|------------|-----------|----------|--------|-------------|-------------|

|                   |    |    |       |      |     |                                                            |
|-------------------|----|----|-------|------|-----|------------------------------------------------------------|
| sens5_data_at_25c | 0  | 15 | false | WARL | 0x0 | Sensor 5 measured data at 25C                              |
| spare             | 15 | 1  | false | WARL | 0x0 | Spare bits                                                 |
| sens5_data_at_85c | 16 | 15 | false | WARL | 0x0 | Sensor 5 measured data at 85C                              |
| lock              | 31 | 1  | false | WARL | 0x0 | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_sens\_data\_cal\_6\_reg

**Address:** 0x0074

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name        | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|-------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| sens6_data_at_25c | 0          | 15        | false    | WARL   | 0x0         | Sensor 6 measured data at 25C                              |
| spare             | 15         | 1         | false    | WARL   | 0x0         | Spare bits                                                 |
| sens6_data_at_85c | 16         | 15        | false    | WARL   | 0x0         | Sensor 6 measured data at 85C                              |
| lock              | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_sens\_data\_cal\_7\_reg

**Address:** 0x0078

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name        | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|-------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| sens7_data_at_25c | 0          | 15        | false    | WARL   | 0x0         | Sensor 7 measured data at 25C                              |
| spare             | 15         | 1         | false    | WARL   | 0x0         | Spare bits                                                 |
| sens7_data_at_85c | 16         | 15        | false    | WARL   | 0x0         | Sensor 7 measured data at 85C                              |
| lock              | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_sens\_data\_cal\_8\_reg

**Address:** 0x007C

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name        | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                   |
|-------------------|------------|-----------|----------|--------|-------------|-------------------------------|
| sens8_data_at_25c | 0          | 15        | false    | WARL   | 0x0         | Sensor 8 measured data at 25C |
| spare             | 15         | 1         | false    | WARL   | 0x0         | Spare bits                    |
| sens8_data_at_85c | 16         | 15        | false    | WARL   | 0x0         | Sensor 8 measured data at 85C |

|      |    |   |       |      |     |                                                            |
|------|----|---|-------|------|-----|------------------------------------------------------------|
| lock | 31 | 1 | false | WARL | 0x0 | Lock Bit, once set write to the MMR is prevented until PoR |
|------|----|---|-------|------|-----|------------------------------------------------------------|

### thub\_sens\_data\_cal\_9\_reg

**Address:** 0x0080

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name        | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|-------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| sens9_data_at_25c | 0          | 15        | false    | WARL   | 0x0         | Sensor 9 measured data at 25C                              |
| spare             | 15         | 1         | false    | WARL   | 0x0         | Spare bits                                                 |
| sens9_data_at_85c | 16         | 15        | false    | WARL   | 0x0         | Sensor 9 measured data at 85C                              |
| lock              | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_sens\_data\_cal\_10\_reg

**Address:** 0x0084

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name         | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|--------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| sens10_data_at_25c | 0          | 15        | false    | WARL   | 0x0         | Sensor 10 measured data at 25C                             |
| spare              | 15         | 1         | false    | WARL   | 0x0         | Spare bits                                                 |
| sens10_data_at_85c | 16         | 15        | false    | WARL   | 0x0         | Sensor 10 measured data at 85C                             |
| lock               | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_sens\_data\_cal\_11\_reg

**Address:** 0x0088

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name         | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|--------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| sens11_data_at_25c | 0          | 15        | false    | WARL   | 0x0         | Sensor 11 measured data at 25C                             |
| spare              | 15         | 1         | false    | WARL   | 0x0         | Spare bits                                                 |
| sens11_data_at_85c | 16         | 15        | false    | WARL   | 0x0         | Sensor 11 measured data at 85C                             |
| lock               | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_sens\_data\_cal\_12\_reg

**Address:** 0x008C

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name         | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|--------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| sens12_data_at_25c | 0          | 15        | false    | WARL   | 0x0         | Sensor 12 measured data at 25C                             |
| spare              | 15         | 1         | false    | WARL   | 0x0         | Spare bits                                                 |
| sens12_data_at_85c | 16         | 15        | false    | WARL   | 0x0         | Sensor 12 measured data at 85C                             |
| lock               | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_sens\_data\_cal\_13\_reg

**Address:** 0x0090

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name         | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|--------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| sens13_data_at_25c | 0          | 15        | false    | WARL   | 0x0         | Sensor 13 measured data at 25C                             |
| spare              | 15         | 1         | false    | WARL   | 0x0         | Spare bits                                                 |
| sens13_data_at_85c | 16         | 15        | false    | WARL   | 0x0         | Sensor 13 measured data at 85C                             |
| lock               | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_sens\_data\_cal\_14\_reg

**Address:** 0x0094

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name         | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|--------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| sens14_data_at_25c | 0          | 15        | false    | WARL   | 0x0         | Sensor 14 measured data at 25C                             |
| spare              | 15         | 1         | false    | WARL   | 0x0         | Spare bits                                                 |
| sens14_data_at_85c | 16         | 15        | false    | WARL   | 0x0         | Sensor 14 measured data at 85C                             |
| lock               | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

### thub\_sens\_data\_cal\_15\_reg

**Address:** 0x0098

**Size:** 32 bits

**Reset Value:** 0x00000000

**Description:** THub sens data calibration 0 Reg

| Field Name         | Bit Offset | Bit Width | Volatile | Access | Reset Value | Description                                                |
|--------------------|------------|-----------|----------|--------|-------------|------------------------------------------------------------|
| sens15_data_at_25c | 0          | 15        | false    | WARL   | 0x0         | Sensor 15 measured data at 25C                             |
| spare              | 15         | 1         | false    | WARL   | 0x0         | Spare bits                                                 |
| sens15_data_at_85c | 16         | 15        | false    | WARL   | 0x0         | Sensor 15 measured data at 85C                             |
| lock               | 31         | 1         | false    | WARL   | 0x0         | Lock Bit, once set write to the MMR is prevented until PoR |

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